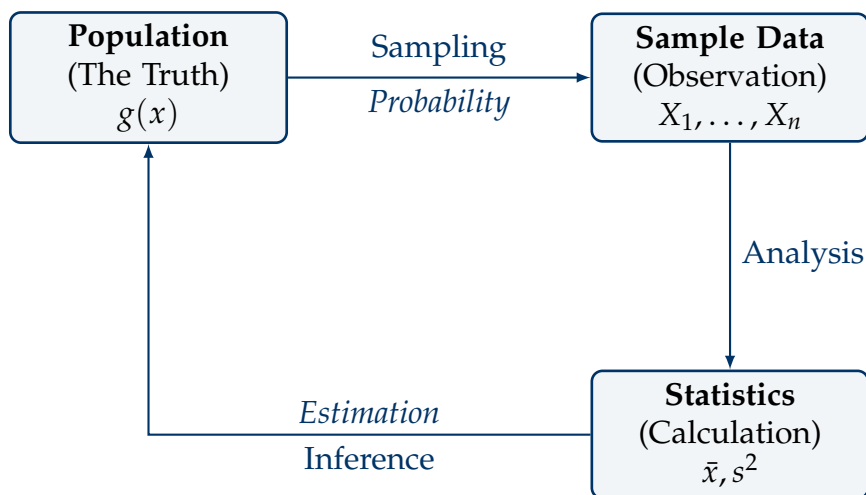

Statistical Inference Notes

May 6, 2026

1 Introduction to Inference

1.1 The Inference Cycle

The fundamental process of statistics flows between the **Population** and the **Sample**. We use the partial information derived from the sample to understand the total information of the population.



Definition 1.1: Statistical Inference

The method of inferring properties about the population based on the sample is called **Statistical Inference**.

Definition 1.2: Random Variable

Mathematically, the "data" comes from a function that maps the sample space (Ω) to real numbers ($\mathbb{R}$):

$$f : \Omega \rightarrow \mathbb{R}$$

This function is called a **Random Variable**.

1.2 Example Calculation

Given a sample dataset: $S = \{11, 2, 139\}$.

Example: Basic Statistics

- **Sample Mean ($\bar{x}$):**

$$\bar{x} = \frac{11 + 2 + 139}{3} = \frac{152}{3} \approx 50.67$$

- **Sample Range:**

$$\text{Max} - \text{Min} = 139 - 2 = 137$$

2 Frequency and Distributions

2.1 Definitions

- **Frequency:** The count of occurrences of an event.
- **Relative Frequency:**

$$\frac{\text{Frequency}}{\text{Total Count}} \approx \text{Probability} \quad (\text{Ratio between 0 and 1})$$

2.2 Population Example: Heights

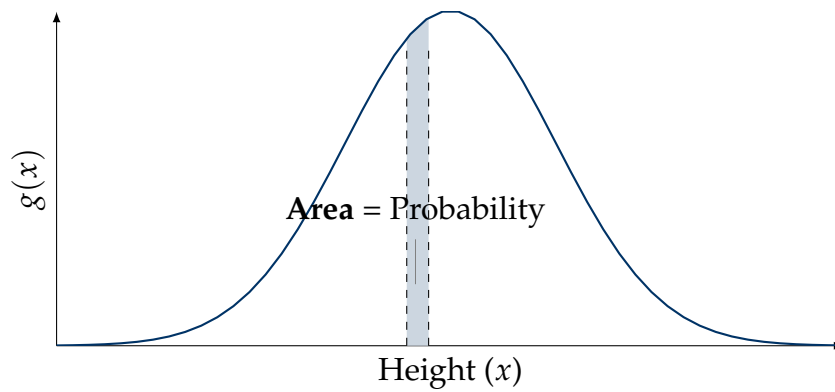
Consider the population of **all 18-year-old boys in India in 2025**. Let their height be represented by a variable X . We can characterize this population using a function $g(x)$, known as the **Probability Density Function (PDF)**.

Properties of $g(x)$:

1. $g(x) \geq 0$ (Non-negative).
2. $\int_{-\infty}^{\infty} g(x) dx = 1$ (Total probability is 1).

To find the proportion of boys with height between 4 and 4.25 units:

$$P(4 \leq X \leq 4.25) = \int_4^{4.25} g(x) dx$$



3 Properties of Distribution Functions

3.1 Connection between Population and PDF

If $g(x)$ is the PDF, we define key population parameters:

- **Median ($\tilde{m}$):** The point that splits the area in half.

$$\int_{-\infty}^{\tilde{m}} g(x) dx = 0.5$$

- **Mean (μ):** The weighted average (center of mass).

$$\mu = \int_{-\infty}^{\infty} x \cdot g(x) dx$$

3.2 Key Question

*If we know only the function of the graph (the theoretical model), can we derive properties of the data?
Will these match the calculations from the observed data?*

Answer

Yes, we can derive theoretical properties (Parameters like μ, σ^2) from the function $g(x)$. However, these will **not be exactly the same** as the values calculated from a finite sample.

- Values from $g(x)$ are **Parameters** (Fixed, True values).
- Values from Data are **Statistics** (Estimates).

4 Sampling Distributions

4.1 Population (P) vs. Sample

Population Characteristics:

1. A complete collection of measurements.
2. **Non-Random** (The population truth is fixed).
3. Characterized by $g(x)$ (PDF or PMF).

Sample Definition: A sample of size n consists of random variables $X_1, X_2, \dots, X_n$ that are **i.i.d.** (Independent and Identically Distributed).

$$X_i \sim g(x)$$

5 Inference on Parameters

In real-world scenarios, the population function is usually unknown. We assume a "shape" (a parametric family) but do not know the specific values.

Example: Assume Population $\mathbb{P} \sim N(\mu, \sigma^2)$.

- The shape is Normal (Bell curve).
- The parameters μ (mean) and σ^2 (variance) are **unknown**.

5.1 The Goal

We have a sample $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$. We need to infer (μ, σ^2) .

Estimators

To find these unknown parameters, we use **Estimators** derived from our sample data:

1. Estimator for Mean (μ): We use the Sample Mean ($\bar{X}$):

$$\hat{\mu} = \bar{X} = \frac{1}{n} \sum_{i=1}^n X_i$$

2. Estimator for Variance (σ^2): We use the Sample Variance (S^2). Note the division by $n - 1$ to make it an unbiased estimator:

$$\hat{\sigma}^2 = S^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X})^2$$

6 Parametric Inference

Definition 6.1: Parametric Inference

When it is assumed that samples $X_1, X_2, \dots, X_n$ are i.i.d. according to some distribution $G(\theta)$, where:

- $G(\cdot)$ is the known form of the CDF.
- θ is the vector of **unknown parameters**.

The task of inferring θ based on the sample is called **Parametric Inference**.

Definition 6.2: Modelling

The task of identifying the form of $G(\theta)$ (the distribution family) is called **Modelling**.

7 Example 1: Agricultural Experiment

Scenario: A 25-acre land is split into 10,000 plots. A seed is planted in each. We want to find the proportion of germination.

7.1 Methodology

We examine a small sample of $n = 15$ plots.

- Notation: 1 = Germinated, 0 = Not Germinated.

Data Collection:

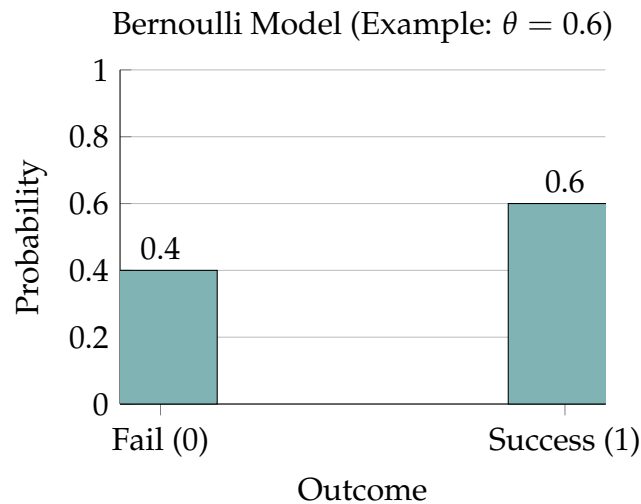
Plot #	1	2	...	15
Variable	X_1	X_2	...	X_{15}
Outcome	0	1	...	1

7.2 The Model (Bernoulli)

Let θ be the true proportion of germinated seeds.

$$X_i \sim \text{Bernoulli}(\theta); \quad \theta \in [0, 1]$$

$$P(X_i = 1) = \theta, \quad P(X_i = 0) = 1 - \theta$$



7.3 Estimation

$$\text{Estimate of } \theta \approx \bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$$

8 Random Samples vs. Realizations

! Important: CRITICAL DISTINCTION

1. **Random Sample** ($X_1, \dots, X_n$): These are discussed **BEFORE** the experiment. They are random variables. We talk about their probability distributions.
2. **Realizations** ($x_1, \dots, x_n$): These are obtained **AFTER** the experiment. These are fixed numbers. We talk about their specific values.

9 Example 2: Physics Experiment

Objective: Estimate the gravitational constant ' g '.

Physical Law: $g \approx \frac{4\pi^2 l}{T^2}$ (where $l = 0.75\text{m}$).

9.1 Data

We take 4 repeated measurements of period T .

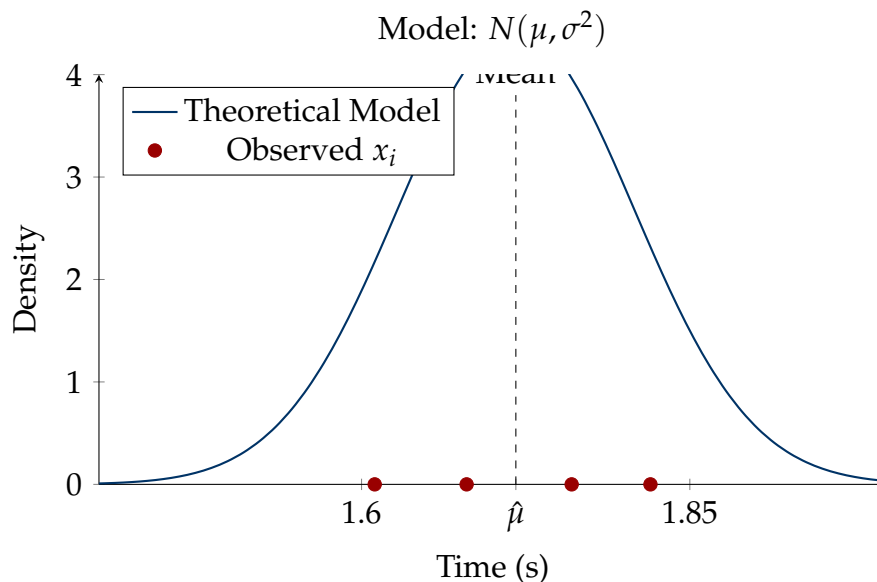
$$\text{Data } \mathbf{x} = \{1.82, 1.68, 1.61, 1.76\} \quad (n = 4)$$

9.2 Modelling Assumptions

We assume the measurement error is random and normally distributed.

$$X_i \stackrel{iid}{\sim} N(\mu, \sigma^2)$$

Here, μ represents the **true** period T_{true} .



9.3 Calculations (Inference)

Step 1: Estimate the Period ($\hat{T}$) We estimate the true period μ using the sample mean $\bar{x}$:

$$\hat{T} = \bar{x} = \frac{1.82 + 1.68 + 1.61 + 1.76}{4} = \frac{6.87}{4} = \mathbf{1.7175 \text{ sec}}$$

Step 2: Estimate the Uncertainty (Variance)

$$s^2 = \frac{1}{4-1} \sum_{i=1}^4 (x_i - 1.7175)^2 \approx \mathbf{0.0082}$$

Step 3: Estimate Gravity ($\hat{g}$) Using $l = 0.75$ meters and $\hat{T} = 1.7175$ s:

$$\hat{g} = \frac{4\pi^2(0.75)}{(1.7175)^2} = \frac{29.6088}{2.9498} \approx \mathbf{10.03 \text{ m/s}^2}$$

Conclusion

Based on our sample of 4 measurements, our point estimate for gravity is 10.03 m/s^2 . The difference from the standard 9.81 m/s^2 is likely due to measurement error (variance) or systematic bias in the pendulum length.

10 Statistical Models and Statistics

Model:

$$X_1, X_2, \dots, X_n \sim G(\theta)$$

where θ is a **parameter** (Unknown Constant).

Goal: Need to infer θ based on $\mathbf{X} = (X_1, X_2, \dots, X_n)$.

Example: Bernoulli Statistic Example

- θ : Population proportion of germinated seeds.
- Statistic $T_n(\mathbf{X}) = \frac{\# \text{ 1s in } X_1 \dots X_n}{n} = \frac{1}{n} \sum X_i$ (Sample Proportion).

Definition 10.1: Statistic

Any function of the samples (only) $\{X_1, X_2, \dots, X_n\}$ is called a **Statistic**.

$$T : \mathcal{X} \rightarrow \mathbb{R}$$

It must **not** depend on any unknown parameters.

Examples:

1. $T_1(\mathbf{X}) = \bar{X}_n$ (Valid)
2. $T_2(\mathbf{X}) = \max_i X_i - \min_i X_i$ (Valid)
3. [Critical Example] $T_3(\mathbf{X}; \theta) = \sum_{i=1}^n (X_i - \theta)$
 - This is **NOT** a statistic.
 - Reason: It depends on the unknown parameter θ .

11 Sample Space and Properties

Sample Space ($\mathcal{X}$): All possible values $\mathbf{X}$ can take.

- Example (Germination): $\mathcal{X} = \{(0, \dots, 0), \dots, (1, \dots, 1)\}$.
- $\mathcal{X}$ has 2^n components.

Properties of Statistic $T(\mathbf{X})$: 1. $T(\mathbf{X})$ is random. 2. $T(\mathbf{X})$ has a probability distribution.

Example:

$$T(\mathbf{X}) = 0 \quad \text{with prob.} \quad P(T(\mathbf{X}) = 0) = P(X_1 = 0, X_2 = 0, \dots, X_n = 0)$$

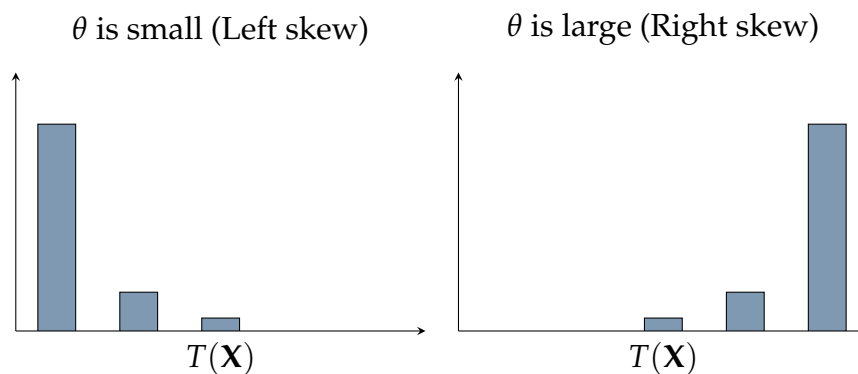
12 Sampling Distribution

The distribution of $T(\mathbf{X})$ is called the "**Sampling Distribution**".

Example 1: Bernoulli Model

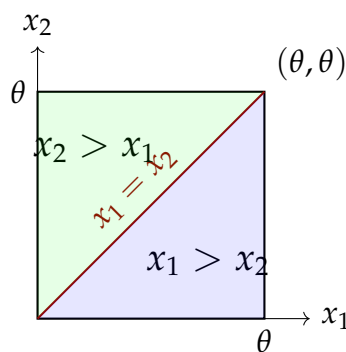
$$X_i \stackrel{iid}{\sim} \text{Ber}(\theta), \quad P(X_i = 1) = \theta, \quad P(X_i = 0) = 1 - \theta$$

Consider the histograms of $T(\mathbf{X})$ (Sample Proportion) for different θ :



12.1 Uniform Distribution Example ($n = 2$)

Let $X_i \stackrel{iid}{\sim} \text{Uniform}(0, \theta)$ for $i = 1, 2$. We visualize the sample space as a square of side θ :



Define Statistic $T(\mathbf{X})$ based on order:

$$T(\mathbf{X}) = \begin{cases} 1 & \text{if } X_1 \geq X_2 \quad (\text{Blue Region}) \\ 2 & \text{if } X_2 > X_1 \quad (\text{Green Region}) \end{cases}$$

Definition 12.1: Estimator

A statistic which is used to estimate $g(\theta)$ is called an **estimator** of $g(\theta)$.

13 Important Results (MGF Technique)

Example: Oscillation Time $X_i \stackrel{iid}{\sim} N(T_0, \sigma^2)$.

$$T(\mathbf{X}) = \bar{X}_n \sim N\left(T_0, \frac{\sigma^2}{n}\right)$$

Result: Linear Combination of Normals

If $X_i \stackrel{iid}{\sim} N(\mu_i, \sigma_i^2)$ for $i = 1, \dots, n$, then the linear combination $T(\mathbf{X}) = a_1X_1 + \dots + a_nX_n$ is distributed as:

$$T(\mathbf{X}) \sim N\left(\sum_{i=1}^n a_i\mu_i, \sum_{i=1}^n a_i^2\sigma_i^2\right)$$

Proof (MGF Technique): The Moment Generating Function (MGF) of X is

$$M_X(t) = E[e^{tX}] = e^{\mu t + \sigma^2 t^2 / 2}.$$

Let $T = \sum_{i=1}^n a_i X_i$.

$$M_T(t) = E[e^{tT}] = E\left[e^{t \sum a_i X_i}\right]$$

Product Law of Expectation

Since X_i are independent:

$$E\left[\prod e^{ta_i X_i}\right] = \prod E[e^{ta_i X_i}]$$

$$\begin{aligned} M_T(t) &= M_{X_1}(a_1 t) \cdot M_{X_2}(a_2 t) \dots M_{X_n}(a_n t) \\ &= e^{a_1 \mu_1 t + a_1^2 \sigma_1^2 t^2 / 2} \dots e^{a_n \mu_n t + a_n^2 \sigma_n^2 t^2 / 2} \\ &= e^{t(\sum a_i \mu_i) + \frac{t^2}{2}(\sum a_i^2 \sigma_i^2)} \end{aligned}$$

By recognizing the form of the Normal MGF, this implies:

$$T \sim N\left(\sum a_i \mu_i, \sum a_i^2 \sigma_i^2\right)$$

14 Order Statistics

More Summation Results:

- $X_i \stackrel{iid}{\sim} \text{Bin}(m_i, \theta) \implies \sum X_i \sim \text{Bin}(\sum m_i, \theta)$

- $X_i \stackrel{ind}{\sim} \text{Poisson}(\lambda_i) \implies \sum X_i \sim \text{Poisson}(\sum \lambda_i)$
- $X_i \stackrel{ind}{\sim} \text{Gamma}(\alpha_i, \beta) \implies \sum X_i \sim \text{Gamma}(\sum \alpha_i, \beta)$

Order Statistics Definitions: Let $X_1, \dots, X_n \stackrel{iid}{\sim} G(\theta)$.

- $T(\mathbf{X}) = X_{(n)} = \max\{X_1, \dots, X_n\}$ (Highest order statistic).
- $S(\mathbf{X}) = X_{(1)} = \min\{X_1, \dots, X_n\}$ (Lowest order statistic).

Derivation of CDF of $T(\mathbf{X})$ (The Max):

$$\begin{aligned} G_T(t) &= P(T(\mathbf{X}) \leq t) = P(\max\{X_1 \dots X_n\} \leq t) \\ &= P(X_1 \leq t, X_2 \leq t, \dots, X_n \leq t) \\ &= P(X_1 \leq t) \cdot P(X_2 \leq t) \dots P(X_n \leq t) \quad (\text{by independence}) \\ &= [F_X(t)]^n \end{aligned}$$

Example: Uniform Max Let $X_i \stackrel{iid}{\sim} U(0, \theta)$. The CDF is $F_X(t) = \frac{t}{\theta}$ for $0 \leq t \leq \theta$.

$$G_T(t) = \left(\frac{t}{\theta}\right)^n$$

PDF of T:

$$g_T(t) = \frac{d}{dt} G_T(t) = \frac{d}{dt} \left[\frac{t^n}{\theta^n} \right] = \frac{nt^{n-1}}{\theta^n}, \quad 0 \leq t \leq \theta$$

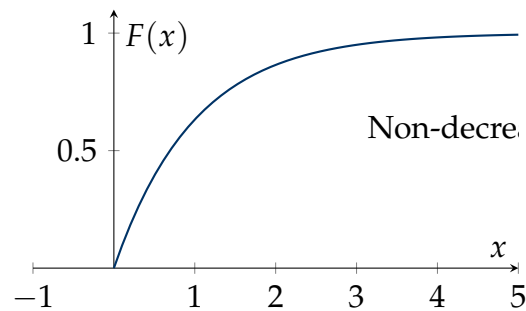
15 Supplementary Notes: Populations and Modeling

Definition: Population

A **Population** is a collection of measures. It is non-random and represents the complete set of data we are interested in.

The population is mathematically represented by the **Cumulative Distribution Function (CDF)**, denoted as $F(x)$.

- $F(-\infty) = 0$
- $F(+\infty) = 1$
- F is a **non-decreasing** function.



Example: Let the population $\mathbb{P}$ follow a Gamma distribution: $\mathbb{P} = \text{Gamma}(2,5)$. The relative frequency of $\mathbb{P}$ in the interval $(4,4.5)$ is given by the integral of the PDF:

$$\int_4^{4.5} \text{constant} \times x^{2-1} e^{-5x} dx$$

16 Sampling and Identical Distributions (Recap)

Since the entire population is rarely known, we rely on samples.

Concept: The Sample

A **Sample** consists of representative candidates of $\mathbb{P}$ to be investigated. Before observing the data point X_i , its value is unknown.

While we don't know the exact value, we calculate the probability of the value falling in a range using the CDF:

$$P(X_i \in [a, b]) = F(b) - F(a)$$

The I.I.D. Assumption

We typically assume the sample $X_1, \dots, X_n$ are:

1. **Identically Distributed:** They follow the same distribution (e.g., $X \sim \mathcal{N}(0,1)$).
2. **Independent:** One observation does not influence another.

17 Advanced Statistical Modeling

Often, the population function F is unknown. We use domain knowledge to guess the shape. This "art of deriving a shape" is called **Modeling**.

Model: Mixture Model Example: Income

Consider "Income per hour". There is a probability p of zero income (unemployment) and a probability $(1 - p)$ of positive income following a Gamma distribution.

$$X \sim p \cdot \delta(\cdot) + (1 - p)f(\cdot \mid \text{Gamma})$$

This implies a "spike" at 0 and a continuous tail for $x > 0$.

A **Parametric Model** assumes a form $f(x; \theta)$ with a finite number of unknown constants (parameters), such as α, β, p .

18 Summary: Parametric Inference

Definition: Inference

The task of parametric inference is to estimate the parameters (or functions of parameters) from the sample $X_1, \dots, X_n$.

Statistics and Sampling Distributions

A function of the sample $X_1, \dots, X_n$ is called a **Statistic** (e.g., the sample mean $\hat{\theta}$).

$$\hat{\theta}_n : \text{Sample Space} \rightarrow \mathbb{R}$$

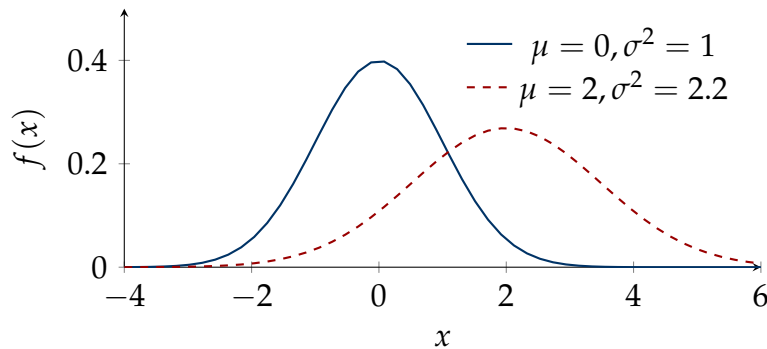
Crucial Note: Since the samples are random, a **statistic is also a random variable**. The distribution of this statistic is called the **Sampling Distribution**.

19 Finding Sampling Distributions

The distribution of a statistic $T_n(\mathbf{X})$ is called its **sampling distribution**. Since the distribution of the data $\mathbf{X}$ depends on parameters θ , the sampling distribution of T_n also typically depends on θ .

Example: Visualizing Parameter Dependence

Consider $X \sim N(\mu, \sigma^2)$. The shape is determined by $\theta = (\mu, \sigma^2)$.



Just as the PDF changes shape, the distribution of any statistic derived from X will change.

We generally use three main techniques to find sampling distributions:

1. **CDF Technique** (Distribution Function Method)
2. **MGF Technique** (Moment Generating Function)
3. **Transformation Technique** (Jacobian Method)

19.1 Technique I: The CDF Technique

This is a fundamental method involving first finding the Cumulative Distribution Function (CDF) and then differentiating it to find the PDF.

Steps:

1. Define the statistic $T = g(X)$.
2. Express the CDF: $F_T(t) = P(T \leq t) = P(g(X) \leq t)$.
3. Invert the inequality to express the probability in terms of X .
4. Integrate $f_X(x)$ over the region to find $F_T(t)$.
5. Differentiate: $f_T(t) = F'_T(t)$.

Example: Ex 3: Cauchy to Beta Transformation

Let $X \sim \text{Cauchy}(0, 1)$, i.e., $f_X(x) = \frac{1}{\pi(1+x^2)}$, $x \in \mathbb{R}$.

Consider the transformation $T = \frac{1}{1+X^2}$. Find the PDF of T .

1. Support: Since $x^2 \geq 0$, we have $0 < T \leq 1$.

2. CDF Derivation: For $0 < t < 1$:

$$\begin{aligned} F_T(t) &= P\left(\frac{1}{1+X^2} \leq t\right) = P\left(1+X^2 \geq \frac{1}{t}\right) \\ &= P\left(X^2 \geq \frac{1}{t} - 1\right) \\ &= 1 - P\left(X^2 < \frac{1}{t} - 1\right) \end{aligned}$$

By symmetry of the Cauchy distribution around 0:

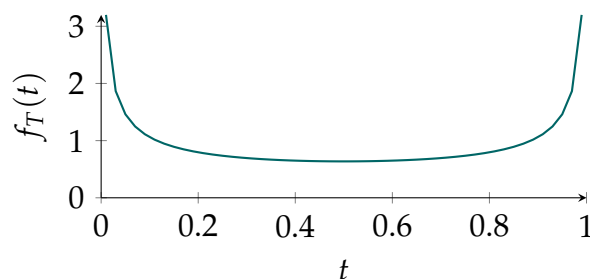
$$\begin{aligned} F_T(t) &= 1 - P\left(-\sqrt{\frac{1}{t}-1} < X < \sqrt{\frac{1}{t}-1}\right) \\ &= 1 - 2 \int_0^{\sqrt{1/t-1}} \frac{1}{\pi(1+x^2)} dx \\ &= 1 - \frac{2}{\pi} \left[\tan^{-1}(x) \right]_0^{\sqrt{1/t-1}} \\ &= 1 - \frac{2}{\pi} \tan^{-1}\left(\sqrt{\frac{1-t}{t}}\right) \end{aligned}$$

3. PDF Derivation: Differentiating $F_T(t)$ w.r.t t :

$$f_T(t) = \frac{d}{dt} F_T(t) = \frac{1}{\pi \sqrt{t(1-t)}}, \quad 0 < t < 1$$

Observation: This is the PDF of the **Beta** $\left(\frac{1}{2}, \frac{1}{2}\right)$ distribution (Arcsine distribution).

Resulting Distribution: Beta(0.5, 0.5)



Result: Example 4: Probability Integral Transform (PIT)

Let X be a continuous random variable with CDF $F_X(x)$.

Define $T = F_X(X)$. Then:

$$T \sim \text{Uniform}(0,1)$$

Proof:

$$P(T \leq t) = P(F_X(X) \leq t) = P(X \leq F_X^{-1}(t)) = F_X(F_X^{-1}(t)) = t$$

Since the CDF is t , the variable is $\text{Uniform}(0,1)$.

Example: Ex 5: Minimum of Exponentials

Let $X_i \stackrel{iid}{\sim} \text{Exp}(\lambda)$ for $i = 1, \dots, n$. ($f(x) = \lambda e^{-\lambda x}$).

Let $T_n = \min\{X_1, \dots, X_n\}$.

Using the CDF technique, it is often easier to work with the **Survival Function**

$P(T_n > t)$:

$$\begin{aligned} P(T_n > t) &= P(\min(X_i) > t) \\ &= P(X_1 > t \cap X_2 > t \cdots \cap X_n > t) \\ &= \prod_{i=1}^n P(X_i > t) \quad (\text{by independence}) \\ &= \prod_{i=1}^n e^{-\lambda t} = (e^{-\lambda t})^n \\ &= e^{-n\lambda t} \end{aligned}$$

The CDF is $F_{T_n}(t) = 1 - e^{-(n\lambda)t}$.

Differentiating gives the PDF: $f_{T_n}(t) = n\lambda e^{-(n\lambda)t}$.

Conclusion: The minimum of n i.i.d. exponentials is $\text{Exp}(n\lambda)$.

! Important: Next Steps

The CDF technique is powerful, but for sums of random variables (like convolution), other methods are often faster. We will next explore:

- **MGF Technique:** Useful for sums of independent variables ($T = \sum X_i$).
- **Transformation Technique:** Using the Jacobian Matrix for one-to-one mappings.

19.2 Technique II: The Transformation (Jacobian) Technique

While the CDF technique is universal, it can be cumbersome. For continuous random variables with smooth, one-to-one mappings, we use the **Transformation Technique** (also known as the Change of Variable method).

Definition 19.1: The Jacobian Formula

Let X be a continuous random variable with PDF $f_X(x)$.

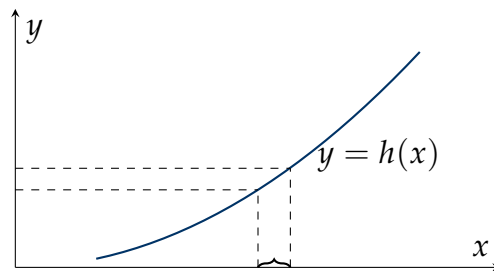
Let $Y = h(X)$, where h is a **one-to-one** (monotonic) and differentiable function.

The PDF of Y is given by:

$$f_Y(y) = f_X(x) \cdot \left| \frac{dx}{dy} \right|$$

where $x = h^{-1}(y)$ is expressed in terms of y , and $J = \frac{dx}{dy}$ is the **Jacobian** of the transformation.

Mapping Probability Mass: $f_X(x)dx = f_Y(y)dy$



Example: Ex 6: Sine Transformation

Let $X \sim \text{Uniform}(-\pi/2, \pi/2)$. The PDF is $f_X(x) = \frac{1}{\pi}$.

Consider the transformation $Y = \sin(X)$.

1. Support: Since $x \in [-\frac{\pi}{2}, \frac{\pi}{2}]$, we have $y \in [-1, 1]$.

2. Jacobian:

$$y = \sin x \implies x = \sin^{-1} y$$

$$\frac{dx}{dy} = \frac{d}{dy}(\sin^{-1} y) = \frac{1}{\sqrt{1-y^2}}$$

3. Apply Formula:

$$f_Y(y) = f_X(x) \cdot \left| \frac{dx}{dy} \right| = \frac{1}{\pi} \cdot \frac{1}{\sqrt{1-y^2}}, \quad -1 < y < 1$$

Note: This is the Arcsine Distribution.

4. Further Transformation (Connection to Beta): Let $Z = \frac{1+Y}{2}$. This maps $[-1, 1]$ to $[0, 1]$. Using the linear transformation rule ($z = \frac{1+y}{2} \implies \frac{dy}{dz} = 2$):

$$\begin{aligned} f_Z(z) &= f_Y(2z-1) \cdot |2| \\ &= \frac{1}{\pi \sqrt{1-(2z-1)^2}} \cdot 2 \\ &= \frac{2}{\pi \sqrt{1-(4z^2-4z+1)}} = \frac{2}{\pi \sqrt{4z-4z^2}} \\ &= \frac{2}{2\pi \sqrt{z(1-z)}} = \frac{1}{\pi \sqrt{z(1-z)}} \end{aligned}$$

Conclusion: $Z \sim \text{Beta}(1/2, 1/2)$. This confirms the result we saw earlier with the Cauchy distribution!

20 Expectation of Functions

Often we do not need the full distribution of $Y = h(X)$, but only its expected value $E[Y]$.

! Important: Law of the Unconscious Statistician (LOTUS)

There are two ways to calculate $E[h(X)]$:

Method 1 (Via derived distribution): Find $f_Y(y)$ first, then calculate:

$$E[Y] = \int_{-\infty}^{\infty} y \cdot f_Y(y) dy$$

Method 2 (Directly via X):

$$E[h(X)] = \int_{-\infty}^{\infty} h(x) \cdot f_X(x) dx$$

Example: Ex 7: Expectation of $\sin(X)$

Let $X \sim U(-\pi/2, \pi/2)$. Find $E[\sin X]$.

Using **Method 2** (LOTUS):

$$E[\sin X] = \int_{-\pi/2}^{\pi/2} \sin(x) \cdot \frac{1}{\pi} dx$$

Since $\sin(x)$ is an **odd function** and the domain is symmetric around 0, the integral is exactly **0**.

21 Introduction to Point Estimation

We often have multiple statistics (estimators) that could serve as a guess for a parameter θ .

- Example: To estimate center θ , we could use Sample Mean $\bar{X}$ or Sample Median $\tilde{X}$.

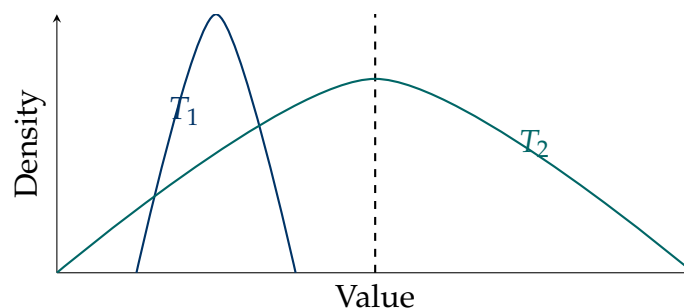
How do we choose? We need a metric for "accuracy".

Definition 21.1: Mean Squared Error (MSE)

The Mean Squared Error of an estimator T_n for a parameter θ is defined as:

$$\text{MSE}(T_n) = E[(T_n - \theta)^2]$$

It measures the average squared distance between the estimator and the truth.



22 Important Sampling Distributions

When sampling from Normal populations, three derived distributions appear repeatedly. These form the backbone of hypothesis testing and interval estimation.

22.1 The Chi-Square Distribution (χ^2)

Definition 22.1: Definition

Let $Z_1, Z_2, \dots, Z_n \stackrel{iid}{\sim} N(0, 1)$. The sum of their squares is distributed as **Chi-Square with n degrees of freedom**:

$$Y = \sum_{i=1}^n Z_i^2 \sim \chi_{(n)}^2$$

Properties:

- **Relation to Gamma:** $\chi_{(n)}^2 \equiv \text{Gamma}\left(\frac{n}{2}, \frac{1}{2}\right)$.
- **PDF:** $f(x) = \frac{1}{2^{n/2}\Gamma(n/2)} x^{n/2-1} e^{-x/2}, \quad x > 0$.
- **Moments:**

$$E[Y] = n \quad \text{and} \quad \text{Var}(Y) = 2n$$

22.2 The F-Distribution

Used primarily for comparing variances (ANOVA, F-tests).

Definition 22.2: Definition

Let $U \sim \chi_{(n)}^2$ and $V \sim \chi_{(m)}^2$ be independent. The random variable

$$F = \frac{U/n}{V/m} \sim F_{n,m}$$

follows an F-distribution with degrees of freedom (n, m) .

Property: If $Z \sim F_{n,m}$, then $\frac{1}{Z} \sim F_{m,n}$.

22.3 The Student's t-Distribution

Used for inference on the mean when variance is unknown.

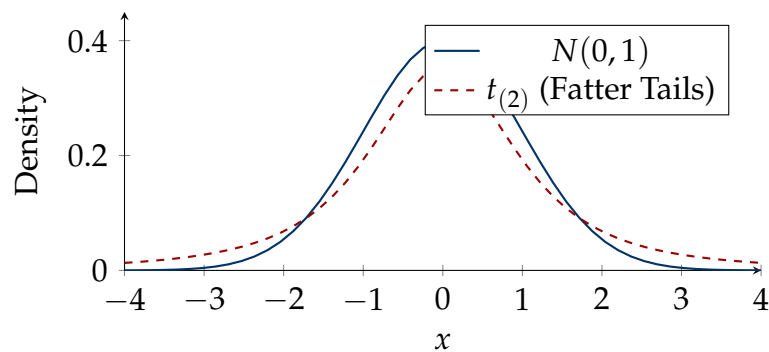
Definition 22.3: Definition

Let $Z \sim N(0, 1)$ and $U \sim \chi_{(n)}^2$ be independent.

$$T = \frac{Z}{\sqrt{U/n}} \sim t_{(n)}$$

Key Relationships:

- The PDF is symmetric around 0 ($E[T] = 0$ for $n > 1$).
- $T^2 \sim F_{1,n}$.
- As $n \rightarrow \infty$, $t_{(n)} \rightarrow N(0,1)$.



23 Case Study: Estimating Variance

Let $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$. We wish to estimate the population variance $g(\theta) = \sigma^2$.

23.1 Two Competing Estimators

1. The MLE (Biased):

$$S_n^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2$$

2. Sample Variance (Unbiased):

$$S_n^{*2} = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2$$

Result: Sampling Distribution of Sum of Squares

$$\frac{\sum (X_i - \bar{X}_n)^2}{\sigma^2} \sim \chi_{(n-1)}^2$$

Let $Y \sim \chi_{(n-1)}^2$. Then $\sum (X_i - \bar{X}_n)^2 = \sigma^2 Y$.

23.2 Expectation and Bias

Using $E[Y] = n - 1$:

- $E[S_n^2] = E\left[\frac{\sigma^2 Y}{n}\right] = \frac{\sigma^2}{n} E[Y] = \frac{n-1}{n} \sigma^2.$
- $E[S_n^{*2}] = E\left[\frac{\sigma^2 Y}{n-1}\right] = \frac{\sigma^2}{n-1} E[Y] = \sigma^2.$

Conclusion: S_n^2 is biased (underestimates variability), while S_n^{*2} is unbiased.

24 The Bias-Variance Decomposition

Which estimator is "better"? We use the **Mean Squared Error (MSE)**.

Definition 24.1: MSE Decomposition

$$\text{MSE}(\hat{\theta}) = E[(\hat{\theta} - \theta)^2] = \text{Var}(\hat{\theta}) + [\text{Bias}(\hat{\theta})]^2$$

24.1 Comparing MSEs

1. MSE of Unbiased Estimator (S_n^{*2}): Since bias is 0, MSE is just variance. Using $\text{Var}(Y) = 2(n-1)$:

$$\text{MSE}(S_n^{*2}) = \text{Var}\left(\frac{\sigma^2 Y}{n-1}\right) = \frac{\sigma^4}{(n-1)^2} \text{Var}(Y) = \frac{\sigma^4}{(n-1)^2} \cdot 2(n-1) = \frac{2\sigma^4}{n-1}$$

2. MSE of Biased Estimator (S_n^2): Here, $\text{Bias} = E[S_n^2] - \sigma^2 = \frac{n-1}{n} \sigma^2 - \sigma^2 = -\frac{\sigma^2}{n}.$

$$\begin{aligned} \text{MSE}(S_n^2) &= \text{Var}(S_n^2) + (\text{Bias})^2 \\ &= \text{Var}\left(\frac{\sigma^2 Y}{n}\right) + \frac{\sigma^4}{n^2} \\ &= \frac{\sigma^4}{n^2} (2(n-1)) + \frac{\sigma^4}{n^2} \\ &= \frac{\sigma^4}{n^2} (2n - 2 + 1) = \frac{2n-1}{n^2} \sigma^4 \end{aligned}$$

The Verdict

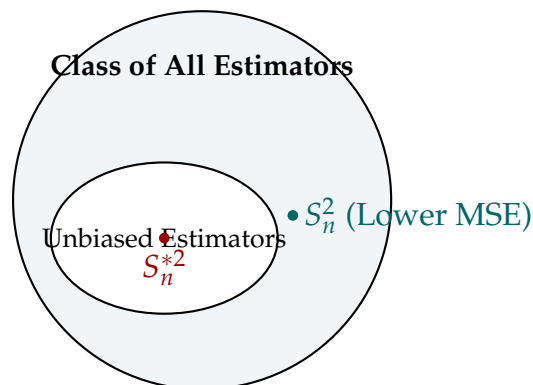
We compare $\frac{2n-1}{n^2}$ vs $\frac{2}{n-1}$.

$$\frac{2n-1}{n^2} < \frac{2}{n-1} \iff (2n-1)(n-1) < 2n^2$$

$$2n^2 - 3n + 1 < 2n^2 \iff 1 < 3n$$

This is true for all $n \geq 1$.

Conclusion: The **Biased Estimator** (S_n^2) has a smaller MSE than the Unbiased Estimator. In the context of MSE, minimizing variance is often more valuable than eliminating bias perfectly.



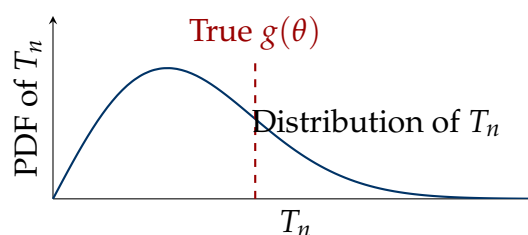
The best estimator often lies outside the unbiased class.

25 Evaluating Estimators: The Quest for the Best

25.1 Estimators vs. Estimates

When dealing with parameters θ , we often wish to estimate a specific function of the parameter, denoted as $g(\theta)$.

- **Estimator** (T_n): A function of the random sample $T(X_1, X_2, \dots, X_n)$ defined *before* the experiment. It is a **random variable** and has a sampling distribution that depends on θ .
- **Estimate** (t_n): A specific numerical value obtained by applying the estimator to a specific realization of the data $T(x_1, x_2, \dots, x_n)$.



25.2 Error and Mean Squared Error (MSE)

Let $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Bernoulli}(\theta)$. We use $T_n = \sum_{i=1}^n X_i \sim \text{Binomial}(n, \theta)$.

We define the **Error** as $\Psi_T(\theta) = T_n - \theta$, and the **Squared Error** as $\Psi_T^2(\theta) = (T_n - \theta)^2$. The distributions can be tabulated as follows:

Value of T_n	0	1	...	n
Error $(T_n - \theta)$	$-\theta$	$1 - \theta$	...	$n - \theta$
Squared Error $(T_n - \theta)^2$	θ^2	$(1 - \theta)^2$	...	$(n - \theta)^2$
Probability $P(T_n = t)$	$(1 - \theta)^n$	$n\theta(1 - \theta)^{n-1}$	...	θ^n

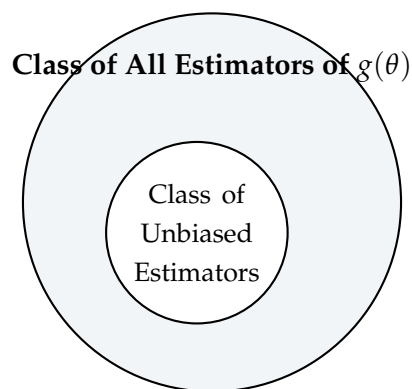
Definition 25.1: Mean Squared Error

The expectation of the squared error is called the Mean Squared Error (MSE) of the estimator T_n in estimating $g(\theta)$:

$$\text{MSE}_{T_n}(g(\theta)) = E[(T_n - g(\theta))^2]$$

The Selection Problem: An estimator with a smaller MSE is preferred. The one with the smallest MSE in a given class is the "best" estimator. However, the class of *all* estimators is infinitely large, making it impossible to find a global minimum for all θ . Therefore, we restrict our focus to a smaller, sensible sub-class: **Unbiased Estimators**.

26 Unbiased Estimators and UMVUE



Definition 26.1: Unbiased Estimator

An estimator $T_n = T(X_1, \dots, X_n)$ is called **unbiased** for $g(\theta)$ if its expected value matches the parameter for all possible values of θ :

$$E_\theta[T_n] = g(\theta) \quad \text{for all } \theta$$

Example: Checking Unbiasedness

Let $X_1, \dots, X_n \sim N(\theta, 1)$. Consider two estimators for θ :

$$1. T_{1n} = \bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i.$$

$$E[T_{1n}] = \frac{1}{n} \sum E[X_i] = \frac{1}{n}(n\theta) = \theta$$

$\Rightarrow T_{1n}$ is unbiased.

$$2. T_{2n} = \frac{2}{n} \sum_{i=2}^n (X_i - X_{i-1}).$$

$$E[T_{2n}] = \frac{2}{n} \sum_{i=2}^n (E[X_i] - E[X_{i-1}]) = \frac{2}{n} \sum_{i=2}^n (\theta - \theta) = 0$$

$\Rightarrow T_{2n}$ is **NOT unbiased** (it only equals θ if $\theta = 0$, not for all θ).

26.1 The UMVUE

Recall the MSE decomposition: $\text{MSE}_{T_n}(g(\theta)) = \text{Var}_\theta(T_n) + [\text{Bias}_{T_n}(\theta)]^2$.

If T_n is unbiased, $\text{Bias} = 0$, so $\text{MSE}_{T_n}(g(\cdot)) = \text{Var}_\theta(T_n)$. The best unbiased estimator is simply the one with the minimum variance.

Definition 26.2: UMVUE

An estimator T_n is called the **Uniformly Minimum Variance Unbiased Estimator (UMVUE)** of $g(\theta)$ if:

1. T_n is unbiased for $g(\theta)$.
2. For any other unbiased estimator S_n of $g(\theta)$, we have:

$$\text{Var}_\theta(T_n) \leq \text{Var}_\theta(S_n) \quad \text{for all } \theta$$

27 Comparing Unbiased Estimators

Example: Variance Showdown

Let $X_1, \dots, X_n \sim \text{Bernoulli}(\theta)$, where sample size $n = 2m$ is even. We have two unbiased estimators for θ :

1. $T_{1n} = \bar{X}_n = \frac{1}{n} \sum_{i=1}^n X_i$ (Uses all data)
2. $T_{2n} = \frac{X_2 + X_4 + \dots + X_{2m}}{m}$ (Uses only even-indexed data)

Both have Bias = 0. Let's compare their variances (noting X_i are independent, so covariance terms are 0):

$$\text{Var}_\theta(T_{1n}) = \text{Var}\left(\frac{\sum X_i}{n}\right) = \frac{1}{n^2} \sum \text{Var}(X_i) = \frac{n\theta(1-\theta)}{n^2} = \frac{\theta(1-\theta)}{n}$$

$$\text{Var}_\theta(T_{2n}) = \text{Var}\left(\frac{\sum_{\text{even}} X_i}{m}\right) = \frac{1}{m^2} \sum_{\text{even}} \text{Var}(X_i) = \frac{m\theta(1-\theta)}{m^2} = \frac{\theta(1-\theta)}{m}$$

Since $n = 2m$, $\frac{1}{m} = \frac{2}{n}$, so:

$$\text{Var}_\theta(T_{2n}) = \frac{2\theta(1-\theta)}{n}$$

Conclusion: $\text{Var}(T_{1n}) < \text{Var}(T_{2n})$. Using all the data (T_{1n}) provides a lower variance, making it a better estimator than T_{2n} .

Definition 27.1: Relative Efficiency

Let T_{1n} and T_{2n} be two unbiased estimators of $g(\theta)$. The **Relative Efficiency** of T_{2n} compared to T_{1n} is:

$$\text{RE}(T_{1n}, T_{2n}) = \frac{\text{Var}_\theta(T_{2n})}{\text{Var}_\theta(T_{1n})}$$

- If $\text{RE} > 1$, T_{1n} is the better estimator.
- If $\text{RE} < 1$, T_{2n} is the better estimator.

28 Finding the UMVUE via Sufficiency

Comparing two estimators is straightforward, but proving one is better than *all possible* unbiased estimators requires advanced theory.

Result: The Sufficiency Approach (Lehmann-Scheffé)

To obtain the UMVUE, we focus exclusively on **Complete Sufficient Statistics (CSS)**. Suppose T_n^* is a Complete Sufficient Statistic for θ . If we can find a function $h(T_n^*)$ such that:

$$E_\theta[h(T_n^*)] = g(\theta)$$

Then $h(T_n^*)$ **must be** the unique UMVUE for $g(\theta)$.

Example: Let $X_i \stackrel{iid}{\sim} \text{Bernoulli}(\theta)$. We know that the sum $Y = \sum_{i=1}^n X_i$ is a Complete Sufficient Statistic (CSS) for θ .

What is the UMVUE for θ ? We know $E[Y] = n\theta$. If we define $h(Y) = \frac{Y}{n} = \bar{X}_n$, then $E[h(Y)] = \theta$.

Because $\bar{X}_n$ is an unbiased function of the CSS, $\bar{X}_n$ **is the UMVUE for θ** .

29 Finding the UMVUE: Advanced Theory

Previously, we found that the UMVUE is the unbiased estimator with the smallest possible variance. However, checking the variance against an infinite number of other estimators is impractical. We need a constructive method to find it.

29.1 Sufficiency and Completeness

Definition 29.1: Sufficient Statistic

A statistic $T^*(\mathbf{X})$ is **sufficient** for θ if the conditional distribution of the sample $\mathbf{X}$ given T^* is free of θ . Intuitively, T^* captures all the information in the sample regarding the parameter.

Definition 29.2: Complete Statistic

A statistic $T^*(\mathbf{X})$ is **complete** for a family of distributions if, for any measurable function h :

$$E_\theta[h(T^*)] = 0 \quad \text{for all } \theta \implies P_\theta(h(T^*) = 0) = 1 \quad (\text{almost everywhere})$$

Intuition: Completeness ensures there are no "redundant" unbiased estimators. If T^* is complete, there is exactly one unique unbiased function that can be formed from it.

If T^* satisfies both properties, it is called a **Complete Sufficient Statistic (CSS)**. **Important property:** Any one-to-one function of a CSS is also a CSS.

Result: The Lehmann-Scheffé Theorem

Let T^* be a Complete Sufficient Statistic (CSS) for θ . If there exists a function $h(T^*)$ such that it is unbiased for $g(\theta)$, i.e.,

$$E_{\theta}[h(T^*)] = g(\theta)$$

Then $h(T^*)$ is the **unique UMVUE** of $g(\theta)$.

Example: Example: Uniform Distribution (Irregular Family)

Let $X_1, \dots, X_n \stackrel{iid}{\sim} U(0, \theta)$. We are given that the maximum order statistic $T^* = X_{(n)}$ is a CSS for θ .

The PDF of T^* is $f_{T^*}(t) = \frac{n}{\theta^n} t^{n-1}$ for $0 < t < \theta$.

Let's find its expectation to check for bias:

$$E_{\theta}[T^*] = \int_0^{\theta} t \cdot \frac{n}{\theta^n} t^{n-1} dt = \frac{n}{\theta^n} \left[\frac{t^{n+1}}{n+1} \right]_0^{\theta} = \frac{n}{n+1} \theta$$

Since $E[X_{(n)}] \neq \theta$, it is biased. We correct the bias by multiplying by the reciprocal:

$$E_{\theta} \left[\frac{n+1}{n} X_{(n)} \right] = \theta$$

Because $\frac{n+1}{n} X_{(n)}$ is an unbiased function of the CSS, it is the **UMVUE** for θ .

30 The Exponential Family

Proving Completeness and Sufficiency from scratch is mathematically rigorous. The **Exponential Family** provides a structural template. If a distribution fits this template, we can identify the CSS instantly.

Definition 30.1: The Exponential Family Form

A distribution $f_{\theta}(x)$ belongs to the k -parameter Exponential Family if the joint PDF/PMF of a sample $X_1, \dots, X_n$ can be expressed as:

$$f_{\mathbf{X}}(\mathbf{x}; \boldsymbol{\theta}) = \exp \left\{ A(\boldsymbol{\theta}) + H(\mathbf{x}) + \sum_{j=1}^k C_j(\boldsymbol{\theta}) T_j(\mathbf{x}) \right\}$$

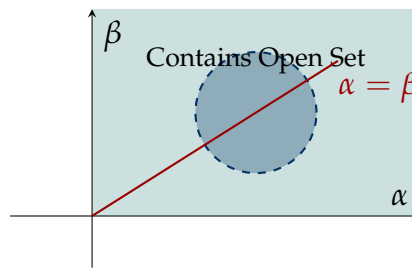
where:

- $A(\boldsymbol{\theta})$ and $C_j(\boldsymbol{\theta})$ are functions of the parameters only.
- $H(\mathbf{x})$ and $T_j(\mathbf{x})$ are functions of the data only.

! Important: Crucial Regularity Conditions

1. **Support Restriction:** The support (range of valid X values) **must not depend on the parameter θ** . Distributions like Uniform or Pareto fail this condition.
2. **Open Set Condition:** The parameter space $\Theta \subseteq \mathbb{R}^k$ must contain an *open set* (e.g., an open 2D region, not just a line). This ensures the analytical properties needed for *completeness*.

Open Set Condition in $\mathbb{R}^2$



Result: CSS from the Exponential Family

If a distribution meets the regularity conditions above, then the statistics:

$$\{T_1(\mathbf{X}), T_2(\mathbf{X}), \dots, T_k(\mathbf{X})\}$$

are **Jointly Complete Sufficient Statistics (CSS)**.

30.1 Examples and Non-Examples

1. Distributions NOT in the Exponential Family:

- **Uniform**($0, \theta$): Support depends on θ .
- **Cauchy**($\theta, 1$): $f(x) \propto \frac{1}{1+(x-\theta)^2}$. Cannot be factored into the required exponential form. (Note: No CSS exists other than the full order statistics).
- **Student's t-distribution**: Cannot be factored.

2. Binomial Distribution ($X_i \sim \text{Bin}(m, \theta)$):

$$f_{\mathbf{X}}(\mathbf{x}) = \exp \left\{ \underbrace{\sum \log \binom{m}{x_i}}_{H(\mathbf{x})} + \underbrace{mn \log(1-\theta)}_{A(\theta)} + \underbrace{\log \left(\frac{\theta}{1-\theta} \right)}_{C(\theta)} \underbrace{\sum x_i}_{T(\mathbf{x})} \right\}$$

CSS: $\sum_{i=1}^n X_i$.

3. Poisson Distribution ($X_i \sim \text{Poisson}(\lambda)$):

$$f_{\mathbf{X}}(\mathbf{x}) = \prod_{i=1}^n \frac{e^{-\lambda} \lambda^{x_i}}{x_i!} = \exp \left\{ -n\lambda - \sum \log(x_i!) + \log(\lambda) \sum x_i \right\}$$

CSS: $\sum X_i$.**UMVUE for λ :** Since $E[\bar{X}_n] = \lambda$, $\bar{X}_n$ is the UMVUE.**4. Gamma Distribution ($X_i \sim \text{Gamma}(\alpha, \beta)$):** Support $x > 0$ is independent of $\theta = (\alpha, \beta)$.

$$f_{\mathbf{X}}(\mathbf{x}) = \exp \left\{ n\alpha \log \beta - n \log \Gamma(\alpha) - \sum \log x_i + \alpha \sum \log x_i - \beta \sum x_i \right\}$$

Joint CSS: $\{\sum_{i=1}^n \log X_i, \sum_{i=1}^n X_i\}$.**5. Beta Distribution ($X_i \sim \text{Beta}(\alpha, \beta)$):**

$$f_{\mathbf{X}}(\mathbf{x}) = \exp \left\{ -n \log B(\alpha, \beta) + (\alpha - 1) \sum \log x_i + (\beta - 1) \sum \log(1 - x_i) \right\}$$

Joint CSS: $\{\sum \log X_i, \sum \log(1 - X_i)\}$.**6. Inverse Gaussian ($X_i \sim IG(\mu, \lambda)$):**

$$f_{\mathbf{X}}(\mathbf{x}) = \exp \left\{ \frac{n}{2} \log \lambda - \frac{n\lambda}{\mu} - \sum \frac{1}{2} \log(2\pi x_i^3) - \frac{\lambda}{2\mu^2} \sum x_i - \frac{\lambda}{2} \sum \frac{1}{x_i} \right\}$$

Joint CSS: $\{\sum X_i, \sum \frac{1}{X_i}\}$.**30.2 Finding UMVUEs for the Normal Distribution**

Let $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$, with parameters $\mu \in \mathbb{R}, \sigma^2 > 0$. Taking the logarithm of the joint PDF and moving it to the exponent:

$$\begin{aligned} f_{\mathbf{X}}(\mathbf{x}) &= \exp \left\{ -\frac{n}{2} \log(2\pi\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i^2 - 2\mu x_i + \mu^2) \right\} \\ &= \exp \left\{ \underbrace{-\frac{n}{2} \log(2\pi\sigma^2) - \frac{n\mu^2}{2\sigma^2}}_{A(\theta)} + \underbrace{\left(\frac{\mu}{\sigma^2}\right)}_{C_1} \underbrace{\sum x_i}_{T_1} + \underbrace{\left(-\frac{1}{2\sigma^2}\right)}_{C_2} \underbrace{\sum x_i^2}_{T_2} \right\} \end{aligned}$$

From this, the statistics $T_1 = \sum X_i$ and $T_2 = \sum X_i^2$ are **Jointly CSS**.

Because any 1-to-1 transformation of a CSS is also a CSS, we can apply algebraic manipulations to find more convenient sufficient statistics:

$$\left\{ \sum_{i=1}^n X_i, \sum_{i=1}^n X_i^2 \right\} \equiv \left\{ \bar{X}_n, S_n^2 = \frac{1}{n-1} \sum_{i=1}^n (X_i - \bar{X}_n)^2 \right\}$$

The Sample Mean and Sample Variance are jointly CSS!

Conclusion: UMVUEs for Normal

By the Lehmann-Scheffé Theorem:

1. We want to estimate $g_1(\theta) = \mu$. We know $E[\bar{X}_n] = \mu$. Since $\bar{X}_n$ is an unbiased function of the CSS, **$\bar{X}_n$ is the UMVUE for μ .**
2. We want to estimate $g_2(\theta) = \sigma^2$. We proved earlier that $E[S_n^2] = \sigma^2$. Since S_n^2 is an unbiased function of the CSS, **S_n^2 is the UMVUE for σ^2 .**

31 Methods of Point Estimation

31.1 Recap: The Estimation Problem

Recall our fundamental setup: We have a random sample $X_1, X_2, \dots, X_n \sim F_\theta$, where θ is the unknown population parameter. Our job is to estimate θ (or a function of it, $g(\theta)$) to understand the population.

We define an estimator $T_n = T(X_1, X_2, \dots, X_n)$.

- **Unbiasedness:** If $E[T_n] = g(\theta)$, the estimator is unbiased.
- **Mean Squared Error (MSE):** $E[(T_n - g(\theta))^2] = \text{Bias}^2 + \text{Var}(T_n)$. For unbiased estimators, $\text{MSE} = \text{Var}(T_n)$.

Example: Recall: Exponential Family & UMVUE

If $X_1, \dots, X_n \sim \text{Gamma}(\alpha, \beta)$, let $\theta = \begin{pmatrix} \alpha \\ \beta \end{pmatrix}$. We want to estimate $g(\theta) = \alpha$. The PDF is:

$$f(x) = \frac{\beta^\alpha}{\Gamma(\alpha)} x^{\alpha-1} e^{-\beta x}$$

From the Exponential Family form, the Joint Complete Sufficient Statistics (CSS) are:

$$\left\{ \sum_{i=1}^n \log X_i, \quad \sum_{i=1}^n X_i \right\}$$

Concept: The Next Step

What if F_θ does **not** belong to the exponential family (like the Uniform distribution $U(0, \theta)$)? Or what if finding a function of the CSS is algebraically too difficult? We need general, systematic methods to obtain estimators for $g(\theta)$.

The two primary methods are:

1. **Method of Moments (MoM)**
2. **Maximum Likelihood Estimation (MLE)**

32 The Method of Moments (MoM)

The Method of Moments is an intuitive approach: we estimate population parameters by equating theoretical population moments to the observed sample moments.

32.1 Defining Moments

Population Moments ($X \sim F_\theta$)

- **r^{th} order Raw Moment:**
 $\mu'_r(\theta) = E[X^r] \quad \text{for } r = 0, 1, 2, \dots$
- **r^{th} order Central Moment:**
 $\mu_r(\theta) = E[(X - \mu'_1)^r]$
Note: $\mu'_1 = \mu$ (Mean), $\mu_2 = \sigma^2$ (Variance).

Sample Moments ($X_1, \dots, X_n$)

- **r^{th} order Sample Raw Moment:**
 $m'_r = \frac{1}{n} \sum_{i=1}^n X_i^r \quad \text{for } r = 0, 1, 2, \dots$
- **r^{th} order Sample Central Moment:**
 $m_r = \frac{1}{n} \sum_{i=1}^n (X_i - m'_1)^r$

Definition 32.1: The MoM Principle

To obtain the Method of Moments estimator for a parameter vector $\theta \in \mathbb{R}^k$, we solve the system of k equations created by setting the sample raw moments equal to the population raw moments:

$$m'_r = \mu'_r(\theta) \quad \text{for } r = 1, 2, \dots, k$$

! Important: The Plug-in Property

If we want to estimate a function of the parameters $\Psi(\theta) = \Psi(\alpha, \beta)$, the MoM estimator is simply $\Psi(\hat{\theta}_{\text{MoM}})$.

Example: If we want to estimate $(\alpha + \beta)$, the MoM estimator is $\hat{\alpha}_{\text{MoM}} + \hat{\beta}_{\text{MoM}}$.

33 Examples of MoM Estimators

Example: Example 1: Gamma Distribution

Let $X_i \stackrel{iid}{\sim} \text{Gamma}(\alpha, \beta)$. We have two parameters $\theta = (\alpha, \beta)$. We need two equations ($r = 1, 2$).

Equation 1 (First Moment):

$$m'_1 = \mu'_1(\theta) \implies \bar{X}_n = \frac{\alpha}{\beta}$$

Equation 2 (Second Moment): Using the variance identity $E[X^2] = \text{Var}(X) + (E[X])^2$:

$$m'_2 = \mu'_2(\theta) \implies \frac{1}{n} \sum_{i=1}^n X_i^2 = \frac{\alpha}{\beta^2} + \left(\frac{\alpha}{\beta}\right)^2$$

Solving the System: Notice that $\left(\frac{\alpha}{\beta}\right)^2 = (m'_1)^2$. Substituting this into Equation 2:

$$m'_2 = \frac{\alpha}{\beta^2} + (m'_1)^2 \implies m'_2 - (m'_1)^2 = \frac{1}{\beta} \left(\frac{\alpha}{\beta}\right)$$

Substitute $\frac{\alpha}{\beta} = m'_1$:

$$m'_2 - (m'_1)^2 = \frac{m'_1}{\beta} \implies \hat{\beta}_{\text{MoM}} = \frac{m'_1}{m'_2 - (m'_1)^2}$$

Now find $\hat{\alpha}_{\text{MoM}}$ using Equation 1 ($\alpha = \beta m'_1$):

$$\hat{\alpha}_{\text{MoM}} = m'_1 \hat{\beta}_{\text{MoM}} = \frac{(m'_1)^2}{m'_2 - (m'_1)^2}$$

Example: Example 2: Normal Distribution

Let $X_i \stackrel{iid}{\sim} N(\mu, \sigma^2)$. Find MoM estimators for μ and σ^2 .

1. Estimate for μ :

$$m'_1 = \mu'_1 \implies \bar{X}_n = \mu \implies \hat{\mu}_{\text{MoM}} = \bar{X}_n$$

2. Estimate for σ^2 :

$$m'_2 = \sigma^2 + \mu^2 \implies \frac{1}{n} \sum_{i=1}^n X_i^2 = \sigma^2 + (\bar{X}_n)^2$$

$$\hat{\sigma}_{\text{MoM}}^2 = \frac{1}{n} \sum_{i=1}^n X_i^2 - \bar{X}_n^2 = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X}_n)^2 = s_n^2$$

Conclusion: The MoM estimator yields the sample mean and the **biased** sample variance. (**Valid/Logical Estimator**)

34 The Flaw of MoM: The Uniform Case

Example: Example 3: Uniform Distribution $U(0, \theta)$

Let $X_i \stackrel{iid}{\sim} U(0, \theta)$ for $i = 1, 2, \dots, n$.

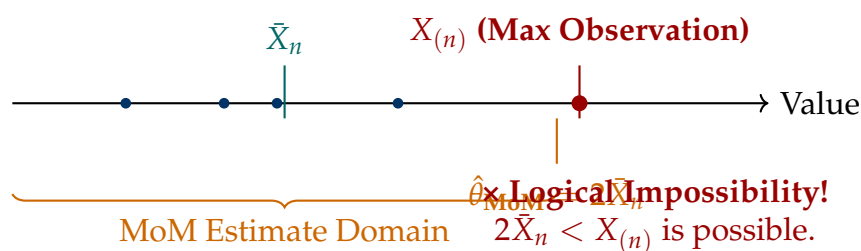
The MoM Equation:

$$m'_1 = \mu'_1 \implies \bar{X}_n = \frac{\theta}{2}$$

$$\hat{\theta}_{\text{MoM}} = 2\bar{X}_n$$

Comparing to UMVUE: Recall from earlier that the UMVUE for θ is derived from the maximum order statistic: $\hat{\theta}_{\text{UMVUE}} = \left(\frac{n+1}{n}\right) X_{(n)}$. The MSE of $2\bar{X}_n$ is significantly higher than the MSE of $\left(\frac{n+1}{n}\right) X_{(n)}$.

The Logical Contradiction: Because θ is the absolute maximum possible value of the distribution, it **must** be true that $\theta \geq X_{(n)}$ (the true parameter cannot be smaller than the largest observed data point). However, the MoM estimator $2\bar{X}_n$ can sometimes yield a value strictly less than $X_{(n)}$.



! Important: Why MoM is imperfect

The Method of Moments does not utilize the joint distribution or the sufficient statistics (like $X_{(n)}$). It merely forces summary averages to match. Because of this, it can produce parameter estimates that are logically impossible given the observed sample space. This leads us to a more robust methodology: Maximum Likelihood Estimation.

35 Introduction to Maximum Likelihood Estimation (MLE)

If MoM is prone to logical errors, how do we find a better estimate? We ask a different question: *Which parameter value makes our observed data the most mathematically probable?*

Example: Intuitive Example: Exponential Likelihood

Let $X_i \sim \text{Exp}(\theta)$, where $E[X_i] = \theta$. Suppose we know that the true parameter θ must be one of three discrete values: $\theta \in \{1, 3, 6\}$.

We take a sample of $n = 4$ observations:

$$x_1 = 4.5, \quad x_2 = 2.5, \quad x_3 = 3.2, \quad x_4 = 4.0$$

The sum of our observations is $\sum x_i = 14.2$.

The joint PDF (which we call the **Likelihood Function** when evaluating it for varying θ) is:

$$f_{\theta}(x_1, x_2, x_3, x_4) = \prod_{i=1}^4 \frac{1}{\theta} e^{-x_i/\theta} = \frac{1}{\theta^4} e^{-\frac{1}{\theta} \sum x_i}$$

Let's plug in our data ($\sum x_i = 14.2$) and evaluate the likelihood of each possible parameter:

1. If $\theta = 1$:

$$f_1(\mathbf{x}) = \frac{1}{1^4} e^{-14.2/1} \approx 6.8 \times 10^{-7} \quad (\text{Rank 3: Least Likely})$$

2. If $\theta = 3$:

$$f_3(\mathbf{x}) = \frac{1}{3^4} e^{-14.2/3} = \frac{1}{81} e^{-4.733} \approx 0.000108 \quad (\text{Rank 1: Most Likely})$$

3. If $\theta = 6$:

$$f_6(\mathbf{x}) = \frac{1}{6^4} e^{-14.2/6} = \frac{1}{1296} e^{-2.366} \approx 0.000072 \quad (\text{Rank 2})$$

Conclusion: The parameter $\theta = 3$ yields the highest probability density for the specific sample we observed. Therefore, the Maximum Likelihood Estimate (MLE) among the choices provided is 3.

36 Formalizing Maximum Likelihood Estimation (MLE)

In our previous intuitive example, we saw that MLE asks a fundamental question: "Out of all possible parameter values, which one makes the data we actually observed the most probable?"

Let's formalize this powerful concept.

Definition 36.1: The Likelihood Function

Let $X_1, X_2, \dots, X_n \stackrel{iid}{\sim} F_\theta$, having a Joint PDF (or PMF) denoted by $f_\theta(\mathbf{x})$. When we view this joint density not as a function of the data $\mathbf{x}$ (which is fixed after the experiment), but rather as a function of the unknown parameter θ , we call it the **Likelihood Function**. It is denoted by $L(\theta; \mathbf{x})$:

$$L(\theta; \mathbf{x}) = \prod_{i=1}^n f_\theta(x_i)$$

Definition 36.2: Maximum Likelihood Estimator (MLE)

Let Θ be the parameter space (the set of all valid values for θ). The Maximum Likelihood Estimator, denoted as $\hat{\theta}_{ML}$, is the value of $\theta \in \Theta$ that maximizes the likelihood function:

$$\hat{\theta}_{ML} = \operatorname{argmax}_{\theta \in \Theta} L(\theta; \mathbf{x})$$

36.1 The Log-Likelihood Trick

Products ($\prod$) are notoriously difficult to differentiate when applying calculus. To find the maximum of $L(\theta; \mathbf{x})$, we rely on a brilliant mathematical convenience: the natural logarithm.

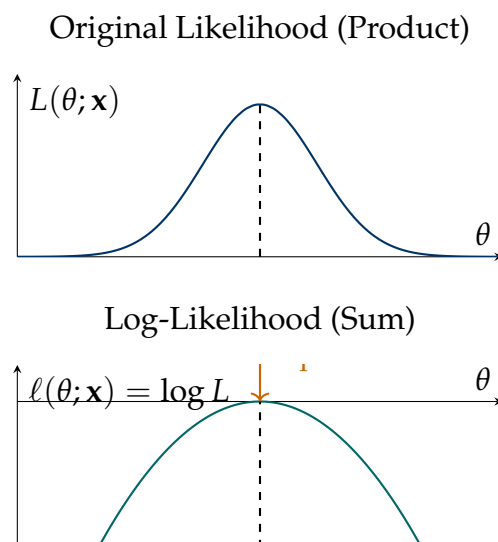
Concept: Log-Likelihood Function

We define the **Log-Likelihood Function**, denoted by $\ell(\theta; \mathbf{x})$, as:

$$\ell(\theta; \mathbf{x}) = \log L(\theta; \mathbf{x}) = \sum_{i=1}^n \log f_\theta(x_i)$$

Crucial Remarks on MLE:

- [1] **It is a true statistic:** As $\hat{\theta}_{ML}$ is the argmax of the likelihood, it depends purely on the observed sample $\mathbf{x}$ and not on any unknown parameters.
- [2] **Validity:** Because the search is restricted to $\theta \in \Theta$, the resulting estimator is guaranteed to lie within the valid parameter space (unlike MoM, which sometimes falls outside it).
- [3] **Maximization Equivalence:** Because the natural logarithm ($\log x$) is a **monotonically strictly increasing function**, the θ that maximizes the log-likelihood $\ell(\theta)$ is exactly the same θ that maximizes the original likelihood $L(\theta)$.



37 Finding MLE via Calculus (F.O.C.)

If the likelihood function is differentiable and the maximum does not lie on the boundary of the parameter space, we can find the MLE by finding the roots of the First Order Condition (F.O.C.) and verifying concavity using the Second Order Condition.

Example: Example: Generalized Exponential MLE

Let $X_1, \dots, X_n \sim \text{Exp}(\theta)$, where $f(x) = \frac{1}{\theta}e^{-x/\theta}$ and $\theta \in (0, \infty)$. Let's find $\hat{\theta}_{ML}$.

1. Formulate the Log-Likelihood:

$$L(\theta; \mathbf{x}) = \prod_{i=1}^n \frac{1}{\theta} e^{-x_i/\theta} = \frac{1}{\theta^n} \exp \left\{ -\frac{1}{\theta} \sum_{i=1}^n x_i \right\}$$

$$\ell(\theta; \mathbf{x}) = \log(\theta^{-n}) - \frac{1}{\theta} \sum_{i=1}^n x_i$$

$$= -n \log \theta - \theta^{-1} \sum x_i$$

2. First Order Condition (F.O.C.): Take the derivative with respect to θ and set it to 0.

$$\frac{\partial \ell}{\partial \theta} = -\frac{n}{\theta} + \theta^{-2} \sum x_i = 0$$

$$\frac{n}{\theta} = \frac{\sum x_i}{\theta^2}$$

$$n\theta = \sum x_i \implies \hat{\theta}_{ML} = \frac{\sum x_i}{n} = \bar{x}_n$$

3. Verify the Maximum (Second Derivative): We must ensure this critical point is a peak, not a valley.

$$\frac{\partial^2 \ell}{\partial \theta^2} = \frac{n}{\theta^2} - 2\theta^{-3} \sum x_i$$

Evaluate at $\hat{\theta} = \bar{x}$:

$$\left. \frac{\partial^2 \ell}{\partial \theta^2} \right|_{\theta=\bar{x}} = \frac{n}{\bar{x}^2} - \frac{2n\bar{x}}{\bar{x}^3} = \frac{n}{\bar{x}^2} - \frac{2n}{\bar{x}^2} = -\frac{n}{\bar{x}^2}$$

Since $n > 0$ and $\bar{x}^2 > 0$, the second derivative is strictly **negative** (< 0). Therefore, $\hat{\theta}_{ML} = \bar{x}_n$ is indeed the global maximum!

38 Multi-Parameter MLE: The Normal Distribution

When estimating a vector of parameters $\boldsymbol{\theta} = [\theta_1, \theta_2]^T$, we must solve a system of partial derivatives. To verify it is a maximum, we cannot just check a single second derivative; we must verify that the multi-dimensional curvature forms a peak. We do this by checking if the **Hessian Matrix** is Negative Definite.

Example: MLE for $N(\mu, \sigma^2)$

Let $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$. Here, $\theta = \begin{pmatrix} \mu \\ \sigma^2 \end{pmatrix}$.

1. The Likelihood and Log-Likelihood:

$$L(\mu, \sigma^2; \mathbf{x}) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} \exp \left\{ -\frac{(x_i - \mu)^2}{2\sigma^2} \right\}$$

$$n = (2\pi\sigma^2)^{-n/2} \exp \left\{ -\frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2 \right\}$$

$$\ell(\mu, \sigma^2; \mathbf{x}) = -\frac{n}{2} \log(2\pi) - \frac{n}{2} \log(\sigma^2) - \frac{1}{2\sigma^2} \sum_{i=1}^n (x_i - \mu)^2$$

2. First Order Conditions (Set Partial Derivatives to 0): For μ :

$$\frac{\partial \ell}{\partial \mu} = -\frac{1}{2\sigma^2} \times 2 \sum (x_i - \mu)(-1) = \frac{1}{\sigma^2} \sum (x_i - \mu) = 0 \implies \sum x_i = n\mu \implies \hat{\mu} = \bar{x}$$

For σ^2 (Treat σ^2 as a single variable block):

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} - \left(-\frac{1}{2(\sigma^2)^2} \right) \sum (x_i - \mu)^2 = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum (x_i - \mu)^2 = 0$$

Multiply by $2\sigma^4$:

$$-n\sigma^2 + \sum (x_i - \mu)^2 = 0 \implies \hat{\sigma}^2 = \frac{1}{n} \sum (\mathbf{x}_i - \hat{\mu})^2$$

Note: As seen previously with MoM, the MLE for variance is the **biased** sample variance s_n^2 .

38.1 Checking the Hessian Matrix

To confirm $(\hat{\mu}, \hat{\sigma}^2)$ is a maximum, we calculate the Hessian matrix $\mathbf{H}$, which holds all second-order partial derivatives:

$$\mathbf{H}(\mu, \sigma^2) = \begin{bmatrix} \frac{\partial^2 \ell}{\partial \mu^2} & \frac{\partial^2 \ell}{\partial \mu \partial \sigma^2} \\ \frac{\partial^2 \ell}{\partial \sigma^2 \partial \mu} & \frac{\partial^2 \ell}{\partial (\sigma^2)^2} \end{bmatrix}$$

Evaluating the second derivatives:

$$\frac{\partial^2 \ell}{\partial \mu^2} = -\frac{n}{\sigma^2}$$

$$\frac{\partial^2 \ell}{\partial (\sigma^2)^2} = \frac{n}{2\sigma^4} - \frac{1}{\sigma^6} \sum (x_i - \mu)^2$$

$$\frac{\partial^2 \ell}{\partial \mu \partial \sigma^2} = -\frac{1}{\sigma^4} \sum (x_i - \mu)$$

Now, we evaluate the Hessian exactly at our critical point $(\hat{\mu}, \hat{\sigma}^2)$. Since $\sum(x_i - \hat{\mu}) = 0$, the cross-diagonal terms vanish. Furthermore, substituting $\sum(x_i - \hat{\mu})^2 = n\hat{\sigma}^2$ into the bottom right term gives $\frac{n}{2\hat{\sigma}^4} - \frac{n\hat{\sigma}^2}{\hat{\sigma}^6} = -\frac{n}{2\hat{\sigma}^4}$.

$$\mathbf{H}(\hat{\mu}, \hat{\sigma}^2) = \begin{bmatrix} -\frac{n}{\hat{\sigma}^2} & 0 \\ 0 & -\frac{n}{2\hat{\sigma}^4} \end{bmatrix}$$

Result: Negative Definite Condition

For a 2×2 diagonal matrix $M = \begin{bmatrix} a & 0 \\ 0 & d \end{bmatrix}$ to be Negative Definite (which guarantees a local maximum), we require:

1. $a < 0$
2. The Determinant $|M| = ad - bc > 0$

Here, $a = -n/\hat{\sigma}^2$ which is strictly < 0 .

The determinant is $(-n/\hat{\sigma}^2) \times (-n/2\hat{\sigma}^4) = \frac{n^2}{2\hat{\sigma}^6}$, which is strictly > 0 .

Thus, the Hessian is **Negative Definite**, proving that $(\hat{\mu}, \hat{\sigma}^2)$ is indeed the global MLE.

39 MLE at the Boundaries (When Calculus Fails)

The calculus-based approach only works when the maximum lies in the *interior* of the parameter space Θ . But what if the support of the distribution depends entirely on the parameter itself?

! Important: The Trap of the Uniform Distribution

If you blindly apply calculus to $U(0, \theta)$, you will take the derivative of the likelihood, set it to zero, and find that it has **no roots**. The calculus approach fundamentally fails here. Let's see why, and how to logically deduce the MLE.

Example: MLE for the Uniform Distribution

Let $X_1, \dots, X_n \stackrel{iid}{\sim} U(0, \theta)$. The parameter is $\theta \in (0, \infty)$. The PDF is $f(x) = \frac{1}{\theta}$ if $x \in [0, \theta]$, and 0 otherwise.

1. The Indicator Function Likelihood: The likelihood is the product of the PDFs. It is only non-zero if *every single observation* x_i is $\leq \theta$. If even a single data point is greater than θ , the entire product collapses to 0 (because the data would be impossible under that model).

$$L(\theta; \mathbf{x}) = \begin{cases} \frac{1}{\theta^n} & \text{if } x_i \leq \theta \quad \forall i \\ 0 & \text{otherwise} \end{cases}$$

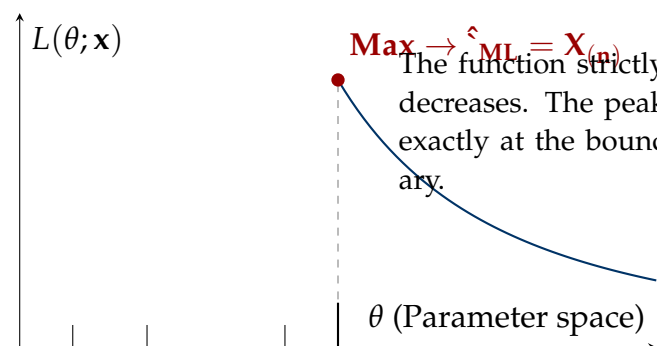
2. The Logical Constraint: Saying "every $x_i \leq \theta$ " is mathematically identical to saying "the *maximum* observed value is $\leq \theta$ ". Therefore, $\theta \geq X_{(n)}$.

3. Maximizing the Likelihood: We want to maximize $L(\theta) = \frac{1}{\theta^n}$ subject to the constraint $\theta \geq X_{(n)}$. Because θ is in the denominator, the function $\frac{1}{\theta^n}$ is a **strictly decreasing function**. To make a fraction as large as possible, we must make the denominator (θ) as *small* as possible.

What is the absolute smallest value θ can take without violating our constraint? It is $X_{(n)}$.

$$\implies \hat{\theta}_{\text{ML}} = X_{(n)}$$

Likelihood Function for $U(0, \theta)$



Generalizing the Uniform Bounds

Consider $X_1, \dots, X_n \stackrel{iid}{\sim} U(\alpha, \beta)$ where $\alpha < \beta$. We must simultaneously estimate both bounds. The likelihood is $\frac{1}{(\beta - \alpha)^n}$ provided that all data falls within the bounds.

Constraints: 1. All $x_i \geq \alpha \implies \alpha \leq X_{(1)}$ (The minimum) 2. All $x_i \leq \beta \implies \beta \geq X_{(n)}$ (The maximum)

To maximize $\frac{1}{(\beta - \alpha)^n}$, we must make the interval $(\beta - \alpha)$ as small as possible.

- We make α as large as possible: $\hat{\alpha}_{\text{ML}} = X_{(1)}$.
- We make β as small as possible: $\hat{\beta}_{\text{ML}} = X_{(n)}$.

40 Advanced MLE: Two-Parameter Uniform Distribution

In our previous discussion, we looked at $U(0, \theta)$. Let us elevate this and look at the general Uniform distribution where *both* boundaries are unknown.

Example: MLE for $U(\alpha, \beta)$

Let $X_1, X_2, \dots, X_n \stackrel{iid}{\sim} U(\alpha, \beta)$ where $\alpha < \beta$. Our parameter vector is $\theta = (\alpha, \beta)$. The PDF for a single observation is:

$$f_{\theta}(x) = \begin{cases} \frac{1}{\beta - \alpha} & \text{if } \alpha \leq x \leq \beta \\ 0 & \text{otherwise} \end{cases}$$

The Joint Likelihood Function demands that *every single observation* falls within the interval $[\alpha, \beta]$. If even one observation falls outside, the entire product becomes zero. We can express this using the order statistics $X_{(1)}$ (the minimum) and $X_{(n)}$ (the maximum):

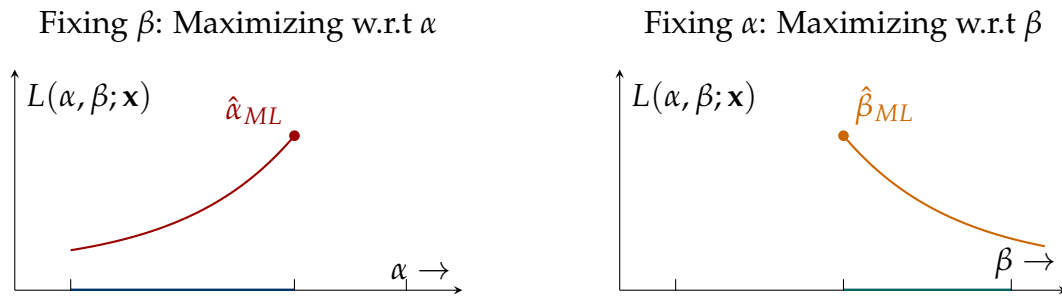
$$L(\alpha, \beta; \mathbf{x}) = \begin{cases} \frac{1}{(\beta - \alpha)^n} & \text{if } \alpha \leq X_{(1)} \leq X_{(n)} \leq \beta \\ 0 & \text{otherwise} \end{cases}$$

To truly understand how to maximize this, let's map out the likelihood space based on the constraints.

β condition α condition	$\beta < X_{(n)}$	$\beta \geq X_{(n)}$
$\alpha \leq X_{(1)}$	0	$\frac{1}{(\beta - \alpha)^n}$
$\alpha > X_{(1)}$	0	0

The Optimization Logic: To maximize $\frac{1}{(\beta - \alpha)^n}$, we must make the denominator $(\beta - \alpha)$ as small as legally possible.

- **Fixing β :** If we hold β constant, we need to make α as *large* as possible. The upper bound for α is the smallest data point, $X_{(1)}$. Thus, $\hat{\alpha}_{ML} = X_{(1)}$.
- **Fixing α :** If we hold α constant, we need to make β as *small* as possible. The lower bound for β is the largest data point, $X_{(n)}$. Thus, $\hat{\beta}_{ML} = X_{(n)}$.



41 MLE for the Double Exponential (Laplace) Distribution

Now, let's examine a distribution where the calculus works, but requires careful handling of absolute values. This is a classic example that proves *why* the median is statistically significant.

Let $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Double Exponential}(\mu, \beta)$. The parameter vector is $\theta = \begin{pmatrix} \mu \\ \beta \end{pmatrix}$, where $\mu \in \mathbb{R}$ and $\beta > 0$.

$$f_{\theta}(x) = \frac{1}{2\beta} \exp \left\{ -\frac{|x - \mu|}{\beta} \right\}$$

1. Formulate the Log-Likelihood:

$$L(\mu, \beta; \mathbf{x}) = \prod_{i=1}^n \frac{1}{2\beta} \exp \left\{ -\frac{|x_i - \mu|}{\beta} \right\}$$

$$\ell(\mu, \beta; \mathbf{x}) = -n \log 2 - n \log \beta - \frac{1}{\beta} \sum_{i=1}^n |x_i - \mu|$$

2. Maximize with respect to β : Taking the partial derivative with respect to β and setting it to 0 (First Order Condition):

$$\frac{\partial \ell}{\partial \beta} = -\frac{n}{\beta} + \frac{1}{\beta^2} \sum_{i=1}^n |x_i - \mu| = 0$$

$$\implies \hat{\beta}_{ML} = \frac{1}{n} \sum_{i=1}^n |x_i - \mu| \quad [\text{Solution to F.O.C.}]$$

Check Second Derivative (S.O.C.):

$$\frac{\partial^2 \ell}{\partial \beta^2} = \frac{n}{\beta^2} - \frac{2}{\beta^3} \sum_{i=1}^n |x_i - \mu|$$

Plugging in $\hat{\beta} = \frac{1}{n} \sum_{i=1}^n |x_i - \mu|$, we get $\frac{n}{\hat{\beta}^2} - \frac{2n\hat{\beta}}{\hat{\beta}^3} = -\frac{n}{\hat{\beta}^2} < 0$. This confirms it is a maximum.

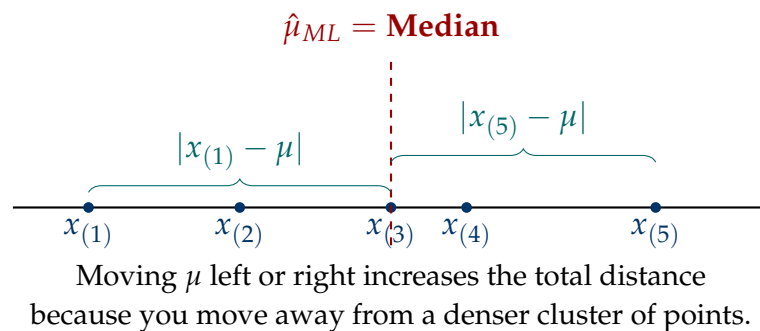
3. Maximize with respect to μ : Observe the log-likelihood function. To maximize ℓ with respect to μ (holding β fixed), we must **minimize the sum of absolute deviations**: $\sum_{i=1}^n |x_i - \mu|$.

Concept: Why the Median?

What value of μ minimizes the total absolute distance to all points? Calculus tells us that the derivative of $|x_i - \mu|$ is $-\text{sgn}(x_i - \mu)$. If we set the sum of derivatives to zero:

$$\sum_{i=1}^n \text{sgn}(x_i - \mu) = 0$$

This equation implies we need an equal number of positive and negative signs. In other words, an equal number of data points must lie to the left of μ as to the right of μ . This is the exact definition of the **Median**!



Therefore, the MLEs for the Double Exponential are:

$$\hat{\mu}_{ML} = \text{Median}\{X_1, X_2, \dots, X_n\}$$

$$\hat{\beta}_{ML} = \frac{1}{n} \sum_{i=1}^n |x_i - \hat{\mu}_{ML}|$$

41.1 Constrained Parameter Spaces

Suppose we have prior knowledge that the dispersion parameter is bounded, e.g., $\mathbf{fi} \geq 1$. How does this modify our result?

If our calculated $\hat{\beta}_{unconstrained} = \frac{1}{n} \sum |x_i - \hat{\mu}|$ happens to be less than 1, we cannot use it. Because the likelihood function is concave (shaped like a downward parabola), if the unconstrained peak is outside our valid region, the maximum *within* the valid region occurs exactly at the boundary.

$$\hat{\beta}_{ML} = \max \left\{ 1, \frac{1}{n} \sum_{i=1}^n |x_i - \hat{\mu}_{ML}| \right\}$$

42 Critical Remarks on MLE Properties

As powerful as MLE is, it is not flawless. We must recognize its limitations mathematically:

! Important: 1. MLE is not necessarily unbiased.

- Example: For $X_i \sim N(\mu, \sigma^2)$, we proved earlier that $\hat{\sigma}_{ML}^2 = s_n^2 = \frac{1}{n} \sum (x_i - \bar{x})^2$, which is **biased**.
- Example: For $X_i \sim U(0, \theta)$, we proved $\hat{\theta}_{ML} = X_{(n)}$. However, $E[X_{(n)}] = \frac{n}{n+1}\theta \neq \theta$, making it **biased**.

! Important: 2. MLE may not be unique.

- Example: Consider the Double Exponential distribution with an *even* sample size (e.g., $n = 4$). Any value between $X_{(2)}$ and $X_{(3)}$ serves as the median and perfectly minimizes the absolute distance. Thus, there are infinitely many valid MLEs for μ in this specific case.

! Important: 3. MLE may not exist.

- Example: Let $X_i \sim U([0, \theta))$. Notice the interval is **open** at θ . Because the data can never literally equal θ , the parameter θ must be strictly greater than $X_{(n)}$. The likelihood function increases as $\theta \rightarrow X_{(n)}$, but since θ cannot equal $X_{(n)}$, the supremum is never reached within the valid parameter space. Thus, a strict maximum does not exist.

43 The Invariance Property of MLE

One of the greatest algebraic advantages of Maximum Likelihood Estimation is its invariance to transformations. This makes calculating estimates for complex functions of parameters incredibly easy.

Result: Invariance Property

Suppose θ is a parameter and $g(\theta)$ is a function of θ . Let $\psi = g(\theta)$. If the MLE of θ is $\hat{\theta}_{ML}$, then the MLE of ψ is simply:

$$\hat{\psi}_{ML} = g(\hat{\theta}_{ML})$$

Example: Application: Estimating a Probability in Poisson

Let $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Poisson}(\theta)$ for $i = 1, \dots, n$. We know the PDF is $f(x) = \frac{e^{-\theta} \theta^x}{x!}$.

Step 1: Find MLE of θ

$$L(\theta; \mathbf{x}) = \prod_{i=1}^n \frac{e^{-\theta} \theta^{x_i}}{x_i!}$$

$$\ell(\theta; \mathbf{x}) = -n\theta + \left(\sum x_i\right) \log \theta - \sum \log(x_i!)$$

$$\frac{\partial \ell}{\partial \theta} = -n + \frac{\sum x_i}{\theta} = 0 \implies \hat{\theta}_{ML} = \bar{X}_n$$

(Checking S.O.C: $\frac{\partial^2 \ell}{\partial \theta^2} = -\frac{\sum x_i}{\theta^2} < 0$, so it is a maximum).

Step 2: Estimate a Function Suppose we want to estimate the probability that exactly 1 event occurs:

$$\psi = P_\theta(X = 1) = \frac{e^{-\theta} \theta^1}{1!} = \theta e^{-\theta}$$

By the Invariance Property, the MLE for this probability is trivially:

$$\hat{\psi}_{ML} = \hat{\theta}_{ML} e^{-\hat{\theta}_{ML}} = \bar{X}_n e^{-\bar{X}_n}$$

Professor's Note: While easy to calculate, $\bar{X}_n e^{-\bar{X}_n}$ is **not** an unbiased estimator for ψ ! If we wanted an unbiased estimator, we could define an indicator function $T = I(\{X_1 = 1\})$. Since $E[T] = 1 \cdot P(X_1 = 1) + 0 = \psi$, T is unbiased.

44 Fundamental Probability Inequalities

As we move toward understanding the asymptotic (large sample) behaviors of these estimators, we must build a bridge using mathematical inequalities. These inequalities allow us to bound probabilities even when we do not know the exact distribution!

44.1 1. Markov's Inequality

This inequality gives an upper bound on the probability that a non-negative random variable is exceptionally large.

Result: Markov's Inequality

If X is a non-negative random variable ($X \geq 0$), then for any $t > 0$:

$$P(X \geq t) \leq \frac{E(X)}{t}$$

Proof (for Continuous Case): Since X is non-negative, its PDF $f(x) = 0$ for $x < 0$. By

definition of Expectation:

$$\begin{aligned} E(X) &= \int_0^{\infty} xf(x) dx \\ &= \int_0^t xf(x) dx + \int_t^{\infty} xf(x) dx \end{aligned}$$

Since X is non-negative, the first integral $\int_0^t xf(x) dx \geq 0$. Therefore, dropping it makes the right side smaller:

$$E(X) \geq \int_t^{\infty} xf(x) dx$$

Inside this integral, the smallest value x can take is t . If we replace x with the constant t , we make the integral smaller still:

$$\begin{aligned} E(X) &\geq \int_t^{\infty} tf(x) dx \\ E(X) &\geq t \int_t^{\infty} f(x) dx \\ E(X) &\geq t \cdot P(X \geq t) \implies P(X \geq t) \leq \frac{E(X)}{t} \quad \blacksquare \end{aligned}$$

44.2 2. Chebyshev's Inequality

While Markov relies only on the mean, Chebyshev uses the variance to provide a much tighter bound regarding how far data strays from the center.

Result: Chebyshev's Inequality

If X is a random variable with finite expectation $E(X) = \mu$ and finite variance $\text{Var}(X) = \sigma^2$, then for any $t > 0$:

$$P(|X - \mu| > t) \leq \frac{\sigma^2}{t^2}$$

Proof (Using Markov's): Let $Y = (X - \mu)^2$. Because it is squared, Y is a non-negative random variable. Note that the event $\{|X - \mu| > t\}$ is mathematically identical to $\{(X - \mu)^2 > t^2\}$. Apply Markov's Inequality to Y :

$$\begin{aligned} P(Y > t^2) &\leq \frac{E(Y)}{t^2} \\ P((X - \mu)^2 > t^2) &\leq \frac{E[(X - \mu)^2]}{t^2} \end{aligned}$$

By definition, $E[(X - \mu)^2] = \text{Var}(X) = \sigma^2$. Therefore:

$$P(|X - \mu| > t) \leq \frac{\sigma^2}{t^2} \quad \blacksquare$$

Extension: This can be extended to higher moments. $P(|X - \mu| > t) \leq \frac{E[|X - \mu|^r]}{t^r}$ for $r = 1, 2, 3, \dots$ provided the r^{th} moment exists.

45 Applying Chebyshev: Sample Size Calculation

Inequalities are immensely practical. We can use them to guarantee estimation precision without knowing the underlying distribution.

Example: Ensuring Precision via Sample Size

Let $X_1, \dots, X_n \stackrel{iid}{\sim} F_\theta$. Assume $E(X_i) = \mu$ and $\text{Var}(X_i) = 1$. We are using the sample mean $\bar{X}_n$ to estimate μ .

Question: What is the minimum sample size (n) that ensures our estimate is within 1 unit of the true mean with at least 95% probability? Mathematically: Find n such that $P(|\bar{X}_n - \mu| \leq 1) \geq 0.95$.

Solution: First, we analyze the sampling distribution of our estimator $\bar{X}_n$:

- $E[\bar{X}_n] = E\left[\frac{\sum X_i}{n}\right] = \frac{n\mu}{n} = \mu$
- $\text{Var}(\bar{X}_n) = \text{Var}\left[\frac{\sum X_i}{n}\right] = \frac{1}{n^2} \sum \text{Var}(X_i) = \frac{n \cdot 1}{n^2} = \frac{1}{n}$

Using Chebyshev's Inequality on $\bar{X}_n$ with $t = 1$:

$$P(|\bar{X}_n - \mu| > 1) \leq \frac{\text{Var}(\bar{X}_n)}{1^2} = \frac{1}{n}$$

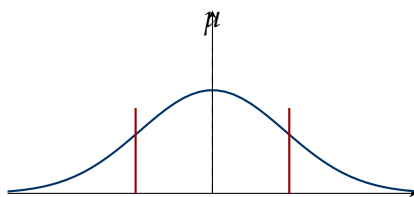
We want the probability of being *close* (≤ 1) to be ≥ 0.95 . This means the probability of being *far* (> 1) must be ≤ 0.05 . Since Chebyshev guarantees the "far" probability is bounded by $\frac{1}{n}$, we simply set:

$$\frac{1}{n} = 1 - 0.95 = 0.05$$

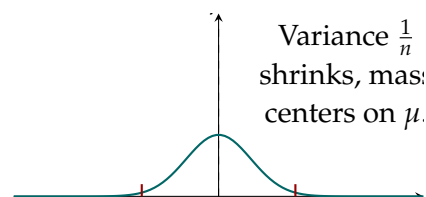
$$n = \frac{1}{0.05} = 20$$

To be safe, we always take the upper integer bound. Thus, a sample size of **$n = 20$** mathematically guarantees our desired precision, *regardless of the shape of the population distribution*.

Distribution of $\bar{X}_n$ for $n = 5$



Distribution of $\bar{X}_n$ for $n = 25$



46 MLE for Discrete Parameters

When the parameter θ we are trying to estimate belongs to a discrete space (e.g., $\theta \in \mathbb{N}$), the calculus-based First Order Condition ($\frac{\partial \ell}{\partial \theta} = 0$) completely fails. You cannot differentiate a function with respect to an integer.

Concept: The Ratio Method (Discrete Analog of Derivatives)

Instead of finding where the slope is zero, we look for the point where the likelihood function stops increasing and starts decreasing. We evaluate the ratio of consecutive likelihoods:

$$\frac{L(\theta; \mathbf{x})}{L(\theta - 1; \mathbf{x})}$$

- If the ratio > 1 , the likelihood is **increasing**.
- If the ratio < 1 , the likelihood is **decreasing**.

The maximum occurs at the integer θ where the ratio transitions from ≥ 1 to < 1 .

Example: Estimating the Number of Trials in a Binomial

Suppose we observe a single random variable $X = x$, where $X \sim \text{Binomial}(\theta, p_0)$. Here, the probability of success p_0 is known, but the total number of trials θ is unknown. We know $\theta \in \{x, x+1, x+2, \dots\}$ because there must be at least as many trials as observed successes.

1. The Likelihood Function:

$$L(\theta; x) = \binom{\theta}{x} p_0^x (1 - p_0)^{\theta - x}$$

2. Formulate the Ratio: We want to find when $\frac{L(\theta; x)}{L(\theta-1; x)} \geq 1$.

$$\frac{L(\theta; x)}{L(\theta-1; x)} = \frac{\frac{\theta!}{x!(\theta-x)!} p_0^x (1-p_0)^{\theta-x}}{\frac{(\theta-1)!}{x!(\theta-1-x)!} p_0^x (1-p_0)^{\theta-1-x}}$$

Most terms cancel out! Notice that $\frac{\theta!}{(\theta-1)!} = \theta$, and $\frac{(\theta-1-x)!}{(\theta-x)!} = \frac{1}{\theta-x}$, and $\frac{(1-p_0)^{\theta-x}}{(1-p_0)^{\theta-1-x}} = (1-p_0)$.

$$\frac{L(\theta; x)}{L(\theta-1; x)} = \left(\frac{\theta}{\theta-x} \right) (1-p_0)$$

3. Solve the Inequality:

$$\begin{aligned} (1-p_0) \frac{\theta}{\theta-x} &\geq 1 \\ \theta(1-p_0) &\geq \theta-x \\ \theta - \theta p_0 &\geq \theta - x \\ -\theta p_0 &\geq -x \implies \theta \leq \frac{x}{p_0} \end{aligned}$$

This profound result tells us that the likelihood is increasing as long as $\theta < \frac{x}{p_0}$, and it starts decreasing when $\theta > \frac{x}{p_0}$.

46.1 Two Cases for the MLE

The exact MLE depends on whether $\frac{x}{p_0}$ is a perfect integer.

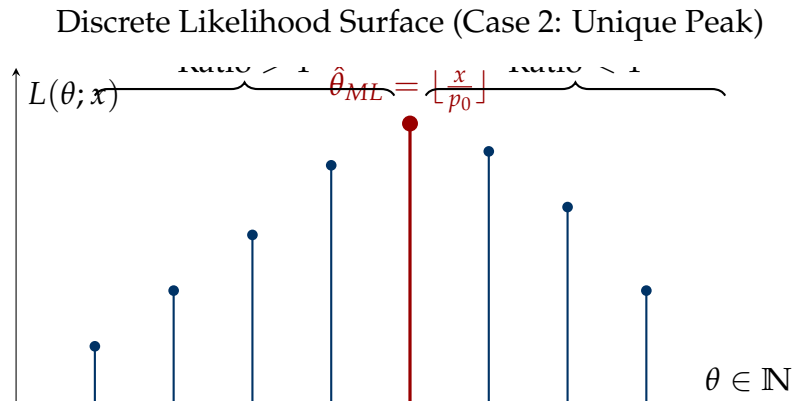
Case 1: When $\frac{x}{p_0}$ is an integer. If $\frac{x}{p_0}$ is an integer, say k , then at exactly $\theta = k$, the ratio is exactly 1. This means $L(k) = L(k-1)$. The likelihood has two identical peaks.

$$\hat{\theta}_{ML} \in \left\{ \frac{x}{p_0} - 1, \frac{x}{p_0} \right\} \quad (\text{Not unique})$$

Case 2: When $\frac{x}{p_0}$ is NOT an integer. The likelihood function strictly peaks at the greatest

integer less than $\frac{x}{p_0}$. We use the floor function $\lfloor \cdot \rfloor$:

$$\hat{\theta}_{ML} = \lfloor \frac{x}{p_0} \rfloor$$



Homework / Self-Study: Try applying this same ratio method to the Hyper-Geometric distribution for a sample of $n = 1$. You will see the exact same algebraic cancellations occur!

47 Inequalities and Large Sample Theory

We now transition from calculating estimators to evaluating their asymptotic behavior. What makes an estimator "good" when we collect a massive amount of data?

Let's recall the variances of estimators we've seen:

- Sample Mean: $\text{Var}(\bar{X}_n) = \frac{\sigma^2}{n}$. As $n \uparrow$, Variance $\downarrow$.
- Sample Variance: $\text{Var}(S_n^{*2}) = \frac{2\sigma^4}{n-1}$. As $n \uparrow$, Variance $\downarrow$.

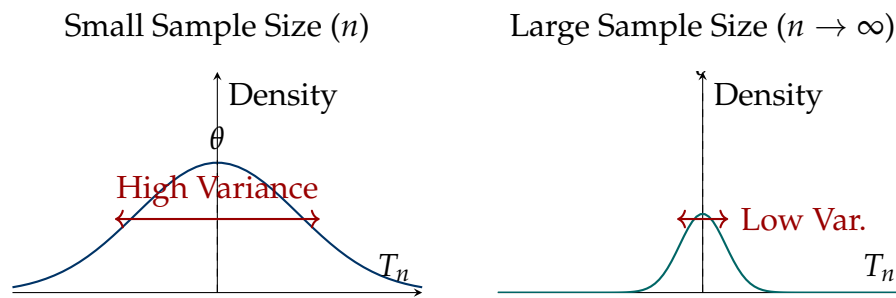
! Important: The Core Intuition of Asymptotics

When is an estimator T_n considered "good" in view of large sample properties? As n becomes very large ($n \rightarrow \infty$), a good estimator T_n must **concentrate** tightly around the true parameter θ . The variance must vanish.

We can visualize this concentration of measure. Recall Chebyshev's Inequality:

$$P(|X - E(X)| > t) \leq \frac{\text{Var}(X)}{t^2}$$

As n grows, $\text{Var}(T_n)$ approaches 0, meaning the probability of the estimator straying further than t from the truth drops to 0.



48 Modes of Convergence

In calculus, we say a sequence of real numbers $a_n \rightarrow a$ if, for any $\epsilon > 0$, $|a_n - a| < \epsilon$ for all $n \geq N_\epsilon$. But T_n is a **sequence of random variables**. It doesn't have a single deterministic value. How do we define convergence for randomness?

We define the distance between the estimator and the truth as a new random variable: Let $A_n = |T_n - \theta|$. We want the "size" of A_n to approach 0. There are multiple ways to measure this "size", leading to different modes of convergence.

48.1 I. Convergence in Mean (Lr Convergence)

Definition 48.1: Convergence in r -th Mean ($T_n \xrightarrow{L^r} \theta$)

Let $r \geq 1$. An estimator T_n converges in r -th mean to θ if the expected value of the r -th power of the absolute error approaches 0 as $n \rightarrow \infty$:

$$E(|T_n - \theta|^r) \rightarrow 0 \quad \text{as } n \rightarrow \infty$$

Note: When $r = 1$, this is called Convergence in Mean (L^1). When $r = 2$, this is called Convergence in Mean Square (L^2). Mean Square Error (MSE) going to zero is exactly L^2 convergence!

Example: Proving Convergence in Mean

Let $X_1, \dots, X_n \stackrel{iid}{\sim} U(0, \theta)$. Let our estimator be $T_n = X_{(n)}$ (the sample maximum). Does T_n converge in mean (L^1) to θ ?

1. Define the Error: We need to evaluate $E(|T_n - \theta|)$. Because the maximum of a $U(0, \theta)$ sample can never physically exceed θ , we know that $T_n \leq \theta$ always. Therefore, $|T_n - \theta| = \theta - T_n$.

2. Calculate the Expectation:

$$E(|T_n - \theta|) = E(\theta - T_n) = \theta - E(T_n)$$

We previously derived the expected value of the maximum order statistic for a uniform distribution. We know the PDF of T_n is $f_{T_n}(t) = \frac{nt^{n-1}}{\theta^n}$ for $0 < t < \theta$. This gives us $E(T_n) = \frac{n}{n+1}\theta$.

3. Evaluate the Limit:

$$\begin{aligned} E(|T_n - \theta|) &= \theta - \frac{n}{n+1}\theta \\ &= \theta \left(1 - \frac{n}{n+1}\right) \\ &= \theta \left(\frac{n+1-n}{n+1}\right) = \frac{\theta}{n+1} \end{aligned}$$

As $n \rightarrow \infty$, the denominator explodes, so:

$$\lim_{n \rightarrow \infty} \frac{\theta}{n+1} = 0$$

Conclusion: Because $E(|T_n - \theta|) \rightarrow 0$, we definitively state that $T_n \xrightarrow{L^1} \theta$. Even though $X_{(n)}$ is a biased estimator for any finite n , as we collect infinite data, the bias vanishes and the estimator locks onto the true parameter.

48.2 Properties of L^r Convergence

Let's dive deeper into the implications of Convergence in r -th mean ($T_n \xrightarrow{L^r} T$).

Result: Hierarchical Nature of Means

- 1. Expectation Convergence:** If $T_n \xrightarrow{L^r} T$ for $r \geq 1$, then $E(T_n) \rightarrow E(T)$ as $n \rightarrow \infty$.
- 2. Downward implication:** If a sequence converges in a higher moment, it automatically converges in all lower moments. If $T_n \xrightarrow{L^r} T$, then $T_n \xrightarrow{L^s} T$ for any $0 < s < r$.
Example: Convergence in Mean Square (L^2) guarantees Convergence in Mean (L^1).

! Important: The Converse is NOT True

Convergence in a lower mean does **not** guarantee convergence in a higher mean. Let us prove this with a classic counterexample.

Example: Counterexample: $L^1 \not\Rightarrow L^2$

Let X_n be a sequence of random variables defined as:

$$X_n = \begin{cases} \sqrt{n} & \text{with probability } \frac{1}{n} \\ 0 & \text{with probability } 1 - \frac{1}{n} \end{cases}$$

Check L^1 convergence to 0:

$$E(|X_n - 0|) = E(X_n) = \sqrt{n} \cdot \left(\frac{1}{n}\right) + 0 \cdot \left(1 - \frac{1}{n}\right) = \frac{1}{\sqrt{n}}$$

As $n \rightarrow \infty$, $\frac{1}{\sqrt{n}} \rightarrow 0$. Thus, $X_n \xrightarrow{L^1} 0$.

Check L^2 convergence to 0:

$$E(|X_n - 0|^2) = E(X_n^2) = (\sqrt{n})^2 \cdot \left(\frac{1}{n}\right) + 0 = n \cdot \frac{1}{n} = 1$$

As $n \rightarrow \infty$, $1 \not\rightarrow 0$. Thus, $X_n \not\xrightarrow{L^2} 0$.

48.3 II) Convergence in Probability

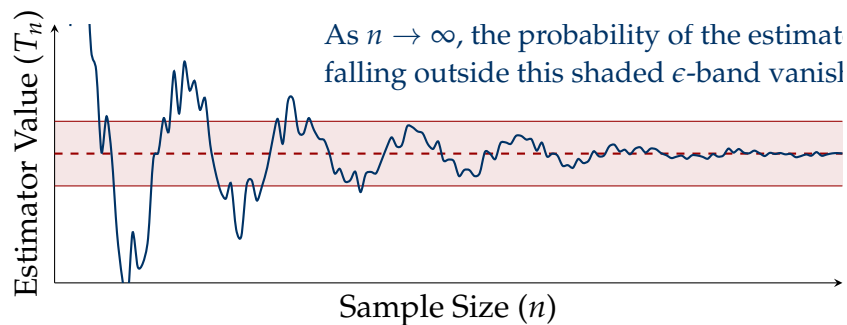
While L^r convergence is strong, it requires expectations (moments) to exist. Sometimes, estimators have infinite variance or undefined means. We need a broader, more universally applicable definition of convergence.

Definition 48.2: Convergence in Probability ($T_n \xrightarrow{P} T$)

A sequence of random variables T_n converges in probability to a target T if, for any strictly positive margin of error $\epsilon > 0$:

$$\lim_{n \rightarrow \infty} P(|T_n - T| > \epsilon) = 0$$

Equivalently: $\lim_{n \rightarrow \infty} P(|T_n - T| \leq \epsilon) = 1$.

Visualizing Convergence in Probability**Example: Proving $X_{(n)} \xrightarrow{P} \theta$ for Uniform Distribution**

Let $X_i \stackrel{iid}{\sim} U(0, \theta)$. Let $T_n = X_{(n)}$. Fix an arbitrary $\epsilon > 0$. We need to evaluate $P(|T_n - \theta| \leq \epsilon)$. Since $X_{(n)}$ can never exceed θ , we can drop the absolute value and just look at the lower bound:

$$P(|T_n - \theta| \leq \epsilon) = P(\theta - \epsilon \leq T_n \leq \theta)$$

Assuming $0 < \epsilon < \theta$, we integrate the PDF of the maximum order statistic $f(t) = \frac{nt^{n-1}}{\theta^n}$:

$$\begin{aligned} P(\theta - \epsilon \leq T_n \leq \theta) &= \int_{\theta - \epsilon}^{\theta} \frac{nt^{n-1}}{\theta^n} dt = \left[\frac{t^n}{\theta^n} \right]_{\theta - \epsilon}^{\theta} \\ &= 1 - \left(\frac{\theta - \epsilon}{\theta} \right)^n = 1 - \left(1 - \frac{\epsilon}{\theta} \right)^n \end{aligned}$$

Since $(1 - \frac{\epsilon}{\theta})$ is a fraction strictly between 0 and 1, raising it to the power of $n \rightarrow \infty$ drives it to 0.

$$\lim_{n \rightarrow \infty} \left[1 - \left(1 - \frac{\epsilon}{\theta} \right)^n \right] = 1 - 0 = 1$$

Therefore, $P(|T_n - \theta| > \epsilon) \rightarrow 0 \implies X_{(n)} \xrightarrow{P} \theta$.

49 Connecting L^r and Probability Convergence

Result: The Master Implication

Convergence in r -th mean is strictly **stronger** than convergence in probability.

$$T_n \xrightarrow{L^r} T \implies T_n \xrightarrow{P} T \quad \text{for any } r \geq 1$$

Proof (Using Markov's Inequality): Fix $\epsilon > 0$. We want to bound $P(|T_n - T| > \epsilon)$. This is equivalent to bounding $P(|T_n - T|^r > \epsilon^r)$. Applying Markov's Inequality:

$$P(|T_n - T|^r > \epsilon^r) \leq \frac{E(|T_n - T|^r)}{\epsilon^r}$$

Since we are given that $T_n \xrightarrow{L^r} T$, we know the numerator $E(|T_n - T|^r) \rightarrow 0$ as $n \rightarrow \infty$. Therefore, the probability bound goes to 0, proving $T_n \xrightarrow{P} T$.

! Important: The Converse is NOT True ($P \not\implies L^r$)

An estimator can converge in probability, yet fail to converge in mean! Consider $X_n = \begin{cases} n & \text{with prob } \frac{1}{n} \\ 0 & \text{with prob } 1 - \frac{1}{n} \end{cases}$

For any $\epsilon > 0$, $P(|X_n| > \epsilon) = P(X_n = n) = \frac{1}{n}$. As $n \rightarrow \infty$, $\frac{1}{n} \rightarrow 0$. So $X_n \xrightarrow{P} 0$.

However, $E(|X_n|) = n(\frac{1}{n}) + 0 = 1 \not\rightarrow 0$. So $X_n \not\xrightarrow{L^1} 0$. (The probability of a massive error shrinks to zero, but the magnitude of that error grows so large that it permanently distorts the expected value!)

50 Consistency of Estimators

Convergence in probability is the mathematical foundation for one of the most critical properties of a good estimator: **Consistency**.

Definition 50.1: Consistent Estimator

An estimator T_n of $g(\theta)$ is said to be **consistent** if:

$$T_n \xrightarrow{P} g(\theta) \text{ as } n \rightarrow \infty$$

Meaning: As the sample size goes to infinity, the probability that the estimator is off by any microscopic amount shrinks to exactly zero.

Result: Sufficient Condition for Consistency

Proving probability convergence directly via integrals is hard. A simpler, sufficient condition is: If as $n \rightarrow \infty$:

1. $E(T_n) \rightarrow g(\theta)$ (The estimator is asymptotically unbiased)
2. $\text{Var}(T_n) \rightarrow 0$ (The variance collapses)

Then $T_n \xrightarrow{P} g(\theta)$.

Proof (via the MSE Decomposition): Fix $\epsilon > 0$. Apply Markov's (Chebyshev's) inequality to the squared error:

$$\begin{aligned} P(|T_n - g(\theta)| > \epsilon) &= P(|T_n - g(\theta)|^2 > \epsilon^2) \\ &\leq \frac{E[(T_n - g(\theta))^2]}{\epsilon^2} \end{aligned}$$

Recall that $E[(T_n - g(\theta))^2] = \text{MSE} = [E(T_n) - g(\theta)]^2 + \text{Var}(T_n)$.

$$P(|T_n - g(\theta)| > \epsilon) \leq \frac{[E(T_n) - g(\theta)]^2 + \text{Var}(T_n)}{\epsilon^2}$$

Since $E(T_n) \rightarrow g(\theta)$, the bias squared term goes to 0. Since $\text{Var}(T_n) \rightarrow 0$, the variance term goes to 0. The entire right side goes to 0, proving consistency!

Example: Proving MLE Variance is Consistent

Let $X_1, \dots, X_n \sim N(\mu, \sigma^2)$. The MLE for variance is $\hat{\sigma}_{ML}^2 = \frac{1}{n} \sum (X_i - \bar{X})^2$. Is it consistent for σ^2 ?

1. **Check Mean:** We proved earlier that $E[\hat{\sigma}_{ML}^2] = \frac{n-1}{n} \sigma^2$. As $n \rightarrow \infty$, $\frac{n-1}{n} \rightarrow 1$. Thus $E \rightarrow \sigma^2$. 2. **Check Variance:** We know $\frac{n\hat{\sigma}_{ML}^2}{\sigma^2} \sim \chi_{(n-1)}^2$, which has variance $2(n-1)$.

$$\text{Var}\left(\frac{n}{\sigma^2} \hat{\sigma}_{ML}^2\right) = \frac{n^2}{\sigma^4} \text{Var}(\hat{\sigma}_{ML}^2) = 2(n-1)$$

$$\implies \text{Var}(\hat{\sigma}_{ML}^2) = \frac{2(n-1)\sigma^4}{n^2}$$

As $n \rightarrow \infty$, the n^2 in the denominator dominates, so $\text{Var} \rightarrow 0$.

Because both conditions are met, $\hat{\sigma}_{ML}^2 \xrightarrow{P} \sigma^2$. It is a consistent estimator!

51 The Weak Law of Large Numbers (WLLN)

Consistency leads us directly to the foundational bedrock of all statistics: The Law of Large Numbers. It guarantees that empirical averages converge to theoretical expectations.

Result: Theorem 1: WLLN (IID Case)

If $X_1, X_2, \dots, X_n$ are independent and identically distributed (IID) with finite expectation $E(X_1) = \mu_\theta$, then the sample mean is consistent for the population mean:

$$\bar{X}_n \xrightarrow{P} \mu_\theta \quad \text{as } n \rightarrow \infty$$

Proof triviality: Assuming finite variance σ^2 , $E(\bar{X}) = \mu$ and $\text{Var}(\bar{X}) = \sigma^2/n \rightarrow 0$. The sufficient condition is instantly met.

51.1 WLLN for Independent (Not Identical) Variables

What if our data points are independent, but come from different distributions? (e.g., measuring with different instruments of varying precision).

Result: Theorem 2: WLLN (Independent Case)

Let X_i be independent variables with $E(X_i) = \mu_i$ and $\text{Var}(X_i) = \sigma_i^2 < \infty$.

Let $\bar{\mu}_n = \frac{1}{n} \sum_{i=1}^n \mu_i$.

If the average variance collapses: $\frac{1}{n^2} \sum_{i=1}^n \sigma_i^2 \rightarrow 0$ as $n \rightarrow \infty$, then:

$$(\bar{X}_n - \bar{\mu}_n) \xrightarrow{P} 0$$

Proof Details: We use Chebyshev's inequality on $(\bar{X}_n - \bar{\mu}_n)$:

$$P(|\bar{X}_n - \bar{\mu}_n| > \epsilon) \leq \frac{\text{Var}(\bar{X}_n)}{\epsilon^2}$$

By properties of variance for independent variables (Covariance terms = 0):

$$\text{Var}(\bar{X}_n) = \text{Var}\left(\frac{1}{n} \sum X_i\right) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{1}{n^2} \sum_{i=1}^n \sigma_i^2$$

Since the condition explicitly states $\frac{1}{n^2} \sum \sigma_i^2 \rightarrow 0$, the bound goes to 0.

Professor's Remark: We do not strictly even need absolute independence. We only need the variables to be **uncorrelated** (i.e., $\text{Cov}(X_i, X_j) = 0$ for all $i \neq j$) for the variance summation to hold.

Example: Evaluating the Variance Condition

Let X_i be independent variables where $X_i \sim N(\mu_i, \sqrt{i})$. Here, the variance of each subsequent measurement is growing: $\sigma_i^2 = \sqrt{i}$. Does the WLLN still hold? Let's check the condition.

We need to evaluate: $\lim_{n \rightarrow \infty} \frac{1}{n^2} \sum_{i=1}^n \sqrt{i}$

The Integral Bounding Trick: A sum of a growing function can be bounded by its integral.

$$\sum_{i=1}^n \sqrt{i} = \sqrt{1} + \sqrt{2} + \cdots + \sqrt{n}$$

Because $f(x) = \sqrt{x}$ is strictly increasing, the discrete sum is bounded above by the continuous area from 0 to $n + 1$, or practically:

$$\sum_{i=1}^n \sqrt{i} \leq n\sqrt{n} \quad (\text{Since every term is } \leq \sqrt{n})$$

For a tighter bound, integrate: $\int_0^n \sqrt{x} dx = \left[\frac{2}{3} x^{3/2} \right]_0^n = \frac{2}{3} n\sqrt{n}$.

Using either bound, let's substitute it back into the limit condition:

$$\frac{1}{n^2} \sum_{i=1}^n \sqrt{i} \leq \frac{1}{n^2} (n\sqrt{n}) = \frac{n\sqrt{n}}{n^2} = \frac{\sqrt{n}}{n} = \frac{1}{\sqrt{n}}$$

As $n \rightarrow \infty$, $\frac{1}{\sqrt{n}} \rightarrow 0$. **Conclusion:** Even though the variance of individual measurements goes to infinity, it grows slowly enough ($\sqrt{n}$) that the n^2 factor in the denominator crushes it. The Weak Law of Large Numbers holds!

52 Convergence in Distribution (Weak Convergence)

We have seen Convergence in Mean (L^r) and Convergence in Probability (P). Both measure how "close" the random variables T_n get to a target T . However, sometimes we don't need the variables themselves to be close; we only need their **probability distributions** to be close. This is the weakest, yet most practically useful, form of convergence.

Definition 52.1: Convergence in Distribution ($T_n \xrightarrow{d} T$)

Let T_n be a sequence of random variables with Cumulative Distribution Functions (CDFs) $F_n(t)$. Let T be a random variable with CDF $F(t)$.

We say T_n converges in distribution to T if:

$$\lim_{n \rightarrow \infty} F_n(t) = F(t)$$

for all t that are continuity points of F .

Concept: Why only at continuity points?

Consider a deterministic sequence $T_n = \frac{1}{n}$. Clearly, $T_n \rightarrow 0$. The CDF of T_n is $F_n(t) = 1$ for $t \geq 1/n$, and 0 otherwise. The target $T = 0$ has CDF $F(t) = 1$ for $t \geq 0$, and 0 otherwise.

If we evaluate exactly at $t = 0$:

$$\lim_{n \rightarrow \infty} F_n(0) = \lim_{n \rightarrow \infty} 0 = 0$$

But $F(0) = 1$. The limit fails at $t = 0$! However, $t = 0$ is a jump (discontinuity) in the target CDF $F(t)$. By restricting our definition to *continuity points*, we mathematically permit $T_n \xrightarrow{d} 0$.

52.1 The Discontinuity Set is Countable

A valid CDF $F(t)$ is non-decreasing and bounded between 0 and 1. A beautiful result from real analysis ensures that the continuity restriction is not a major handicap.

Claim: The set of discontinuity points of any CDF F is at most countable. **Proof:** Let the jump at a point x be defined as $F(x) - F(x^-)$. Define a set D_n containing all points where the jump is greater than $\frac{1}{n}$:

$$D_n = \left\{ x : F(x) - F(x^-) > \frac{1}{n} \right\}$$

Because the total accumulated probability cannot exceed 1, there can be at most $n - 1$ such points in D_n . Thus, D_n is a *finite* set. The set of *all* discontinuities is the union of all these sets for $n = 1, 2, 3, \dots$:

$$\mathcal{D} = \bigcup_{n=1}^{\infty} D_n$$

Since $\mathcal{D}$ is a countable union of finite sets, $\mathcal{D}$ is at most countable. Therefore, the set of continuity points $(\mathbb{R} \setminus \mathcal{D})$ forms almost the entire real line!

53 Examples of Convergence in Distribution

Example: Example 1: Max Order Statistic of Uniform

Let $X_i \stackrel{iid}{\sim} U(0, \theta)$. Let $T_n = X_{(n)}$. We previously showed $T_n \xrightarrow{P} \theta$. Let's prove $T_n \xrightarrow{d} \theta$. The target $T = \theta$ (with probability 1) is a degenerate random variable. Its CDF is a step function:

$$F_T(t) = \begin{cases} 0 & \text{if } t < \theta \\ 1 & \text{if } t \geq \theta \end{cases}$$

The CDF of T_n is:

$$F_n(t) = \begin{cases} 0 & \text{if } t < 0 \\ \left(\frac{t}{\theta}\right)^n & \text{if } 0 \leq t < \theta \\ 1 & \text{if } t \geq \theta \end{cases}$$

Taking the limit as $n \rightarrow \infty$:

- For $t < 0$: $F_n(t) = 0 \rightarrow 0 = F_T(t)$.
- For $0 \leq t < \theta$: Since $t/\theta < 1$, $(t/\theta)^n \rightarrow 0 = F_T(t)$.
- For $t > \theta$: $F_n(t) = 1 \rightarrow 1 = F_T(t)$.

At $t = \theta$, $(1)^n \rightarrow 1$, but $t = \theta$ is a discontinuity point of $F_T(t)$, so we do not check it. Since $F_n(t) \rightarrow F_T(t)$ at all continuity points, $T_n \xrightarrow{d} \theta$.

54 Hierarchy of Convergence & Key Theorems

Result: Convergence Hierarchy

$$T_n \xrightarrow{L^r} T \implies T_n \xrightarrow{P} T \implies T_n \xrightarrow{d} T$$

Convergence in Probability implies Convergence in Distribution.

! Important: The Converse ($d \not\Rightarrow P$)

Convergence in distribution means the *shapes* of the distributions match, not the actual values. **Counterexample:** Let $T_n \sim \text{Bernoulli}(1/2)$ and $T \sim \text{Bernoulli}(1/2)$ be completely independent. Since they have the exact same CDF, $F_n(t) = F_T(t)$ for all t , so trivially $T_n \xrightarrow{d} T$.

However, look at the probability of a large error ($\epsilon = 1/2$):

$$P(|T_n - T| > 1/2) = P(T_n = 1, T = 0) + P(T_n = 0, T = 1) \\ = \left(\frac{1}{2} \times \frac{1}{2}\right) + \left(\frac{1}{2} \times \frac{1}{2}\right) = \frac{1}{2}$$

As $n \rightarrow \infty$, this probability does NOT go to 0. Thus, $T_n \not\xrightarrow{P} T$.

The One Exception: If the target c is a constant, then $T_n \xrightarrow{P} c \iff T_n \xrightarrow{d} c$.

54.1 Continuous Mapping and Slutsky's Lemma

When we manipulate estimators algebraically, how do the limits behave?

Result: Continuous Mapping Theorem (CMT)

If $g(\cdot)$ is a continuous function, then limits pass through the function:

1. If $T_n \xrightarrow{P} T$, then $g(T_n) \xrightarrow{P} g(T)$.
2. If $T_n \xrightarrow{d} T$, then $g(T_n) \xrightarrow{d} g(T)$.

Result: Slutsky's Lemma

If $X_n \xrightarrow{d} X$ and $Y_n \xrightarrow{P} c$ (where c is a constant), then:

1. $X_n + Y_n \xrightarrow{d} X + c$
2. $X_n Y_n \xrightarrow{d} cX$
3. $\frac{X_n}{Y_n} \xrightarrow{d} \frac{X}{c}$ (provided $c \neq 0$)

Note: Y_n must converge to a constant for the joint distribution to behave predictably.

55 The Central Limit Theorem (CLT)

The Weak Law of Large Numbers tells us that $\bar{X}_n \xrightarrow{P} \mu$. The variance of $\bar{X}_n$ shrinks to 0. But if we "zoom in" on $\bar{X}_n$ by multiplying it by an inflating factor ($\sqrt{n}$), the distribution stabilizes into a perfect Normal curve. This is arguably the most famous theorem in statistics.

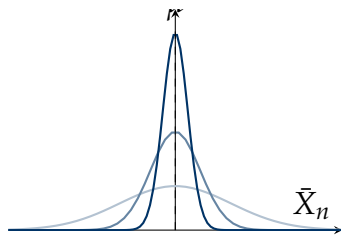
Definition 55.1: The Central Limit Theorem (CLT)

Let $X_1, X_2, \dots, X_n$ be IID random variables from *any* distribution with mean $E(X_i) = \mu$ and finite variance $\text{Var}(X_i) = \sigma^2 < \infty$.

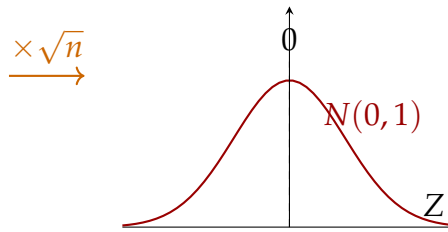
Then, the standardized sample mean converges in distribution to a Standard Normal random variable:

$$\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \xrightarrow{d} Z, \quad \text{where } Z \sim N(0, 1)$$

Unscaled: Variance $\rightarrow 0$



Scaled: Stabilizes to $N(0, 1)$

**Example: Applying the CLT**

Let $X_i \stackrel{iid}{\sim} N(0, \sigma^2)$. Consider the estimator for variance: $T_n = \frac{1}{n} \sum_{i=1}^n X_i^2$. Find the asymptotic distribution of T_n .

1. Define a new sequence: Let $Y_i = X_i^2$. Because X_i are IID, Y_i are also IID. Notice that $T_n = \frac{1}{n} \sum Y_i = \bar{Y}_n$ (The sample mean of the Y 's).

2. Find the Mean and Variance of Y_i :

- Mean: $\mu_Y = E(Y_i) = E(X_i^2) = \text{Var}(X_i) + [E(X_i)]^2 = \sigma^2 + 0 = \sigma^2$.
- Variance: $\sigma_Y^2 = \text{Var}(Y_i) = E(Y_i^2) - [E(Y_i)]^2 = E(X_i^4) - \sigma^4$. Since X_i is Normal, its 4th moment is $3\sigma^4$. $\sigma_Y^2 = 3\sigma^4 - \sigma^4 = 2\sigma^4$.

3. Apply the CLT to $\bar{Y}_n$: By the Central Limit Theorem, $\frac{\sqrt{n}(\bar{Y}_n - \mu_Y)}{\sigma_Y} \xrightarrow{d} N(0, 1)$. Substituting our parameters back in:

$$\frac{\sqrt{n}(T_n - \sigma^2)}{\sqrt{2\sigma^4}} \xrightarrow{d} N(0, 1)$$

This shows that for large samples, our estimator T_n is approximately Normally distributed around σ^2 with variance $\frac{2\sigma^4}{n}$.

56 Formalizing Asymptotic Distributions

We have discussed convergence in distribution ($T_n \xrightarrow{d} T$). In practice, as $n \rightarrow \infty$, the variance of our estimator usually shrinks to 0, causing the distribution to degenerate into a single spike at the true parameter θ .

To study the *shape* of the errors, we must "zoom in" by scaling the difference between the estimator and the truth.

Definition 56.1: Asymptotic Distribution

By the **asymptotic distribution** of an estimator T_n , we mean there exist two sequences of real numbers $\{a_n\}$ and $\{b_n\}$ (where $b_n > 0$) and a *non-degenerate* probability distribution F , such that:

$$\frac{T_n - a_n}{b_n} \xrightarrow{d} Z, \quad \text{where } Z \sim F$$

- a_n is typically the mean (e.g., μ or θ).
- b_n is the shrinking scale factor (e.g., $\sigma/\sqrt{n}$ or $1/n$).

Example (The CLT): Let $X_i \stackrel{iid}{\sim} F$ with $E(X_1) = \mu$ and $\text{Var}(X_1) = \sigma^2$. The Central Limit Theorem states $\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \xrightarrow{d} N(0, 1)$. Here, $a_n = \mu$, $b_n = \frac{\sigma}{\sqrt{n}}$, and the asymptotic distribution is Standard Normal.

57 Asymptotic Distribution of Extreme Order Statistics

The Central Limit Theorem applies to sums and averages (which converge at a rate of $1/\sqrt{n}$). But what about order statistics like the maximum, $X_{(n)}$?

Example: The Maximum of a Uniform Distribution

Let $X_1, \dots, X_n \stackrel{iid}{\sim} U(0, \theta)$. Let $T_n = X_{(n)}$. We know $T_n \xrightarrow{P} \theta$. Because T_n approaches θ much faster than an average does, scaling by $\sqrt{n}$ will not work. We must scale by n . Let us define a new random variable representing the scaled error:

$$W_n = n(\theta - T_n)$$

We want to find the asymptotic distribution of W_n .

1. Formulate the CDF of W_n : For $w \geq 0$:

$$\begin{aligned} F_n(w) &= P(W_n \leq w) = P(n(\theta - T_n) \leq w) \\ &= P\left(\theta - T_n \leq \frac{w}{n}\right) \\ &= P\left(T_n \geq \theta - \frac{w}{n}\right) \\ &= 1 - P\left(T_n < \theta - \frac{w}{n}\right) \end{aligned}$$

2. Substitute the CDF of T_n : The CDF of the maximum order statistic is $F_{T_n}(t) = \left(\frac{t}{\theta}\right)^n$.

$$F_n(w) = 1 - \left(\frac{\theta - w/n}{\theta}\right)^n = 1 - \left(1 - \frac{w}{n\theta}\right)^n$$

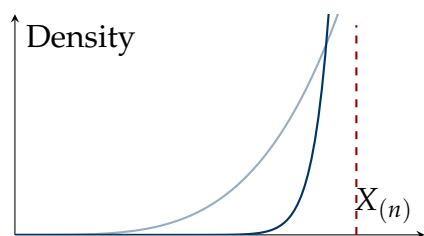
3. Take the Limit as $n \rightarrow \infty$: Using the standard limit definition of Euler's number e , $\lim_{n \rightarrow \infty} \left(1 - \frac{x}{n}\right)^n = e^{-x}$.

$$\lim_{n \rightarrow \infty} F_n(w) = 1 - e^{-w/\theta} \quad \text{for } w \geq 0$$

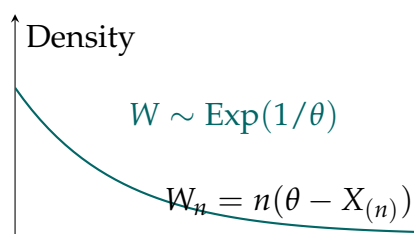
Conclusion: This limit is exactly the Cumulative Distribution Function of an Exponential distribution with mean θ (i.e., rate parameter $\lambda = 1/\theta$).

$$n(\theta - X_{(n)}) \xrightarrow{d} \text{Exp}(1/\theta)$$

Original $X_{(n)}$ (Left-Skewed Spike)



Scaled W_n (Exponential Tail)



Notice the transformation: the dense cluster of probability just to the left of θ is stretched out by the factor of n and reversed in direction, perfectly forming the decaying tail of an exponential distribution!

58 The Practical Power of the CLT: Sample Size Calculation

Why do we care about asymptotic approximations? Let's compare a rigorous mathematical bound (Chebyshev) against our asymptotic approximation (CLT) to see why the CLT is indispensable in practice.

Example: Sample Size: Chebyshev vs. CLT

Let $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Exp}(1/\theta)$ (meaning $E(X_i) = \theta$ and $\text{Var}(X_i) = \theta^2$). We want to find the minimum sample size n that ensures our sample mean $\bar{X}_n$ is within 10% of the true mean θ , with at least 99% probability.

Mathematically, we require:

$$P(|\bar{X}_n - \theta| \leq 0.1\theta) \geq 0.99 \iff P(|\bar{X}_n - \theta| > 0.1\theta) \leq 0.01$$

Method 1: The Chebyshev Bound (Rigorous but Conservative)

$$P(|\bar{X}_n - \theta| > 0.1\theta) \leq \frac{\text{Var}(\bar{X}_n)}{(0.1\theta)^2} = \frac{\theta^2/n}{0.01\theta^2} = \frac{1}{0.01n}$$

To guarantee the error is ≤ 0.01 , we set the bound:

$$\frac{1}{0.01n} \leq 0.01 \implies n \geq \frac{1}{(0.01)(0.01)} \implies n \geq 10,000$$

Method 2: The CLT Approximation (Practical) By the CLT, $\frac{\sqrt{n}(\bar{X}_n - \theta)}{\theta} \xrightarrow{d} N(0, 1)$.

$$\begin{aligned} P(|\bar{X}_n - \theta| \leq 0.1\theta) &= P\left(\frac{\sqrt{n}|\bar{X}_n - \theta|}{\theta} \leq 0.1\sqrt{n}\right) \\ &\approx P(|Z| \leq 0.1\sqrt{n}) \quad \text{where } Z \sim N(0, 1) \end{aligned}$$

We want $P(-0.1\sqrt{n} \leq Z \leq 0.1\sqrt{n}) \geq 0.99$. Using the standard normal CDF Φ :

$$2\Phi(0.1\sqrt{n}) - 1 \geq 0.99 \implies \Phi(0.1\sqrt{n}) \geq 0.995$$

Looking up the Z-table, the critical value for 0.995 is $z \approx 2.576$.

$$0.1\sqrt{n} \geq 2.576 \implies \sqrt{n} \geq 25.76 \implies n \geq 663.5$$

Therefore, we only need $n = 664$!

The Verdict

$664 \ll 10,000$. Chebyshev provides an absolute, unbreakable mathematical guarantee for *any* distribution shape, meaning it must account for the absolute worst-case scenario. The CLT recognizes that averages quickly become bell-shaped, allowing us to achieve the same confidence with a fraction of the data!

59 CLT on Functions of Random Variables

The CLT applies to the sample average of *any* sequence of IID random variables, even if those variables are themselves transformations of other data.

Example: Asymptotic Distribution of the Sample Second Moment

Let $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Exp}(1/\theta)$. What is the asymptotic distribution of $T_n = \frac{1}{n} \sum_{i=1}^n X_i^2$?

1. Treat X_i^2 as a new variable Y_i : Let $Y_i = X_i^2$. Because X_i are IID, Y_i are also IID. Our estimator is simply the sample mean of these new variables: $T_n = \bar{Y}_n$.

2. Find the Mean and Variance of Y_i : For an Exponential distribution with mean θ , the k -th raw moment is $E(X^k) = k!\theta^k$.

- **Mean of Y :** $E(Y_i) = E(X_i^2) = 2!\theta^2 = 2\theta^2$.
- **Variance of Y :** $\text{Var}(Y_i) = E(Y_i^2) - [E(Y_i)]^2 = E(X_i^4) - (2\theta^2)^2$. Using the moment formula, $E(X_i^4) = 4!\theta^4 = 24\theta^4$.

$$\text{Var}(Y_i) = 24\theta^4 - 4\theta^4 = 20\theta^4$$

3. Apply the CLT directly to T_n : Since T_n is the sample mean of the Y_i 's, the CLT states:

$$\frac{\sqrt{n}(T_n - E(Y_i))}{\sqrt{\text{Var}(Y_i)}} \xrightarrow{d} N(0, 1)$$

Substitute our calculated moments:

$$\frac{\sqrt{n} \left(\frac{1}{n} \sum_{i=1}^n X_i^2 - 2\theta^2 \right)}{\sqrt{20\theta^4}} \xrightarrow{d} N(0, 1)$$

We conclude that T_n is asymptotically normal. The asymptotic mean is $2\theta^2$, and the asymptotic variance is $\frac{20\theta^4}{n}$.

60 Testing of Hypotheses: The Foundations

Up to this point, our goal has been to estimate an unknown parameter θ . Now, our goal shifts: we have a preconceived claim about the population, and we must decide whether the sample data supports or refutes this claim.

Definition 60.1: Statistical Hypothesis

A **Statistical Hypothesis** is a formal statement about the probability distribution of a population. In a parametric setup where $X_1, \dots, X_n \stackrel{iid}{\sim} F_\theta$, it is a statement specifically about the parameter θ or a function $g(\theta)$.

60.1 The Null and Alternative Hypotheses

Hypothesis testing is formulated as a decision between two competing statements:

- **Null Hypothesis (H_0):** The statement about the parameter that is actively being tested. It usually represents the status quo, no effect, or a specific baseline parameter (e.g., a coin is fair: $\theta = 1/2$).
- **Alternative Hypothesis (H_1 or H_A):** Any statement that contradicts the null hypothesis. It represents the researcher's new claim or the presence of an effect.

Directionality:

- **Two-Sided:** $H_0 : \theta = \theta_0$ vs. $H_1 : \theta \neq \theta_0$.
- **One-Sided:** $H_0 : \theta = \theta_0$ vs. $H_1 : \theta > \theta_0$ (or $H_1 : \theta < \theta_0$).

61 Simple vs. Composite Hypotheses

Definition 61.1: Simple Hypothesis

Under a hypothesis H , if the probability distribution of the population is **completely specified**, then H is a **Simple Hypothesis**. There are no unknown parameters left.

Definition 61.2: Composite Hypothesis

If the distribution is not completely specified under H , it is a **Composite Hypothesis**.

Example: Identifying Simple vs Composite

1. **Bernoulli Example:** Let $X_i \stackrel{iid}{\sim} \text{Bern}(\theta)$.

- $H_0 : \theta = 0.5 \implies$ **Simple** (We know everything about the distribution).
- $H_1 : \theta > 0.5 \implies$ **Composite** (It could be $\text{Bern}(0.75)$, $\text{Bern}(0.8)$, etc.).

2. **Gamma Example (The Nuisance Parameter):** Let $X_i \stackrel{iid}{\sim} \text{Gamma}(\alpha, \beta)$.

- Assume $\beta = 1$ is known. $H_0 : \alpha = 0.5. \implies$ **Simple** (Distribution is $\text{Gamma}(0.5, 1)$).
- Assume β is *unknown*. $H_0 : \alpha = 0.5. \implies$ **Composite** (Distribution is $\text{Gamma}(0.5, \beta)$. The unknown β makes it incomplete).

62 Test Functions and Regions

To make a decision, we define a mathematical rule that maps our sample data to an action (Accept or Reject H_0).

Definition 62.1: The Test Function (ϕ)

A test function, denoted $\phi(\mathbf{x})$, represents the probability of rejecting H_0 given the observed sample $\mathbf{x}$.

$$\phi : \mathcal{X} \rightarrow [0, 1]$$

For *non-randomized* tests, $\phi(\mathbf{x})$ only takes values of 0 (Accept) or 1 (Reject).

Based on $\phi(\mathbf{x})$, the sample space $\mathcal{X}$ is partitioned into two mutually exclusive regions:

1. **Critical Region ($\mathcal{C}$):** The subset of the sample space where H_0 is rejected.

$$\mathcal{C} = \{\mathbf{x} \in \mathcal{X} : \phi(\mathbf{x}) = 1\}$$

2. **Acceptance Region ($\mathcal{A}$):** The subset where H_0 is accepted.

$$\mathcal{A} = \{\mathbf{x} \in \mathcal{X} : \phi(\mathbf{x}) = 0\}$$

Example: Coin Toss Example

Let $n = 10$ tosses of a coin. $X_i \sim \text{Bern}(\theta)$. We test $H_0 : \theta = 0.5$ vs $H_1 : \theta > 0.5$. Our sample statistic is the total number of heads: $Y = \sum_{i=1}^{10} X_i \in \{0, 1, \dots, 10\}$.

We could propose different tests (rules):

- **Test A:** Reject H_0 if $Y \geq 8$.

$$\phi_A(\mathbf{x}) = \begin{cases} 1 & \text{if } \sum x_i \geq 8 \\ 0 & \text{otherwise} \end{cases} \implies C_A = \{8, 9, 10\}$$

- **Test B:** Reject H_0 if $Y \geq 9$.

$$\phi_B(\mathbf{x}) = \begin{cases} 1 & \text{if } \sum x_i \geq 9 \\ 0 & \text{otherwise} \end{cases} \implies C_B = \{9, 10\}$$

Which test is better? To answer this, we must evaluate the errors they produce.

63 The Two Types of Errors

Because our decision is based on a random sample, we are subject to randomness and will sometimes make the wrong choice. There are exactly two ways to be wrong:

Decision $\rightarrow$	Accept H_0 ($\phi = 0$)	Reject H_0 ($\phi = 1$)
Truth: H_0 is True	Correct Decision	Type I Error
Truth: H_1 is True	Type II Error	Correct Decision

- **Type I Error (α):** Rejecting the Null Hypothesis when it is actually true. (False Positive).

$$P(\text{Type I Error}) = P(\phi(\mathbf{x}) = 1 \mid H_0 \text{ is true})$$

- **Type II Error (β):** Accepting the Null Hypothesis when the Alternative is actually true. (False Negative).

$$P(\text{Type II Error}) = P(\phi(\mathbf{x}) = 0 \mid H_1 \text{ is true})$$

64 Case Study: Normal Distribution Errors

Let $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, 1)$. We want to test:

$$H_0 : \mu = 0 \quad \text{vs.} \quad H_1 : \mu = 1$$

We propose the following test function ϕ^* : Reject H_0 if the sample mean $\bar{X}_n \geq 0.5$.

$$\phi^*(\mathbf{x}) = \begin{cases} 1 & \text{if } \bar{X}_n \geq 0.5 \\ 0 & \text{otherwise} \end{cases}$$

64.1 Calculating Error Probabilities

1. Probability of Type I Error: Assume H_0 is true ($\mu = 0$). The sampling distribution is $\bar{X}_n \sim N(0, \frac{1}{n})$.

$$\begin{aligned} P(\text{Type I Error}) &= P(\text{Reject } H_0 \mid \mu = 0) \\ &= P(\bar{X}_n \geq 0.5 \mid \mu = 0) \end{aligned}$$

Standardizing to $Z \sim N(0, 1)$:

$$\begin{aligned} &= P\left(\frac{\bar{X}_n - 0}{1/\sqrt{n}} \geq \frac{0.5 - 0}{1/\sqrt{n}}\right) \\ &= P(Z \geq 0.5\sqrt{n}) = 1 - \Phi(0.5\sqrt{n}) \end{aligned}$$

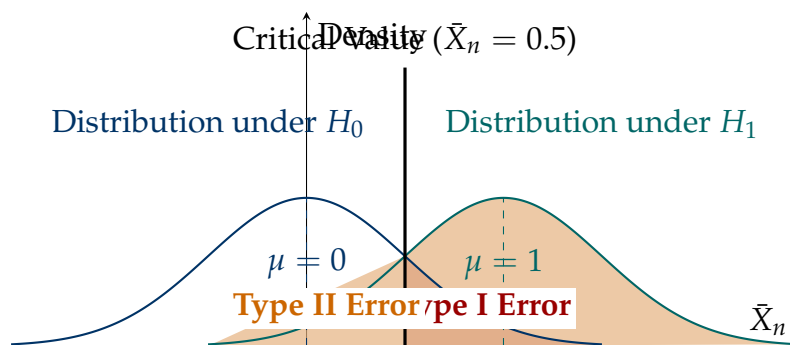
2. Probability of Type II Error: Assume H_1 is true ($\mu = 1$). The sampling distribution is $\bar{X}_n \sim N(1, \frac{1}{n})$.

$$\begin{aligned} P(\text{Type II Error}) &= P(\text{Accept } H_0 \mid \mu = 1) \\ &= P(\bar{X}_n < 0.5 \mid \mu = 1) \end{aligned}$$

Standardizing to $Z \sim N(0, 1)$:

$$\begin{aligned} &= P\left(\frac{\bar{X}_n - 1}{1/\sqrt{n}} < \frac{0.5 - 1}{1/\sqrt{n}}\right) \\ &= P(Z < -0.5\sqrt{n}) \\ &= \Phi(-0.5\sqrt{n}) = 1 - \Phi(0.5\sqrt{n}) \quad (\text{by symmetry}) \end{aligned}$$

Visualizing Type I and Type II Errors ($n = 4$)



! Important: The Role of Sample Size (n)

Notice that both error probabilities depend on n . As $n \rightarrow \infty$, the term $0.5\sqrt{n} \rightarrow \infty$. Therefore, $\Phi(0.5\sqrt{n}) \rightarrow \Phi(\infty) = 1$. Both $P(\text{Type I})$ and $P(\text{Type II})$ will approach $1 - 1 = 0$.

Intuition: As we collect more data, the variance of the sample mean ($1/n$) shrinks. The two bell curves in the diagram above will become infinitely narrow spikes centered perfectly at 0 and 1. They will no longer overlap, and the probability of making an error drops to zero! This is the fundamental power of Large Sample Theory applied to Hypothesis Testing.

65 The Trade-off Between Errors

In an ideal world, we would want both $P(\text{Type I Error}) = 0$ and $P(\text{Type II Error}) = 0$. However, for a fixed sample size n , this is mathematically impossible. Decreasing one error strictly increases the other.

Example: Example: The Exponential Mean Reversal

Let $X_1, \dots, X_n \sim \text{Exp}(\lambda)$. Recall that $E(X_i) = \frac{1}{\lambda}$.

We want to test $H_0 : \lambda = 1$ vs. $H_1 : \lambda > 1$.

Test Structure: Since $\lambda > 1$ implies the true mean is *smaller* than 1, evidence favoring H_1 will manifest as smaller values in our sample. Thus, our rejection rule must be:

$$\phi(\mathbf{x}) = \begin{cases} 1 & \text{if } \bar{X}_n \leq c \\ 0 & \text{otherwise} \end{cases}$$

Example: Example: Normal Distribution Trade-off

Let $X_i \sim N(\mu, 1)$. Test $H_0 : \mu = 0$ vs. $H_1 : \mu = 2$. Let our test be to reject H_0 if $\bar{X}_n \geq c$.

- $P(\text{Type I Error}) = P(\bar{X}_n \geq c \mid \mu = 0) = 1 - \Phi(c\sqrt{n})$
- $P(\text{Type II Error}) = P(\bar{X}_n < c \mid \mu = 2) = \Phi((c - 2)\sqrt{n})$

! Important: The Seesaw Effect

Observe the role of the critical value ' c ':

1. If we take a **larger** c : We make it harder to reject H_0 . This *reduces* $P(\text{Type I Error})$, but it *increases* $P(\text{Type II Error})$.
2. If we take a **smaller** c , we make it easier to reject H_0 . This *increases* $P(\text{Type I Error})$, but it *reduces* $P(\text{Type II Error})$ and makes the trade-off more severe.

66 Level of Significance (alpha)

Because we cannot minimize both errors simultaneously, we establish a hierarchy. In science and law, a Type I error (convicting an innocent person, or publishing a false discovery) is generally considered worse than a Type II error (letting a guilty person go, or missing a discovery).

Definition 66.1: Level of Significance (α)

We impose an **upper permissible limit** on the probability of a Type I error. This limit is denoted by α , known as the Level of Significance.

$$P(\text{Type I Error}) \leq \alpha$$

Commonly used values in practice are $\alpha = 0.05$ or $\alpha = 0.01$.

Result: The Neyman-Pearson Paradigm

Among the class of all tests that satisfy the level condition ($P(\text{Type I}) \leq \alpha$), we should select the test that yields the **smallest possible probability of Type II error**.

66.1 Finding the Best Critical Region

Returning to our Normal example ($H_0 : \mu = 0$): We require $1 - \Phi(c\sqrt{n}) \leq 0.05 \implies \Phi(c\sqrt{n}) \geq 0.95$.

Let $\tau_{0.05}$ be the upper 5% point of the standard normal distribution (we know $\tau_{0.05} \approx 1.645$).

$$c\sqrt{n} \geq 1.645 \implies c \geq \frac{1.645}{\sqrt{n}}$$

Let $c_0 = \frac{1.645}{\sqrt{n}}$. Any $c \in [c_0, \infty)$ is a valid test that restricts Type I error to ≤ 0.05 .

Which c do we choose? We know that to minimize Type II error, we need c to be as *small* as possible. The smallest permissible value in our valid range is exactly c_0 . Thus, the optimal test is to reject H_0 if $\bar{X}_n \geq \frac{1.645}{\sqrt{n}}$.

67 Discrete Distributions and Randomized Tests

The equality $P(\text{Type I}) = \alpha$ is easy to achieve for continuous distributions because we can slide c continuously. But what happens with discrete distributions where probabilities jump in chunks?

Example: The Binomial Dilemma

Let $X_1, \dots, X_{10} \stackrel{iid}{\sim} \text{Bern}(p)$. We want to test at level $\alpha = 0.05$:

$$H_0 : p = 1/2 \quad \text{vs.} \quad H_1 : p = 3/4$$

Our test statistic is $T = \sum X_i \sim \text{Bin}(10, p)$. We reject H_0 if $T \geq c$.

Let's evaluate $P(\text{Type I Error})$ for different integer values of c :

- **If $c = 10$:** $P(T \geq 10 \mid p = 0.5) = \binom{10}{10}(0.5)^{10} = \frac{1}{1024} \approx 0.00097$.
Valid (≤ 0.05), but **Type II Error** = $P(T < 10 \mid p = 0.75) \approx \mathbf{0.9436}$. (Terrible!)
- **If $c = 9$:** $P(T \geq 9 \mid p = 0.5) = \frac{11}{1024} \approx \mathbf{0.0107}$.
Valid (≤ 0.05). **Type II Error** = $P(T < 9 \mid p = 0.75) \approx \mathbf{0.7558}$. (Better).
- **If $c = 8$:** $P(T \geq 8 \mid p = 0.5) = \frac{11+45}{1024} = \frac{56}{1024} \approx \mathbf{0.0547}$.
Invalid! $0.0547 > 0.05$. We have violated the level of significance.

If we restrict ourselves to deterministic rules, the best we can do is $c = 9$. However, 0.0107 is far below our allowed budget of 0.05. We are "leaving power on the table".

67.1 The Randomized Test (phi star)

To hit *exactly* $\alpha = 0.05$ and extract the absolute minimum Type II error, we interpolate using a **Randomized Test**.

Instead of strictly accepting or rejecting at the boundary $T = 8$, we flip a biased coin. Let γ (gamma) be the probability of rejecting H_0 when $T = 8$. (Note: We use γ here instead of β to avoid confusion with the Type II error rate).

$$\phi^*(\mathbf{x}) = \begin{cases} 1 & \text{if } T \in \{9, 10\} \\ \gamma & \text{if } T = 8 \\ 0 & \text{if } T \in \{0, 1, \dots, 7\} \end{cases}$$

Solving for γ : We set the expected Type I error of this test exactly equal to $\alpha = 0.05$:

$$\begin{aligned} E[\phi^*(\mathbf{x}) \mid H_0] &= 1 \cdot P(T \geq 9) + \gamma \cdot P(T = 8) + 0 \cdot P(T \leq 7) \\ 0.05 &= \frac{11}{1024} + \gamma \left(\frac{45}{1024} \right) \end{aligned}$$

Multiply by 1024:

$$\begin{aligned} 51.2 &= 11 + 45\gamma \\ 40.2 &= 45\gamma \\ \gamma &= \frac{40.2}{45} \approx \mathbf{0.8933} \end{aligned}$$

How to execute this test in reality

1. Collect your 10 samples and calculate $T = \sum x_i$.
2. If $T \geq 9$, **Reject** H_0 .
3. If $T \leq 7$, **Accept** H_0 .
4. If $T = 8$ exactly, draw a random number U from a Uniform(0,1) distribution (e.g., using a computer).
 - If $U \leq 0.8933$, **Reject** H_0 .
 - If $U > 0.8933$, **Accept** H_0 .

This achieves an exact 0.05 Type I error rate across infinite repetitions, providing the most mathematically powerful test possible!

68 Size and Power of a Test

We previously discussed establishing a Level of Significance (α), which acts as an upper bound on our Type I error. Let us define the exact metrics we use to evaluate the quality of a test function ϕ .

Definition 68.1: Power Function

Let $\phi(\mathbf{x})$ be a test function. The **Power Function**, often denoted by $E_\theta[\phi(\mathbf{X})]$ or $\beta_\phi(\theta)$, gives the probability of rejecting the null hypothesis H_0 as a function of the true parameter θ .

$$\text{Power Function} = E_\theta[\phi(\mathbf{X})] = P_\theta(\text{Rejecting } H_0)$$

- **Size of a test:** The maximum probability of a Type I error. It is the supremum of the power function evaluated *under the null hypothesis*.

$$\text{Size} = \sup_{\theta \in \Theta_0} E_\theta[\phi(\mathbf{X})]$$

- **Power of a test:** The probability of making the correct decision to reject H_0 when H_1 is actually true.

$$\text{Power} = 1 - P(\text{Type II Error}) = E_{\theta \in \Theta_1}[\phi(\mathbf{X})]$$

Concept: The Goal of Hypothesis Testing

A "good" test must restrict its Size to be $\leq \alpha$, while simultaneously achieving the **highest possible Power**. A test that achieves the maximum power among all valid level- α tests is called the **Most Powerful (MP) level- α test**.

69 The Neyman-Pearson (NP) Lemma

How do we systematically find the Most Powerful test? For testing a *Simple* Null Hypothesis against a *Simple* Alternative Hypothesis, the Neyman-Pearson Lemma gives us the exact mathematical blueprint.

Let $H_0 : X \sim f_0(x)$ versus $H_1 : X \sim f_1(x)$.

Result: Theorem: Neyman-Pearson Lemma

A test function $\phi^*(\mathbf{x})$ is the **Most Powerful level- α test** if there exists a critical constant $k \geq 0$ such that:

1. The Rejection Rule (Likelihood Ratio):

$$\phi^*(\mathbf{x}) = \begin{cases} 1 & \text{if } f_1(\mathbf{x}) > k f_0(\mathbf{x}) \\ \gamma & \text{if } f_1(\mathbf{x}) = k f_0(\mathbf{x}) \\ 0 & \text{if } f_1(\mathbf{x}) < k f_0(\mathbf{x}) \end{cases}$$

2. The Size Condition:

$$E_{H_0}[\phi^*(\mathbf{X})] = \alpha$$

Intuition: We reject H_0 when the data is significantly more likely to have come from f_1 than from f_0 . We sort every possible sample by the ratio $\frac{f_1(\mathbf{x})}{f_0(\mathbf{x})}$ and reject the ones with the highest ratio until we spend our entire α "budget".

70 Applying NP Lemma: Discrete Case

Let us apply the NP lemma to a discrete random variable X with the following probabilities under H_0 and H_1 :

x	1	2	3	4	5	6
$f_0(x)$	0.01	0.01	0.01	0.01	0.01	0.95
$f_1(x)$	0.05	0.04	0.03	0.02	0.01	0.85
$\lambda(x) = \frac{f_1(x)}{f_0(x)}$	5	4	3	2	1	$85/95 \approx 0.89$

Scenario 1: Find the MP test for $\alpha = 0.01$ We want to reject H_0 for the highest values of $\lambda(x)$, stopping when the sum of $f_0(x)$ reaches 0.01. The highest ratio is at $x = 1$ ($\lambda = 5$). The probability under H_0 is $f_0(1) = 0.01$. This exactly matches our α budget!

$$\phi^*(x) = \begin{cases} 1 & \text{if } x = 1 \\ 0 & \text{if } x \in \{2, 3, 4, 5, 6\} \end{cases}$$

Any k strictly between 4 and 5 (e.g., $k = 4.25$) satisfies the NP lemma conditions.

Scenario 2: Find the MP test for $\alpha = 0.05$ We take the highest ratios until the probabilities under f_0 sum to 0.05. Taking $x = 1, 2, 3, 4, 5$, the sum of f_0 is $0.01 + 0.01 + 0.01 + 0.01 + 0.01 = 0.05$.

$$\phi^*(x) = \begin{cases} 1 & \text{if } x \in \{1, 2, 3, 4, 5\} \\ 0 & \text{if } x = 6 \end{cases}$$

We can choose $k = 0.9$ (since $1 > 0.9 > 0.89$).

71 Applying NP Lemma: Continuous Case

Example: Normal vs. Double Exponential

Test $H_0 : X \sim N(0, 1)$ vs. $H_1 : X \sim \text{DE}(0, 1)$ based on a single observation X .

- $f_0(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$
- $f_1(x) = \frac{1}{2} e^{-|x|}$

1. Form the Likelihood Ratio:

$$\lambda(x) = \frac{f_1(x)}{f_0(x)} = \frac{\frac{1}{2} e^{-|x|}}{\frac{1}{\sqrt{2\pi}} e^{-x^2/2}} = \sqrt{\frac{\pi}{2}} e^{-|x| + x^2/2}$$

The NP lemma states we reject if $\lambda(x) > k$.

$$e^{x^2/2 - |x|} > k \sqrt{\frac{2}{\pi}}$$

2. Algebraic Simplification: Take the natural log of both sides:

$$\frac{x^2}{2} - |x| > \ln \left(k \sqrt{\frac{2}{\pi}} \right)$$

Multiply by 2 and complete the square:

$$x^2 - 2|x| > 2 \ln(\dots)$$

$$x^2 - 2|x| + 1 > 2 \ln(\dots) + 1$$

$$(|x| - 1)^2 > k_0 \quad (\text{where } k_0 \text{ is just a new constant})$$

3. Defining the Rejection Region: Taking the square root gives two possibilities:
 $|x| - 1 > \sqrt{k_0} \implies |x| > 1 + \sqrt{k_0}$ $|x| - 1 < -\sqrt{k_0} \implies |x| < 1 - \sqrt{k_0}$

Thus, we reject H_0 if X is very close to 0 (the sharp peak of the Double Exponential) OR if X is extremely far in the tails (where the DE's heavy tails overpower the Normal's thin tails). We set k_0 such that $P_{H_0}((|X| - 1)^2 > k_0) = \alpha$.

72 Correcting the Homework: Normal Mean Shift

HW Correction: $H_0 : \mu = 0$ vs. $H_1 : \mu = -2$

Let $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, 1)$. We want to find the MP test at level α .

1. The Likelihood Ratio:

$$\begin{aligned} \frac{L(-2; \mathbf{x})}{L(0; \mathbf{x})} &= \frac{\exp \left\{ -\frac{1}{2} \sum (x_i - (-2))^2 \right\}}{\exp \left\{ -\frac{1}{2} \sum (x_i - 0)^2 \right\}} = \exp \left\{ -\frac{1}{2} \sum [(x_i + 2)^2 - x_i^2] \right\} \\ &= \exp \left\{ -\frac{1}{2} \sum (4x_i + 4) \right\} = \exp \{ -2 \sum x_i - 2n \} \end{aligned}$$

2. The Rejection Rule: We reject if the ratio $> k$. Since e^y is increasing, this means the exponent must be large.

$$-2 \sum x_i - 2n > \ln(k) \implies -2n \bar{X}_n > \ln(k) + 2n$$

Dividing by $-2n$ flips the inequality sign:

$$\bar{X}_n < c$$

Intuition: Since the alternative mean (-2) is smaller than the null (0), strong evidence for H_1 consists of very small/negative sample means.

3. Finding c (The Source of the Confusion): Under H_0 , $\bar{X}_n \sim N(0, \frac{1}{n})$. We need $P_{H_0}(\bar{X}_n < c) = \alpha$. Standardizing to $Z \sim N(0, 1)$:

$$P \left(Z < \frac{c - 0}{1/\sqrt{n}} \right) = \alpha \implies P(Z < c\sqrt{n}) = \alpha$$

Let $-z_\alpha$ be the critical Z-score (e.g., for $\alpha = 0.05$, $-z_\alpha = -1.645$).

$$c\sqrt{n} = -1.645 \implies c = \frac{-1.645}{\sqrt{n}}$$

Addressing the notes: If $n = 25$, then $c = \frac{-1.645}{5} = -0.329$. You correctly wrote $\bar{X}_n \leq -0.328$ in your notes, but crossed it out with a question mark "-1.67??". You were confusing the unscaled Z-score (-1.645) with the scaled critical value of $\bar{X}_n$. Your original instinct was perfectly correct!

73 Binomial NP Lemma & Randomized Thresholds

Let's formally verify the randomized test for the Binomial example in your notes.

Example: Testing p in a Binomial Sample

Let $X_1, \dots, X_{10} \stackrel{iid}{\sim} \text{Bern}(p)$. $T = \sum x_i \sim \text{Bin}(10, p)$. $H_0 : p = 1/2$ vs. $H_1 : p = 1/4$. Level $\alpha = 0.05$.

1. Likelihood Ratio:

$$\frac{L(1/4; \mathbf{x})}{L(1/2; \mathbf{x})} = \frac{(1/4)^T (3/4)^{10-T}}{(1/2)^{10}} = \left(\frac{1}{2}\right)^T \left(\frac{3}{2}\right)^{10-T} = \frac{3^{10}}{3^T 2^{10}} \left(\frac{1}{2^T}\right) \cdots \propto \left(\frac{1}{3}\right)^T$$

Because $1/3 < 1$, this ratio grows larger as T gets **smaller**. By the NP Lemma, we reject H_0 for small values of $T \leq c$.

2. Calculating cumulative probabilities under H_0 ($p = 0.5$):

- $P(T = 0) = \frac{1}{1024}$
- $P(T = 1) = \frac{10}{1024}$
- $P(T = 2) = \frac{45}{1024}$

$$P(T \leq 1) = 11/1024 \approx 0.0107 \quad (< 0.05)$$

$$P(T \leq 2) = 56/1024 \approx 0.0547 \quad (> 0.05)$$

3. The Randomized Test: We definitively reject if $T \leq 1$, and randomize if $T = 2$ with probability γ .

$$E_{H_0}[\phi(X)] = P(T \leq 1) + \gamma P(T = 2) = 0.05$$

$$\frac{11}{1024} + \gamma \frac{45}{1024} = \frac{51.2}{1024}$$

$$45\gamma = 40.2 \implies \gamma = 0.8933$$

Your stated test function:

$$\phi(\mathbf{x}) = \begin{cases} 1 & \text{if } \sum x_i \in \{0, 1\} \\ 0.89 & \text{if } \sum x_i = 2 \\ 0 & \text{otherwise} \end{cases}$$

is exactly the Most Powerful level-0.05 test by the Neyman-Pearson Lemma!

74 The Power Function and Composite Hypotheses

Until now, we have evaluated tests using a single number for "Power" because our alternative hypothesis H_1 only had one specific value (a Simple Hypothesis). But what happens when H_1 is composite, covering a whole range of values (e.g., $H_1 : \theta > \theta_0$)?

In this case, the power is no longer a single number; it changes depending on exactly what the true parameter θ is.

Definition 74.1: The Power Function

For a given test ϕ and a parameter space Θ , the **Power Function**, denoted $\beta_\phi(\theta)$, gives the probability of rejecting H_0 as a function of the true parameter θ .

$$\beta_\phi(\theta) = E_\theta[\phi(\mathbf{X})] = P_\theta(\text{Reject } H_0)$$

- If $\theta \in \Theta_0$ (Null is true), we want $\beta_\phi(\theta)$ to be as **small** as possible ($\leq \alpha$). This evaluates the **Size**.
- If $\theta \in \Theta_1$ (Alternative is true), we want $\beta_\phi(\theta)$ to be as **large** as possible. This evaluates the **Power**.

Definition 74.2: Uniformly Most Powerful (UMP) Test

Consider testing $H_0 : \theta = \theta_0$ vs $H_1 : \theta \in \Theta_1$.

A test ϕ^* is a **Uniformly Most Powerful (UMP) level- α test** if:

1. It satisfies the level condition: $\sup_{\theta \in \Theta_0} \beta_{\phi^*}(\theta) \leq \alpha$.
2. For **any other** level- α test ϕ , and for **all** $\theta \in \Theta_1$:

$$\beta_{\phi^*}(\theta) \geq \beta_\phi(\theta)$$

75 Extending NP to Find UMP Tests

The Neyman-Pearson Lemma tells us how to find the Most Powerful (MP) test for θ_0 vs a specific θ_1 . If the critical region we find *does not depend* on the specific value of θ_1 , then that exact same test is simultaneously the MP test for *every* θ_1 in the alternative space. This makes it UMP!

Example: Normal Mean Shift (Revisited)

Let $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, 1)$. Test $H_0 : \mu = 0$ vs. $H_1 : \mu < 0$.

Previously, we picked a specific alternative (e.g., $\mu = -2$) and found that the NP test rule was to reject if:

$$\bar{X}_n \leq c \quad \text{where } c = \frac{-z_\alpha}{\sqrt{n}}$$

Notice a profound fact: **The critical value c only depends on α , n , and the null hypothesis $\mu = 0$.** It does NOT depend on the fact that the alternative was -2 .

If we had chosen $\mu = -1$, or $\mu = -5$, the likelihood ratio would still have yielded the exact same rejection rule: $\bar{X}_n \leq c$. Because this single test maximizes power for *every single value* where $\mu < 0$, it is the **UMP level- α test** for $H_1 : \mu < 0$.

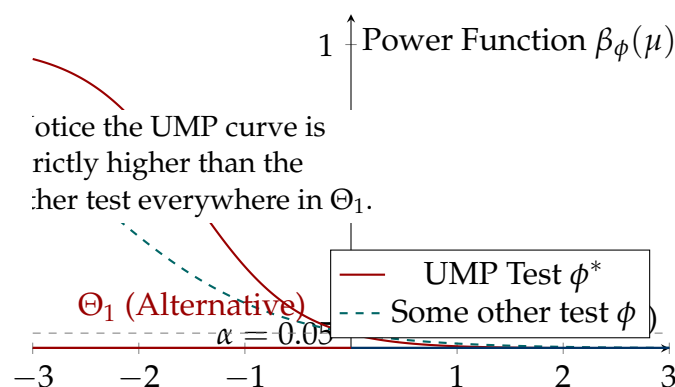
! Important: Does a UMP test always exist?

No! Consider the two-sided alternative $H_1 : \mu \neq 0$.

- To maximize power against $\mu = -2$, we must use the left tail ($\bar{X}_n \leq -c$).
- To maximize power against $\mu = +2$, we must use the right tail ($\bar{X}_n \geq +c$).

There is no single test that can simultaneously be optimal for both sides. Therefore, **a UMP test for a two-sided alternative generally does not exist.**

Visualizing a UMP Test ($H_0 : \mu \geq 0$ vs $H_1 : \mu < 0$)



76 Deep Dive: UMP Randomized Poisson Test

Let us fully flesh out your brilliant homework example to solidify our understanding of discrete UMP tests.

Example: Poisson Parameter Testing

Let $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Poisson}(\lambda)$. We wish to test:

$$H_0 : \lambda = \lambda_0 \quad \text{vs} \quad H_1 : \lambda = \lambda_1 \quad (\text{where } \lambda_1 > \lambda_0)$$

1. The Likelihood Ratio Formulation:

$$\begin{aligned} \frac{L(\lambda_1; \mathbf{x})}{L(\lambda_0; \mathbf{x})} &= \frac{\prod_{i=1}^n \frac{e^{-\lambda_1} \lambda_1^{x_i}}{x_i!}}{\prod_{i=1}^n \frac{e^{-\lambda_0} \lambda_0^{x_i}}{x_i!}} \\ &= \frac{e^{-n\lambda_1} \lambda_1^{\sum x_i}}{e^{-n\lambda_0} \lambda_0^{\sum x_i}} = e^{-n(\lambda_1 - \lambda_0)} \left(\frac{\lambda_1}{\lambda_0} \right)^{\sum x_i} \end{aligned}$$

The NP Lemma dictates we reject if this ratio $> k$. Taking the natural logarithm:

$$-n(\lambda_1 - \lambda_0) + \left(\sum x_i \right) \log \left(\frac{\lambda_1}{\lambda_0} \right) > \log k$$

Because $\lambda_1 > \lambda_0$, the term $\log(\lambda_1/\lambda_0)$ is strictly positive. Therefore, dividing by it **does not flip the inequality**.

$$\sum_{i=1}^n x_i > \frac{\log k + n(\lambda_1 - \lambda_0)}{\log(\lambda_1/\lambda_0)} = k_0$$

Our test statistic is $T_n = \sum X_i \sim \text{Poisson}(n\lambda)$. The optimal rule is to reject if $T_n > k_0$. Because k_0 depends on λ_1 but the *direction* of the rule ($>$) only depends on $\lambda_1 > \lambda_0$, this forms a UMP test!

76.1 Calculating the Randomized Threshold

Assume $n = 5$. Let $H_0 : \lambda = 1$ and $H_1 : \lambda > 1$. Let $\alpha = 0.05$. Under H_0 , $T_5 \sim \text{Poisson}(n\lambda_0) \implies \text{Poisson}(5)$.

We need to find k_0 such that $P_{H_0}(T_5 > k_0) \leq 0.05$. Using Poisson CDF tables (or software):

- If $k_0 = 8$: $P(T_5 > 8) = 1 - P(T_5 \leq 8) \approx 1 - 0.9319 = \mathbf{0.0681} > 0.05$. (Too big).
- If $k_0 = 9$: $P(T_5 > 9) = 1 - P(T_5 \leq 9) \approx 1 - 0.9682 = \mathbf{0.0318} \leq 0.05$. (Valid).

To construct the most powerful level- α test, we set the strict rejection threshold at $T_5 > 9$, and we randomize exactly at the boundary $T_5 = 9$.

$$P_{H_0}(T_5 > 9) + \gamma \cdot P_{H_0}(T_5 = 9) = 0.05$$

We know $P_{H_0}(T_5 = 9) = \frac{e^{-5}5^9}{9!} \approx 0.0363$.

$$0.0318 + \gamma(0.0363) = 0.05$$

$$0.0363\gamma = 0.0182$$

$$\gamma = \frac{0.0182}{0.0363} \approx \mathbf{0.501}$$

Final UMP Test for Poisson(λ)

$$\phi^*(\mathbf{x}) = \begin{cases} 1 & \text{if } \sum x_i \geq 10 \\ 0.501 & \text{if } \sum x_i = 9 \\ 0 & \text{if } \sum x_i \leq 8 \end{cases}$$

This perfectly achieves an exact 0.05 Type I error rate!

77 Formalizing the Power Function

When testing composite hypotheses (e.g., $H_0 : \theta \leq \theta_0$ vs. $H_1 : \theta > \theta_0$), the probability of rejecting H_0 is no longer a single number. It is a continuous curve that depends on the true, unknown value of θ .

Definition 77.1: The Power Function $\beta_\phi(\theta)$

Consider a test ϕ for testing $H_0 : \theta \in \Theta_0$ against $H_1 : \theta \in \Theta_1$.

The **power function** of ϕ , denoted by $\beta_\phi(\theta)$, is the expected value of the test function (which equals the probability of rejecting H_0) evaluated at the parameter θ :

$$\beta_\phi(\theta) = E_\theta[\phi(\mathbf{X})] = P_\theta(\text{Rejecting } H_0)$$

Special Cases:

1. **When $\theta \in \Theta_0$:** $\beta_\phi(\theta) = P_\theta(\text{Rejecting } H_0 \mid H_0 \text{ is true}) = \text{Probability of Type I Error}$.
To maintain a level- α test, we require $\sup_{\theta \in \Theta_0} \beta_\phi(\theta) \leq \alpha$.
2. **When $\theta \in \Theta_1$:** $\beta_\phi(\theta) = \text{Power of the test}$ at the specific alternative θ .

Example: Power Function of the Normal Mean

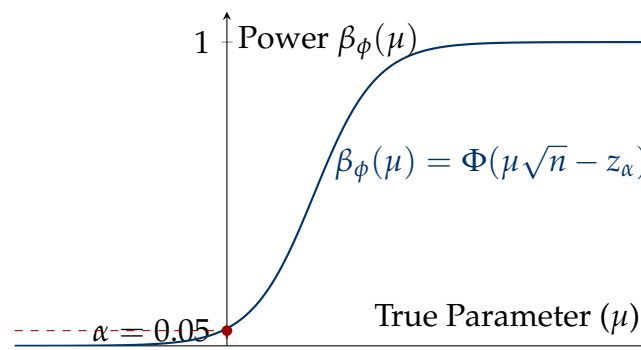
Let $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, 1)$. We test $H_0 : \mu = 0$ (or $\mu \leq 0$) vs. $H_1 : \mu > 0$. Let our test be: Reject H_0 if $\bar{X}_n \geq \frac{z_\alpha}{\sqrt{n}}$ (where $z_{0.05} \approx 1.645$).

$$\phi(\mathbf{x}) = \begin{cases} 1 & \text{if } \bar{X}_n \geq \frac{1.645}{\sqrt{n}} \\ 0 & \text{otherwise} \end{cases}$$

The power function is:

$$\begin{aligned} \beta_\phi(\mu) &= P_\mu \left(\bar{X}_n \geq \frac{1.645}{\sqrt{n}} \right) \\ &= P \left(\frac{\bar{X}_n - \mu}{1/\sqrt{n}} \geq \frac{\frac{1.645}{\sqrt{n}} - \mu}{1/\sqrt{n}} \right) \\ &= P(Z \geq 1.645 - \mu\sqrt{n}) \\ &= 1 - \Phi(1.645 - \mu\sqrt{n}) = \Phi(-\sqrt{n}\mu - 1.645) \quad [\text{By symmetry of Normal}] \end{aligned}$$

Power Function for $H_0 : \mu \leq 0$ vs $H_1 : \mu > 0$



78 UMP Test for the Exponential Distribution

Let us find the UMP test for an Exponential distribution and rigorously prove that its level α condition holds for the entire composite null space.

The Setup: Let $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Exp}(\lambda)$. We want to test:

$$H_0 : \lambda = \lambda_0 \quad \text{vs} \quad H_1 : \lambda > \lambda_0$$

78.1 Step 1: Find the MP test for a specific alternative

Choose a specific alternative $\lambda_1 > \lambda_0$. We find the Most Powerful (MP) test for $H_0 : \lambda = \lambda_0$ vs $H_{11} : \lambda = \lambda_1$. By the Neyman-Pearson Lemma, we reject if $\frac{f_1(\mathbf{x})}{f_0(\mathbf{x})} > k$:

$$\begin{aligned} \frac{\lambda_1^n \exp\{-\lambda_1 \sum x_i\}}{\lambda_0^n \exp\{-\lambda_0 \sum x_i\}} &> k \\ \left(\frac{\lambda_1}{\lambda_0}\right)^n \exp\{-(\lambda_1 - \lambda_0) \sum x_i\} &> k \end{aligned}$$

Take the natural logarithm:

$$n \log\left(\frac{\lambda_1}{\lambda_0}\right) - (\lambda_1 - \lambda_0) \sum x_i > \log k$$

Because $\lambda_1 > \lambda_0$, the term $(\lambda_1 - \lambda_0)$ is strictly positive. When we isolate $\sum x_i$, we divide by this positive number, which **flips** the negative sign and reverses the inequality:

$$\sum_{i=1}^n x_i < \frac{n \log(\lambda_1/\lambda_0) - \log k}{\lambda_1 - \lambda_0} = K_0^*$$

Resulting Rule: $\phi^*(\mathbf{x}) = 1$ if $\sum X_i \leq K_0^*$. *Intuition: The parameter λ is the rate. A higher rate means smaller expected values ($E[X] = 1/\lambda$). Therefore, very small sums provide evidence for the larger λ .*

78.2 Step 2: Argue UMP Status

Observe that the rejection rule ($\sum X_i \leq K_0^*$) completely depends on the fact that $\lambda_1 > \lambda_0$, but the *direction* of the inequality does not depend on the specific numerical value of λ_1 . Thus, ϕ^* is the **Uniformly Most Powerful (UMP)** level- α test for testing $H_0 : \lambda = \lambda_0$ against $H_1 : \lambda > \lambda_0$.

78.3 Step 3: Establish the Exact Critical Region

We need $P_{\lambda_0}(\sum X_i \leq K_0^*) = \alpha$. Recall that the sum of n independent $\text{Exp}(\lambda)$ variables follows a Gamma distribution.

$$T_n = \sum_{i=1}^n X_i \sim \text{Gamma}(n, \lambda)$$

Therefore, K_0^* is exactly the lower α -quantile of the $\text{Gamma}(n, \lambda_0)$ distribution. Let us denote this critical value as $\gamma_{\alpha; n, \lambda_0}$.

79 Extending to a Composite Null Hypothesis

What if our null hypothesis is not a single point, but a region? Consider the updated hypotheses: $H_0^* : \lambda \leq \lambda_0$ vs $H_1 : \lambda > \lambda_0$.

To prove our test ϕ^* is still valid, we must ensure that the probability of a Type I error never exceeds α for *any* value in H_0^* .

$$\sup_{\lambda \leq \lambda_0} \beta_{\phi^*}(\lambda) \leq \alpha$$

Result: Proof via Gamma Transformation

The power function is $\beta_{\phi^*}(\lambda) = P_{\lambda}(T_n \leq \gamma_{\alpha; n, \lambda_0})$. We apply a scale transformation. If $X \sim \text{Exp}(\lambda)$, then $\lambda X \sim \text{Exp}(1)$. By extension, if $T_n \sim \text{Gamma}(n, \lambda)$, then the scaled variable:

$$W_n = \lambda T_n \sim \text{Gamma}(n, 1)$$

Notice that W_n is a **pivotal quantity** — its distribution is completely free of the parameter λ .

We rewrite the power function using W_n :

$$\begin{aligned} \beta_{\phi^*}(\lambda) &= P_{\lambda}(\lambda T_n \leq \lambda \gamma_{\alpha; n, \lambda_0}) \\ &= P(W_n \leq \lambda \gamma_{\alpha; n, \lambda_0}) \end{aligned}$$

Because W_n has a fixed, positive probability density, its CDF is a strictly increasing function. Since the threshold $(\lambda \cdot \gamma_{\alpha; n, \lambda_0})$ increases as λ increases, **the power function $\beta_{\phi^*}(\lambda)$ is a strictly increasing function of λ .**

Conclusion: Because $\beta_{\phi^*}(\lambda)$ strictly increases, its maximum value over the interval $\lambda \leq \lambda_0$ must occur exactly at the boundary $\lambda = \lambda_0$.

$$\sup_{\lambda \leq \lambda_0} \beta_{\phi^*}(\lambda) = \beta_{\phi^*}(\lambda_0) = \alpha$$

Thus, the test perfectly maintains the level- α condition for the entire composite null hypothesis H_0^* .

80 When UMP Tests Do Not Exist (Two-Sided Alternatives)

We have seen how one-sided tests gracefully collapse into a single UMP test. But what happens when we test a two-sided alternative?

Let $X_1, \dots, X_n \sim N(\mu, 1)$. We test $H_0 : \mu = 0$ against the two-sided alternative $H_1 : \mu \neq 0$. The alternative hypothesis is effectively a union of two regions: $H_{11} : \mu > 0$ and $H_{12} : \mu < 0$.

If we treat them separately, we have two different UMP tests:

1. **For $H_{11}(\mu > 0)$:** The UMP test ϕ_1^* rejects if $\bar{X}_n \geq 1.645/\sqrt{n}$. (Captures the right tail).
2. **For $H_{12}(\mu < 0)$:** The UMP test ϕ_2^* rejects if $\bar{X}_n \leq -1.645/\sqrt{n}$. (Captures the left tail).

80.1 The Collision of Critical Regions

If we want a test that is powerful for *both* sides, we might logically try to combine them into a two-tailed test ϕ^{***} :

$$\phi^{***}(\mathbf{x}) = \begin{cases} 1 & \text{if } \bar{X}_n \geq \frac{1.645}{\sqrt{n}} \text{ OR } \bar{X}_n \leq \frac{-1.645}{\sqrt{n}} \\ 0 & \text{otherwise} \end{cases}$$

The Fatal Flaw: What is the size of this combined test?

$$\begin{aligned} \text{Size} &= \beta_{\phi^{***}}(0) = P_0 \left(\bar{X}_n \geq \frac{1.645}{\sqrt{n}} \right) + P_0 \left(\bar{X}_n \leq \frac{-1.645}{\sqrt{n}} \right) \\ &= 0.05 + 0.05 = \mathbf{0.10} \quad (= 2\alpha) \end{aligned}$$

The combined test violates the level- α condition! It is a level-0.10 test, not a level-0.05 test.

80.2 The Power Sacrifice

To force the combined test back down to a valid level- α (0.05), we must shrink the rejection regions by moving the goalposts outward to $z_{\alpha/2} \approx 1.96$:

$$\phi_{\text{valid}}(\mathbf{x}) = \begin{cases} 1 & \text{if } \bar{X}_n \geq \frac{1.96}{\sqrt{n}} \text{ OR } \bar{X}_n \leq \frac{-1.96}{\sqrt{n}} \\ 0 & \text{otherwise} \end{cases}$$

By moving the critical value from 1.645 to 1.96, we made it harder to reject H_0 .

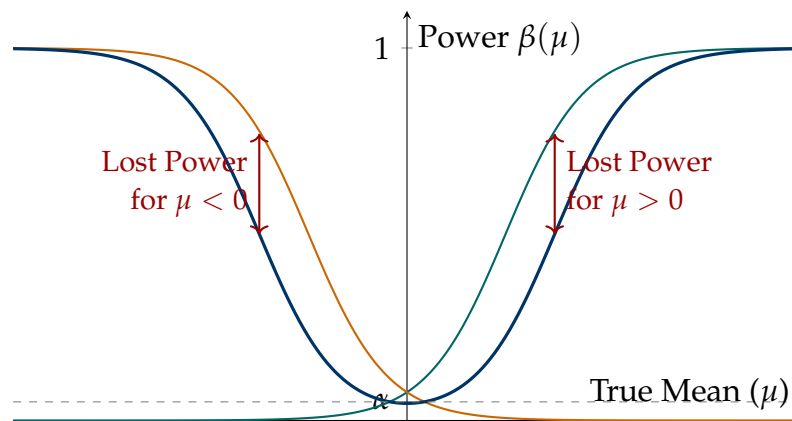
- For any $\mu > 0$, ϕ_{valid} has **strictly less power** than ϕ_1^* .
- For any $\mu < 0$, ϕ_{valid} has **strictly less power** than ϕ_2^* .

The UMP Non-Existence Theorem

Because the valid two-sided test is demonstrably weaker than the right-sided test (for $\mu > 0$) and weaker than the left-sided test (for $\mu < 0$), there is no single test that dominates all others simultaneously across the entire domain of $\mu \neq 0$.

Therefore, a **UMP level- α test does NOT exist for testing $H_0 : \mu = \mu_0$ against $H_1 : \mu \neq \mu_0$.**

The UMP Envelope Paradox (Two-Sided Alternative)



— UMP for $\mu > 0$ (ϕ_1^*) — UMP for $\mu < 0$ (ϕ_2^*) — Valid Two-Sided Test

This visual proof implies: The UMP tests form an unattainable "envelope" of maximum power. The valid two-sided test sags below this envelope on both sides in order to pinch through the constraint $\beta(0) = \alpha$.

81 Power Function Example: Variance

Let us evaluate the power function for a test concerning the variance of a Normal distribution. Let $X_1, \dots, X_n \stackrel{iid}{\sim} N(0, \sigma^2)$. We want to test:

$$H_0 : \sigma \leq 1 \quad \text{vs.} \quad H_1 : \sigma > 1$$

Consider the test function $\phi(\mathbf{x})$ which rejects H_0 if the sample variance S_n^2 is strictly greater than 2:

$$\phi(\mathbf{x}) = \begin{cases} 1 & \text{if } S_n^2 > 2 \\ 0 & \text{otherwise} \end{cases}$$

The power function is defined as the expected value of the test function under the true parameter σ :

$$\begin{aligned} \beta_\phi(\sigma) &= E_\sigma[\phi(\mathbf{X})] \\ &= 1 \cdot P_\sigma(S_n^2 > 2) + 0 \cdot P_\sigma(S_n^2 \leq 2) \\ &= P_\sigma(S_n^2 > 2) \end{aligned}$$

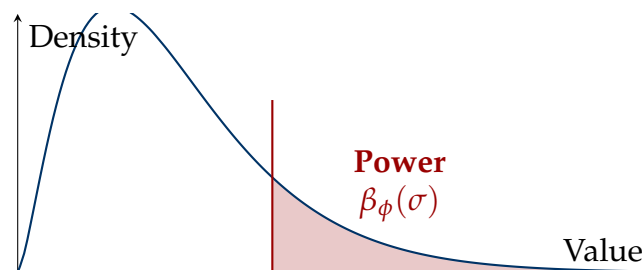
To calculate this probability, we must transform the statistic into a known distribution. We know that for a Normal population, $\frac{nS_n^2}{\sigma^2} \sim \chi_{(n-1)}^2$. We multiply both sides of our inequality by $\frac{n}{\sigma^2}$:

$$\begin{aligned} \beta_\phi(\sigma) &= P\left(\frac{nS_n^2}{\sigma^2} > \frac{2n}{\sigma^2}\right) \\ &= 1 - P\left(\frac{nS_n^2}{\sigma^2} \leq \frac{2n}{\sigma^2}\right) \end{aligned}$$

Let $T_{n-1} \sim \chi_{(n-1)}^2$ with CDF $F_{\chi_{n-1}^2}$. The power function becomes:

$$\beta_\phi(\sigma) = 1 - F_{\chi_{n-1}^2}\left(\frac{2n}{\sigma^2}\right)$$

Sampling Distribution: $T_{n-1} \sim \chi_{(n-1)}^2$



Special Evaluation Points:

- If $\sigma^* \in \Theta_0$ (for example, $\sigma^* \leq 1$), then $\beta_\phi(\sigma^*)$ is the Type I error probability at σ^* .
- If $\sigma^{**} \in \Theta_1$ (for example, $\sigma^{**} > 1$), then $\beta_\phi(\sigma^{**})$ is the power at σ^{**} , equal to $1 - P(\text{Type II error at } \sigma^{**})$.

82 Analyzing the Two-Sided Test and Power Intersections

Consider the problem of testing a Normal mean with known variance. Let $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, 1)$. $H_0 : \mu = 0$ vs. $H_1 : \mu \neq 0$

We split the alternative into two one-sided hypotheses: $H_{11} : \mu > 0$ and $H_{12} : \mu < 0$. Assume we use a critical value constant of 1.67 for our testing regions.

- **Test A (UMP for H_{11}):** $\phi_A(\mathbf{x}) = 1$ if $\bar{X}_n \geq 1.67/\sqrt{n}$.

$$\beta_{\phi_A}(\mu) = P_\mu \left(\bar{X}_n \geq \frac{1.67}{\sqrt{n}} \right) = 1 - \Phi(1.67 - \mu\sqrt{n}) = \Phi(\mu\sqrt{n} - 1.67)$$

- **Test B (UMP for H_{12}):** $\phi_B(\mathbf{x}) = 1$ if $\bar{X}_n \leq -1.67/\sqrt{n}$.

$$\beta_{\phi_B}(\mu) = P_\mu \left(\bar{X}_n \leq \frac{-1.67}{\sqrt{n}} \right) = 1 - \Phi(\mu\sqrt{n} + 1.67)$$

82.1 The Combined Test Failure

If we try to combine these into a two-sided test $\phi_C(\mathbf{x}) = 1$ if $\bar{X}_n \geq \frac{1.67}{\sqrt{n}}$ or $\bar{X}_n \leq \frac{-1.67}{\sqrt{n}}$:

$$\beta_{\phi_C}(\mu) = \Phi(\mu\sqrt{n} - 1.67) + 1 - \Phi(\mu\sqrt{n} + 1.67)$$

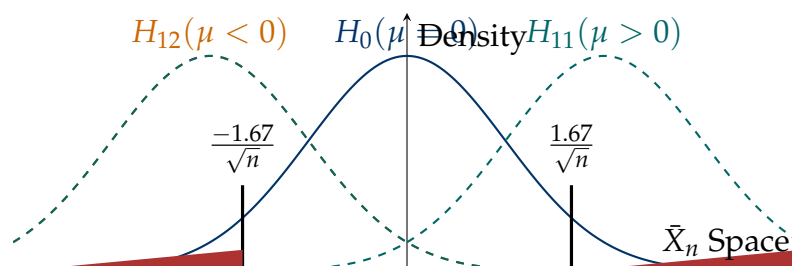
Evaluating the size under the null hypothesis ($\mu = 0$):

$$\beta_{\phi_C}(0) = \Phi(-1.67) + 1 - \Phi(1.67) \approx \alpha + \alpha = 2\alpha$$

Conclusion: ϕ_C is *not* a level- α test; its Type I error rate is doubled.

To correct this, we must reduce the critical region by using a larger threshold (e.g., $c = 2.68$). Let $\phi_C^*(\mathbf{x}) = 1$ if $\bar{X}_n \geq \frac{2.68}{\sqrt{n}}$ or $\bar{X}_n \leq \frac{-2.68}{\sqrt{n}}$. While $\beta_{\phi_C^*}(0) = \alpha$, the power of ϕ_C^* for $\mu > 0$ is strictly less than the power of ϕ_A , and for $\mu < 0$ it is strictly less than ϕ_B . Thus, **there is NO UMP test for H_0 against H_1 .**

Visualizing Rejection Regions under Different True Means



83 The Necessity of Unbiased Tests

What happens if we use the wrong one-sided test? Consider using the right-tailed UMP test ϕ_A when the true parameter is actually $\mu_1 = -5$ (which belongs to H_1 , so we *should* reject H_0).

$$\begin{aligned} P(\text{Type I error of } \phi_A \text{ under } \mu_1) &= P(\text{Accepting } H_0 \text{ using } \phi_A \text{ when } \mu_1 \text{ is true}) \\ &= 1 - \Phi(\mu_1\sqrt{n} - 1.67) = 1 - \Phi(-5\sqrt{n} - 1.67) \end{aligned}$$

Because $\Phi(\text{large negative number}) \approx 0$, this probability evaluates to $1 - 0 = 1$.

This reveals a massive problem with the power of ϕ_A at $\mu = -5$:

$$\text{Power at } \mu_1(-5) \approx 0 < \alpha$$

Because $\text{Power} < \alpha$, we arrive at a paradoxical and undesirable situation:

$$P(\text{Rejecting } H_0 \mid H_1 \text{ is true}) < P(\text{Rejecting } H_0 \mid H_0 \text{ is true})$$

In simple terms: **The probability of making the correct decision is lower than the probability of making the wrong decision!**

Definition 83.1: Unbiasedness Criterion

To prevent this paradox, we demand that a good test should be more likely to reject H_0 when it is false than when it is true. A test ϕ is called an **Unbiased Test** if the power of the test is always larger than or equal to the size of the test for all parameters in the alternative space.

$$\sup_{\theta \in \Theta_0} \beta_\phi(\theta) \leq \inf_{\theta \in \Theta_1} \beta_\phi(\theta)$$

Concept: UMPU Tests

Since a generic UMP test may not exist for two-sided alternatives, we restrict our search space. Among the class of all *unbiased* tests, we search for the one that is most powerful. This is called the **Uniformly Most Powerful Unbiased (UMPU)** level- α test.

84 Introduction to p-values

Up to this point, hypothesis testing has been a binary decision (Accept or Reject) based on a pre-selected level of significance α . However, different researchers might choose different α levels.

To provide a measure of evidence that is independent of a pre-chosen α , we calculate the **p-value**.

Definition 84.1: p-value

Given an observed test statistic x_0 , the **p-value** is the probability of observing a value *at least as extreme* as x_0 , assuming the null hypothesis H_0 is completely true.

Visualizing Extremity: Suppose our test statistic T_n ranges from 0 to 50, and under H_0 , it concentrates around the middle (e.g., 28). If we observe $x_0 = 11$, the "extremity" is the tail probability beyond that point.

$$\text{p-value} = P_{H_0}(T_n \leq 11)$$

If the p-value is extremely small (e.g., 0.002), it means that obtaining a result as low as 11 purely by random chance under H_0 is highly improbable, giving us strong evidence to reject H_0 .

85 Power Function Example: Variance

Let us evaluate the power function for a test concerning the variance of a Normal distribution. Let $X_1, \dots, X_n \stackrel{iid}{\sim} N(0, \sigma^2)$. We want to test:

$$H_0 : \sigma \leq 1 \quad \text{vs.} \quad H_1 : \sigma > 1$$

Consider the test function $\phi(\mathbf{x})$ which rejects H_0 if the sample variance S_n^2 is strictly greater than 2:

$$\phi(\mathbf{x}) = \begin{cases} 1 & \text{if } S_n^2 > 2 \\ 0 & \text{otherwise} \end{cases}$$

The power function is defined as the expected value of the test function under the true parameter σ :

$$\begin{aligned} \beta_\phi(\sigma) &= E_\sigma[\phi(\mathbf{X})] \\ &= 1 \cdot P_\sigma(S_n^2 > 2) + 0 \cdot P_\sigma(S_n^2 \leq 2) \\ &= P_\sigma(S_n^2 > 2) \end{aligned}$$

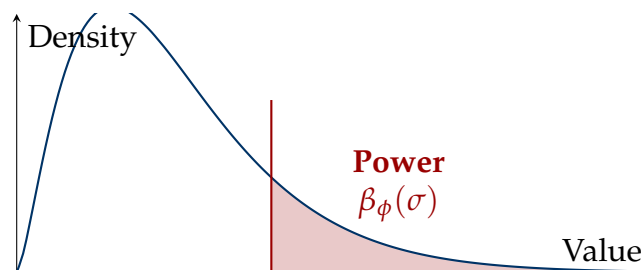
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Let $T_{n-1} \sim \chi_{(n-1)}^2$ with CDF $F_{\chi_{n-1}^2}$. The power function becomes:

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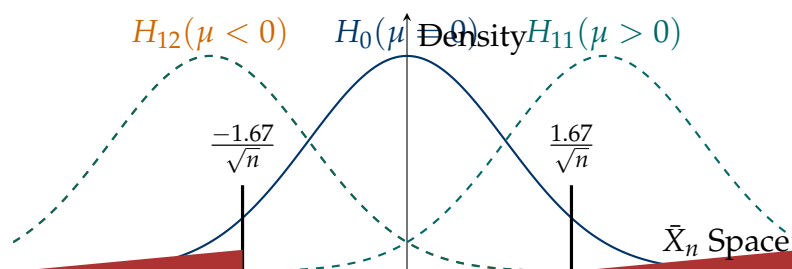
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88 Measuring the Extremity of an Observation

When we conduct a hypothesis test, we establish a critical region before seeing the data. However, simply saying "Reject" or "Accept" doesn't tell us *how strong* the evidence is. We need a continuous measure of extremity for our specific realization of data.

Consider the testing problem: $H_0 : \mu = 0$ vs. $H_1 : \mu > 0$ Based on $n = 100$ samples $X_1, \dots, X_{100} \stackrel{iid}{\sim} N(\mu, 1)$.

The UMP test at a 5% level gives the rule:

$$\phi^*(\mathbf{x}) = \begin{cases} 1 & \text{if } \bar{X}_n \geq \frac{1.645}{\sqrt{100}} = 0.1645 \\ 0 & \text{otherwise} \end{cases}$$

Equivalently, in terms of sums: Reject if $\sum X_i \geq 16.45$.

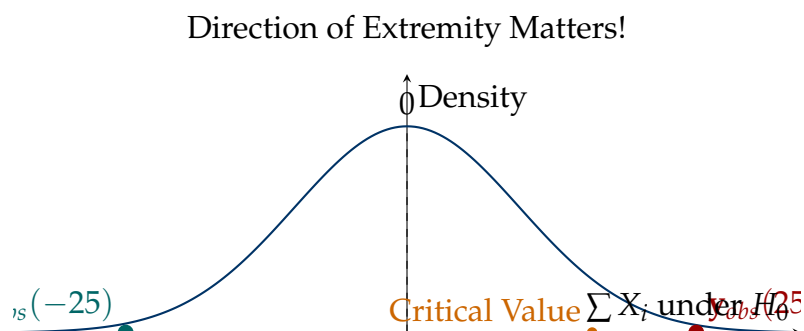
Example: The Problem with PDF Evaluation

Suppose we collect data and observe three different realizations:

- Sample $\mathbf{x}_{obs}$ yields $\sum x_i = 17.25$
- Sample $\mathbf{y}_{obs}$ yields $\sum y_i = 25.7$
- Sample $\mathbf{z}_{obs}$ yields $\sum z_i = -25.0$

How unlikely is it to observe these realizations under H_0 ? We might intuitively think to just evaluate the joint PDF $f_{H_0}(\mathbf{x}_{obs})$. However, this is flawed! Both $f_{H_0}(\mathbf{y}_{obs})$ and $f_{H_0}(\mathbf{z}_{obs})$ will evaluate to extremely small, near-zero probabilities because both 25.7 and -25.0 are far from the mean of 0.

The Catch: While both are rare, $\mathbf{y}_{obs}$ indicates an extreme observation *in favor of* H_1 (since we are testing $\mu > 0$), whereas $\mathbf{z}_{obs}$ actually strongly supports $\mu \leq 0$ and contradicts H_1 !



To accurately measure evidence, we must accommodate the **direction** of rejection. Instead of calculating the probability of the exact point, we calculate the probability of observing a value *at least as extreme* as our realization.

89 Formalizing the p-value

Definition 89.1: The p-value

The **p-value** is the probability of observing data at least as extreme as the realized value, assuming the null hypothesis H_0 is true.

Equivalently, given a realization $\mathbf{x}_{obs}$, the p-value is the infimum (smallest possible value) of the significance level α at which H_0 would be rejected:

$$p(\mathbf{x}_{obs}) = \inf\{\alpha : \text{given } \mathbf{x}_{obs}, H_0 \text{ is rejected against } H_1 \text{ at level } \alpha\}$$

! Important: Calculating p-values

To calculate a p-value, we only require:

1. An appropriate test statistic (e.g., $\bar{X}_n$ or $\sum X_i$).
2. The direction of rejection/extremity (derived from H_1).
 - If $H_1 : \mu > \mu_0$, then $p(\mathbf{x}_{obs}) = P_{H_0}(\bar{X}_n \geq \bar{x}_{obs})$
 - If $H_1 : \mu < \mu_0$, then $p(\mathbf{x}_{obs}) = P_{H_0}(\bar{X}_n \leq \bar{x}_{obs})$

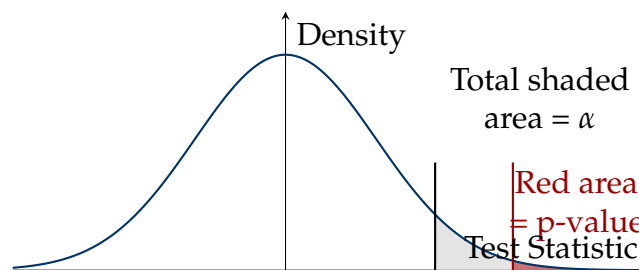
89.1 Testing Using the p-value

The universal rule for hypothesis testing using a p-value is straightforward:

If $p(\mathbf{x}_{obs}) < \alpha$, then we Reject H_0 .

If the p-value is smaller than our error budget (α), it means our data fell firmly inside the critical rejection region.

Visualizing $p(\mathbf{x}_{obs}) < \alpha \implies \text{Reject } H_0$



90 Example: Binomial p-value

Let $X_1, \dots, X_{10} \stackrel{iid}{\sim} \text{Bern}(p)$. We want to test:

$$H_0 : p = 1/4 \quad \text{vs.} \quad H_1 : p > 1/4$$

Consider the specific realization of 10 trials: $\mathbf{x}_{obs} = \{0, 1, 0, 0, 1, 1, 1, 0, 0, 1\} \implies \sum x_{i,obs} = 5$.

Our test statistic is $T_{10} = \sum X_i$. Under H_0 , $T_{10} \sim \text{Bin}(10, 1/4)$. Because H_1 is $p > 1/4$, extreme values are large values. We calculate the p-value:

$$\begin{aligned} p(\mathbf{x}_{obs}) &= P_{H_0}(T_{10} \geq 5) \\ &= P(T_{10} \geq 5 \mid T_{10} \sim \text{Bin}(10, 1/4)) \end{aligned}$$

Using the Binomial PMF $P(T = k) = \binom{10}{k}(1/4)^k(3/4)^{10-k}$, we sum the probabilities for $k = 5, 6, 7, 8, 9, 10$.

$$p(\mathbf{x}_{obs}) \approx \mathbf{0.0781}$$

Conclusion: Suppose our significance level is $\alpha = 0.05$. Because $0.0781 > 0.05$, we **Accept** H_0 given this realization. The evidence is not extreme enough to reject the null hypothesis at the 5% level.

91 p-values for Two-Sided Alternatives

Calculating the p-value becomes slightly more nuanced when the alternative is two-sided ($H_1 : \mu \neq \mu_0$). Because extreme values can occur in *either* tail, we must account for both directions.

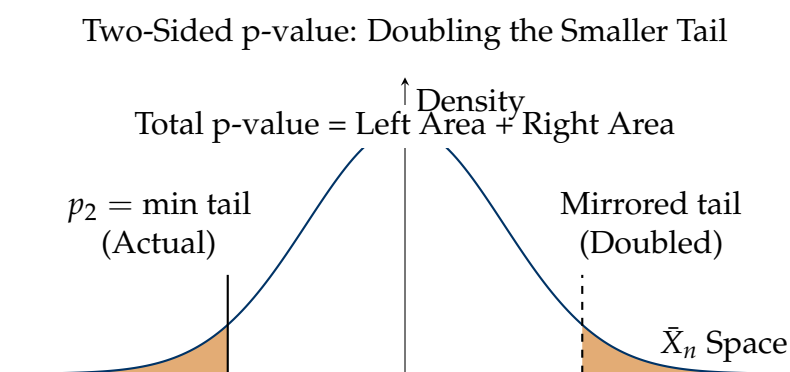
Let $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, 1)$. Test $H_0 : \mu = 0$ vs $H_1 : \mu \neq 0$. Recall that the UMPU test rejects H_0 if $\bar{X}_n \geq c_0$ or $\bar{X}_n \leq -c_0$.

Suppose we observe a realization $\bar{x}_{obs}$. We define two one-sided tail probabilities:

$$\begin{aligned} p_1 &= P_{H_0}(\bar{X}_n \geq \bar{x}_{obs}) \\ p_2 &= P_{H_0}(\bar{X}_n \leq \bar{x}_{obs}) \end{aligned}$$

Because the test is symmetric, an observation is extreme if it is in the smallest tail. To make the p-value comparable to our total α budget, we take the smallest tail probability and double it:

$$p(\mathbf{x}_{obs}) = 2 \min\{p_1, p_2\}$$



Just like the one-sided case, if the combined two-sided $p(\mathbf{x}_{obs}) < \alpha$, we reject H_0 .

92 Interval Estimation (Confidence Intervals)

Up until now, we have focused on **Point Estimation** (where estimators like $\bar{X}_n$ or S_n^2 provide a single, specific number as a guess for the parameter μ or σ^2) and **Hypothesis Testing** (checking if a parameter equals a specific value).

Point estimates are precise but almost certainly wrong in a continuous space (i.e., $P(\bar{X}_n = \mu) = 0$). To reflect the uncertainty inherent in our sample, we use **Interval Estimation**. Instead of a single point, we construct a range of plausible values that is highly likely to contain the true parameter.

Definition 92.1: Interval Estimator

An interval estimator consists of a pair of statistics, $\{L(\mathbf{X}), U(\mathbf{X})\}$, derived from the sample data, such that:

$$L(\mathbf{X}) \leq U(\mathbf{X}) \quad \text{with probability 1.}$$

The interval $[L(\mathbf{X}), U(\mathbf{X})]$ is called the **interval estimate** for the unknown parameter θ .

! Important: Randomness in Intervals

It is crucial to understand where the randomness lies in this framework. The population parameter θ is a fixed, non-random, but unknown constant. The interval bounds $L(\mathbf{X})$ and $U(\mathbf{X})$ are statistics (functions of the random sample), making them **random variables**.

Therefore, it is not the parameter that moves to fall into the interval; it is the interval that moves and breathes with each new random sample, attempting to "catch" the fixed parameter.

92.1 The Confidence Coefficient

How do we measure the quality or reliability of our interval estimator? We evaluate the probability that the random interval successfully contains the true parameter.

Definition 92.2: Confidence Coefficient ($1 - \alpha$)

Let $\{L(\mathbf{X}), U(\mathbf{X})\}$ be an interval estimator for the parameter θ . The **confidence coefficient** of this interval, denoted by $(1 - \alpha)$, is defined as:

$$P_\theta(L(\mathbf{X}) \leq \theta \leq U(\mathbf{X})) \geq 1 - \alpha \quad \text{for all } \theta \in \Theta$$

The resulting interval $[L(\mathbf{x}), U(\mathbf{x})]$ calculated from a specific dataset is called a $100(1 - \alpha)\%$ **Confidence Interval**.

Common choices for $(1 - \alpha)$ include 0.90, 0.95, and 0.99, corresponding to α levels (error rates) of 0.10, 0.05, and 0.01. Just as with hypothesis testing, setting a higher confidence level requires widening the interval, which represents a trade-off between certainty and

precision.

93 Deep Dive: Interval Estimation

While a point estimate provides a single "best guess" for a parameter, it lacks a measure of precision. An **interval estimate** provides a range of values, calculated from the data, that is likely to capture the true parameter.

Example: Motivating Example: Normal Mean

Let $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, 1)$. We know that the sample mean $\bar{X}_n$ is the UMVUE for μ , and its sampling distribution is:

$$\bar{X}_n \sim N\left(\mu, \frac{1}{n}\right)$$

To build an interval around μ , we standardize $\bar{X}_n$ to remove its dependence on μ and n in the variance.

$$T_n = \sqrt{n}(\bar{X}_n - \mu) \sim N(0, 1)$$

Because T_n is Standard Normal, we can easily find two symmetric points (let's call them $-a$ and a) that contain a specific probability, say 90% (0.9). Using the standard normal table, $P(-1.645 \leq Z \leq 1.645) = 0.90$. (Note: The handwritten notes use the approximation 1.67 for this threshold. We will use 1.645 for exactness, but the algebraic process remains identical).

Step 1: Set up the probability statement.

$$P_\mu(-1.645 \leq T_n \leq 1.645) = 0.90$$

Step 2: Substitute T_n and isolate the parameter μ .

$$\begin{aligned} P_\mu(-1.645 \leq \sqrt{n}(\bar{X}_n - \mu) \leq 1.645) &= 0.90 \\ P_\mu\left(\frac{-1.645}{\sqrt{n}} \leq \bar{X}_n - \mu \leq \frac{1.645}{\sqrt{n}}\right) &= 0.90 \end{aligned}$$

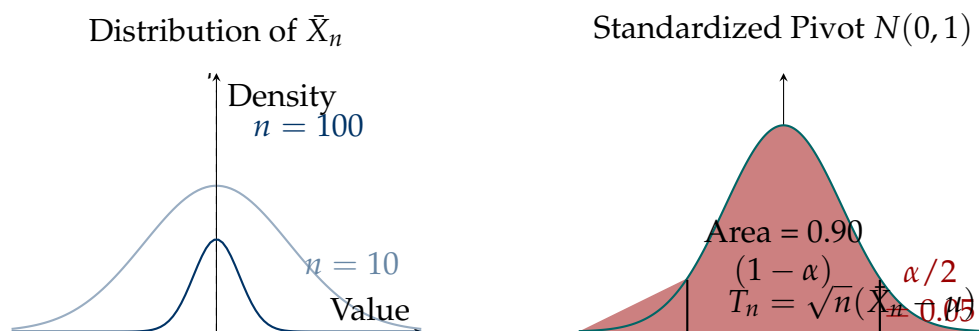
Subtract $\bar{X}_n$ from all sides:

$$P_\mu\left(-\bar{X}_n - \frac{1.645}{\sqrt{n}} \leq -\mu \leq -\bar{X}_n + \frac{1.645}{\sqrt{n}}\right) = 0.90$$

Multiply by -1 (which **flips the inequalities**):

$$P_\mu\left(\bar{X}_n - \frac{1.645}{\sqrt{n}} \leq \mu \leq \bar{X}_n + \frac{1.645}{\sqrt{n}}\right) = 0.90$$

Conclusion: The interval $\left[\bar{X}_n - \frac{1.645}{\sqrt{n}}, \bar{X}_n + \frac{1.645}{\sqrt{n}}\right]$ contains the true mean μ with probability 0.90. The **width of this interval** is $(U - L) = \frac{2 \times 1.645}{\sqrt{n}}$. As sample size n increases, the width shrinks, providing a more precise estimate!



94 Formalizing Confidence Intervals

Definition 94.1: Interval Estimate & Confidence Coefficient

Let $X_1, \dots, X_n \stackrel{iid}{\sim} F_\theta$. An **interval estimate** of a parameter function $g(\theta)$ is a pair of statistics $[L(\mathbf{X}), U(\mathbf{X})]$ such that $L(\mathbf{X}) \leq U(\mathbf{X})$ with probability 1.

The **confidence coefficient (cc)** associated with this interval is a number $\beta = (1 - \alpha) \in [0, 1]$ such that:

$$P_\theta(L(\mathbf{X}) \leq g(\theta) \leq U(\mathbf{X})) \geq 1 - \alpha \quad \text{for all } \theta \in \Theta$$

Equivalently, it is the infimum over all parameter space: $\inf_{\theta \in \Theta} P_\theta(L(\mathbf{X}) \leq g(\theta) \leq U(\mathbf{X})) = 1 - \alpha$.

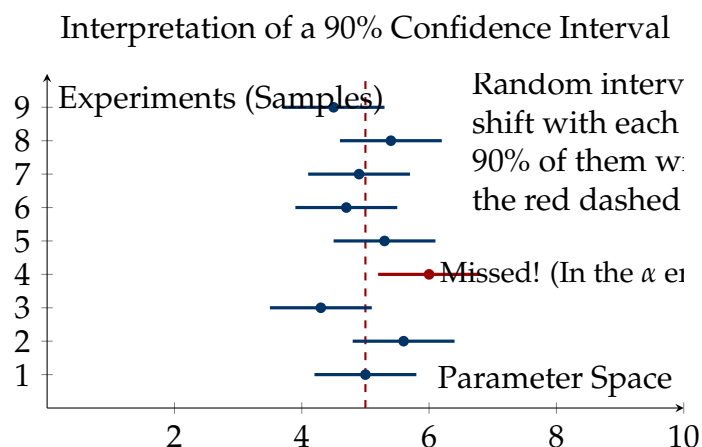
Definition 94.2: Confidence Interval

An interval estimate of $g(\theta)$ and its associated confidence coefficient $(1 - \alpha)$ evaluated together is called a $100(1 - \alpha)\%$ **Confidence Interval (CI)**.

94.1 The Correct Interpretation of a Confidence Interval

It is mathematically incorrect to say "There is a 95% probability that μ is in $[2.1, 4.5]$." Once the data is realized, the interval is fixed. The parameter μ is also fixed. The true probability is either 1 (it's in there) or 0 (it's not).

Frequentist Interpretation: $P_\mu(L(\mathbf{X}) \leq \mu \leq U(\mathbf{X})) = 1 - \alpha$ means that if we were to repeat the experiment N times (where N is very large), drawing new samples and calculating a new interval each time, approximately $N \times (1 - \alpha)$ of those intervals will successfully capture the true, fixed parameter μ .



95 Constructing CIs: Method of Pivots

There are two primary ways to obtain a confidence interval: the **Method of Pivots** and **Test Inversion**. We will focus heavily on the Method of Pivots.

Definition 95.1: Pivot (Pivotal Quantity)

A random variable $T_n(\theta) = g(\mathbf{X}, \theta)$, which is a function of the sample $\mathbf{X}$ and the unknown parameter θ , but whose **probability distribution is completely free of θ** (and all other unknown parameters), is called a **pivot**.

95.1 Examples of Pivots

1. Normal Mean (Variance Known): $X_i \sim N(\mu, \sigma_0^2)$.

$$T_n(\mu) = \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma_0} \sim N(0, 1)$$

This is a valid pivot. It contains data ($\bar{X}_n$), the parameter (μ), and its distribution ($N(0, 1)$) does not depend on μ .

2. Normal Mean (Variance Unknown): $X_i \sim N(\mu, \sigma^2)$. If we try to use $Z = \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma}$, it is **not** a usable pivot because it depends on the unknown nuisance parameter σ . (Completion of notes' '?'): Instead, we replace σ with the sample standard deviation S_n to form the **Student's t-pivot**:

$$T_n(\mu) = \frac{\sqrt{n}(\bar{X}_n - \mu)}{S_n} \sim t_{(n-1)}$$

3. Probability Integral Transform (Uniform Pivot): Let $X_1, \dots, X_n \sim \text{Exp}(\lambda)$. For $n = 1$, the CDF is $F_X(x) = P(X \leq x) = 1 - e^{-\lambda x}$. By the Probability Integral Transform, evaluating any continuous random variable inside its own CDF results in a $\text{Uniform}(0, 1)$ distribution:

$$U_\lambda(X) = 1 - e^{-\lambda X} \sim \text{Uniform}(0, 1)$$

This forms a pivot! For $n > 1$, we use the sufficient statistic $T_n = \sum X_i \sim \text{Gamma}(n, \lambda)$. The CDF of T_n , evaluated at T_n , forms a $U(0, 1)$ pivot.

4. Normal Variance: $X_i \sim N(\mu, \sigma^2)$. We want an interval for $g(\theta) = \sigma^2$.

$$T_n(\sigma^2) = \frac{\sum_{i=1}^n (X_i - \bar{X}_n)^2}{\sigma^2} \sim \chi^2_{(n-1)}$$

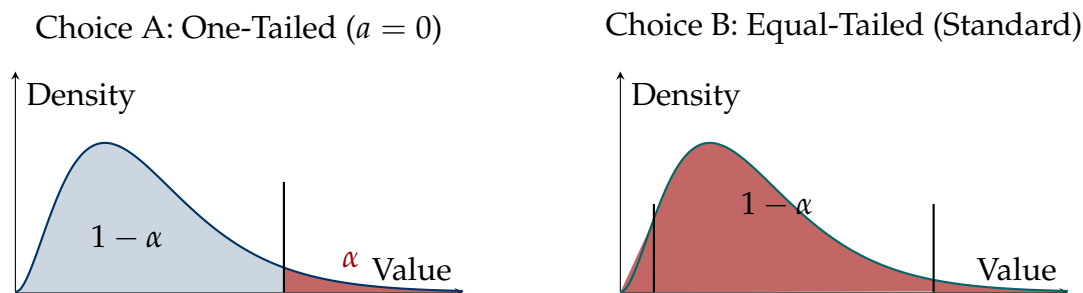
This is a pivot for σ^2 because its distribution (χ^2_{n-1}) is parameter-free.

96 Systematic Steps for the Method of Pivots

To construct a confidence interval using a pivot, follow these 4 rigorous steps:

Step 1: Choose an appropriate pivot. Select a function $T_n(\theta)$ based on a good point estimator. For example, to estimate σ^2 , let $T_n(\sigma^2) = \frac{\sum (X_i - \bar{X}_n)^2}{\sigma^2} \sim \chi^2_{n-1}$.

Step 2: Find constants a, b based on the parameter-free distribution. Since the distribution of $T_n(\theta)$ is known, find points (a, b) such that $P(a \leq T_n(\theta) \leq b) \geq 1 - \alpha$. Because distributions like χ^2 and Uniform are bounded or asymmetric, we have choices:



(Notation: $\chi^2_{\alpha/2; n-1}$ represents the value that leaves $\alpha/2$ area in the right tail. Therefore, the lower bound a that leaves $\alpha/2$ in the left tail is the value that leaves $1 - \alpha/2$ in the right tail, denoted $\chi^2_{1-\alpha/2; n-1}$).

Step 3: Set up the bounds and solve for $g(\theta)$. We use the "Equal-Tailed" choice (Choice B):

$$P\left(\chi^2_{1-\alpha/2; n-1} \leq \frac{\sum (X_i - \bar{X}_n)^2}{\sigma^2} \leq \chi^2_{\alpha/2; n-1}\right) = 1 - \alpha$$

Let the bounds be a and b . We isolate σ^2 :

$$\begin{aligned} a &\leq \frac{\sum (X_i - \bar{X}_n)^2}{\sigma^2} \leq b \\ \frac{1}{b} &\leq \frac{\sigma^2}{\sum (X_i - \bar{X}_n)^2} \leq \frac{1}{a} \quad (\text{Taking reciprocals flips inequalities!}) \\ \frac{\sum (X_i - \bar{X}_n)^2}{b} &\leq \sigma^2 \leq \frac{\sum (X_i - \bar{X}_n)^2}{a} \end{aligned}$$

Substitute a and b back into the bounds. Let these solutions be S_b (lower limit) and S_a (upper limit).

$$S_b = \frac{\sum (X_i - \bar{X}_n)^2}{\chi_{\alpha/2; n-1}^2}, \quad S_a = \frac{\sum (X_i - \bar{X}_n)^2}{\chi_{1-\alpha/2; n-1}^2}$$

Step 4: Formulate the final Interval Estimate. The probability statement is successfully inverted:

$$P(S_b \leq \sigma^2 \leq S_a) = 1 - \alpha$$

Therefore, the $100(1 - \alpha)\%$ confidence interval for the variance σ^2 is:

$$\left[\frac{\sum (X_i - \bar{X}_n)^2}{\chi_{\alpha/2; n-1}^2}, \frac{\sum (X_i - \bar{X}_n)^2}{\chi_{1-\alpha/2; n-1}^2} \right]$$

Notice how the algebraically smaller bound corresponds to the larger χ^2 value in the denominator, and vice versa. This cross-mapping is a standard feature of variance and scale-parameter intervals.

97 Formalizing Interval Estimation

A confidence interval for $g(\theta)$ is an interval $[L(\mathbf{X}), U(\mathbf{X})]$ with $L(\mathbf{X}) \leq U(\mathbf{X})$ with probability 1.

The associated **confidence coefficient** is denoted by $1 - \alpha = \beta$, such that:

$$P_{\theta}(L(\mathbf{X}) \leq g(\theta) \leq U(\mathbf{X})) \geq (1 - \alpha) \quad \forall \theta \in \Theta$$

Or, expressed as an infimum over the parameter space:

$$\inf_{\theta \in \Theta} P_{\theta}(L(\mathbf{X}) \leq g(\theta) \leq U(\mathbf{X})) = (1 - \alpha)$$

We write that $[L(\mathbf{X}), U(\mathbf{X})]$ is a $100(1 - \alpha)\%$ **confidence interval** for $g(\theta)$, or an interval estimate with confidence coefficient $(1 - \alpha)$. *Note: In some cases, $L(\mathbf{X}) = -\infty$ or $U(\mathbf{X}) = \infty$. In such cases, we get a one-sided interval.*

98 The Method of Pivots

Definition 98.1: Pivot

A **pivot** (or pivotal quantity) is a random variable $T_n(\theta) = g(\mathbf{X}, \theta)$, which is a function of the sample $\mathbf{X}$ and the parameter θ , but whose **probability distribution is completely free of θ** .

Let $X_1, \dots, X_n \stackrel{iid}{\sim} F_{\theta}$. We need to find a $100(1 - \alpha)\%$ confidence interval for $g(\theta)$ using the Method of Pivots. This involves four systematic steps:

98.1 Step 1: Find an appropriate pivot

An appropriate pivot $T_n(\theta)$ should be a function of a 'good' estimator of $g(\theta)$ and $g(\theta)$ itself.

- $T_n(\theta)$ should be **solvable** with respect to $g(\theta)$.
- $T_n(\theta)$ should be **monotone** (strictly increasing or decreasing) with respect to $g(\theta)$.

98.2 Step 2: Find a pair of constants

Because the distribution of $T_n(\theta)$ is free of θ and completely known, we can find constants a and b such that:

$$P_{\theta}(a \leq T_n(\theta) \leq b) = 1 - \alpha$$

(Usually, we pick a and b to leave $\alpha/2$ probability in each tail of the distribution of T_n).

98.3 Step 3: Solve the bounds

Set up the equations $T_n(\theta) = a$ and $T_n(\theta) = b$, and solve them for $g(\theta)$. Let the solutions be S_a and S_b , respectively. *Note: Because T_n is a function of the data $\mathbf{X}$ and θ , the solutions S_a and S_b will be functions of $\mathbf{X}$ and a, b . Thus, S_a and S_b are statistics!*

98.4 Step 4: Establish the Inequalities (Monotonicity)

We rewrite the probability statement $P(a \leq T_n(\theta) \leq b) \geq 1 - \alpha$ in terms of $g(\theta)$. The direction of the inequalities depends on the monotonicity of the pivot:

Case A: $T_n(\theta)$ is a strictly increasing function of $g(\theta)$

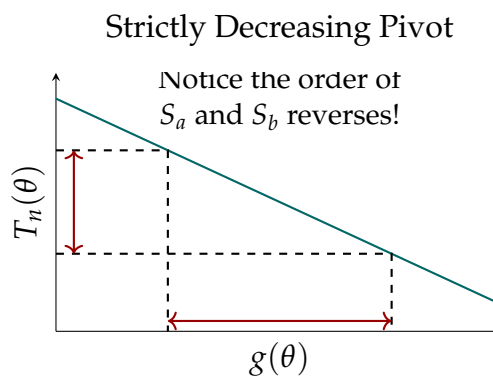
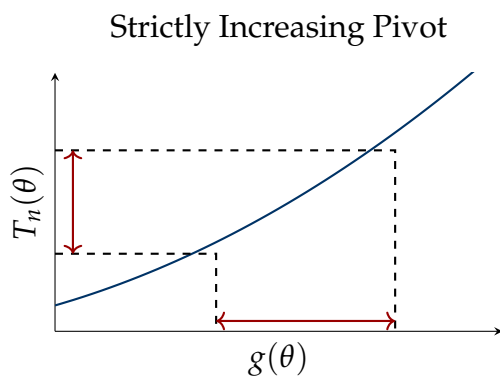
$$a \leq T_n(\theta) \leq b \iff S_a \leq g(\theta) \leq S_b$$

$$P_\theta(S_a \leq g(\theta) \leq S_b) = 1 - \alpha \implies L(\mathbf{X}) = S_a, U(\mathbf{X}) = S_b$$

Case B: $T_n(\theta)$ is a strictly decreasing function of $g(\theta)$

$$a \leq T_n(\theta) \leq b \iff S_b \leq g(\theta) \leq S_a$$

$$P_\theta(S_b \leq g(\theta) \leq S_a) = 1 - \alpha \implies L(\mathbf{X}) = S_b, U(\mathbf{X}) = S_a$$



99 Examples: Applying the Method of Pivots

Example: Example 1: Normal Mean (Known Variance)

Let $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma_0^2)$, where σ_0^2 is known.

Step 1: The standard pivot is $T_n(\mu) = \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma_0} \sim N(0, 1)$.

Step 2: Find bounds (a, b) . We use the standard normal points:

$$a = -z_{\alpha/2} = z_{1-\alpha/2}, \quad b = z_{\alpha/2}$$

So, $P_\mu(z_{1-\alpha/2} \leq T_n(\mu) \leq z_{\alpha/2}) = 1 - \alpha$.

Step 3: Solve for μ .

$$\begin{aligned} a = \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma_0} &\implies S_a = \bar{X}_n - \frac{z_{1-\alpha/2}\sigma_0}{\sqrt{n}} \\ b = \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma_0} &\implies S_b = \bar{X}_n - \frac{z_{\alpha/2}\sigma_0}{\sqrt{n}} \end{aligned}$$

Step 4: Analyze monotonicity. As μ increases, $-\mu$ decreases, so $T_n(\mu)$ is a **monotonically decreasing** function of μ . Therefore, the bounds reverse:

$$P_\mu(S_b \leq \mu \leq S_a) = 1 - \alpha$$

Substituting $z_{1-\alpha/2} = -z_{\alpha/2}$, the confidence interval is:

$$\left[\bar{X}_n - \frac{z_{\alpha/2}\sigma_0}{\sqrt{n}}, \bar{X}_n + \frac{z_{\alpha/2}\sigma_0}{\sqrt{n}} \right]$$

Example: Example 2: Uniform Distribution Maximum

Let $X_1, \dots, X_n \stackrel{iid}{\sim} U(0, \theta)$. We want to estimate $g(\theta) = \theta$. We know the MLE for θ is $X_{(n)}$.

Step 1: An appropriate pivot should involve $X_{(n)}$ and θ . The CDF of $X_{(n)}$ is $F_{X_{(n)}}(t) = (t/\theta)^n$. By the Probability Integral Transform, evaluating a variable at its own CDF yields a $U(0, 1)$ distribution.

$$T_n(\theta) = \left(\frac{X_{(n)}}{\theta} \right)^n \sim U(0, 1)$$

Step 2: Find bounds (a, b) for a $U(0, 1)$ distribution that leave $\alpha/2$ in each tail.

$$a = \alpha/2, \quad b = 1 - \alpha/2$$

Step 3: Solve for θ .

$$\begin{aligned} T_n(\theta) = a &\implies \left(\frac{X_{(n)}}{\theta} \right)^n = \alpha/2 \implies \theta = \frac{X_{(n)}}{(\alpha/2)^{1/n}} = S_a \\ T_n(\theta) = b &\implies \left(\frac{X_{(n)}}{\theta} \right)^n = 1 - \alpha/2 \implies \theta = \frac{X_{(n)}}{(1 - \alpha/2)^{1/n}} = S_b \end{aligned}$$

Step 4: Monotonicity. Since θ is in the denominator, $T_n(\theta)$ is a **monotonically decreasing** function of θ . The bounds reverse: $S_b \leq \theta \leq S_a$.

$$\text{C.I.} = \left[\frac{X_{(n)}}{(1 - \alpha/2)^{1/n}}, \frac{X_{(n)}}{(\alpha/2)^{1/n}} \right]$$

100 Identifying Pivots: Location & Scale Families

Pivots don't have to be guessed; they often arise naturally from the structure of the probability distribution family.

Concept: 1. Location Family

Let $X_1, \dots, X_n \stackrel{iid}{\sim} F_\theta$. The family is a **location family** if we can write:

$$X_i = W_i + \theta$$

where $W_i \stackrel{iid}{\sim} F$, and F is a distribution **completely free of θ** .

Consequence: Any function of $W_1, \dots, W_n$ is free of θ . Therefore, functions of $(X_1 - \theta, \dots, X_n - \theta)$ are valid pivots.

Examples:

- $X_i \sim N(\mu, 1) \implies (X_i - \mu) \sim N(0, 1)$. The pivot is $\sqrt{n}(\bar{X}_n - \mu) = \sqrt{n}\bar{W}_n$.
- $X_i \sim U(\theta - 1/2, \theta + 1/2) \implies W_i = X_i - \theta \sim U(-1/2, 1/2)$.

Concept: 2. Scale Family

The family is a **scale family** if we can write:

$$W_i = \theta X_i \quad \text{or} \quad W_i = X_i / \theta$$

where $W_i \stackrel{iid}{\sim} F$, and F is completely free of θ .

Consequence: Any function of $(\theta X_1, \dots, \theta X_n)$ or $(X_1/\theta, \dots, X_n/\theta)$ is a valid pivot.

Examples:

- $X \sim \text{Exp}(\lambda)$ with PDF $f_X(x) = \lambda e^{-\lambda x}$. Here, $W = \lambda X \sim \text{Exp}(1)$.
- $X \sim N(0, \sigma^2)$ with PDF $f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{x^2}{2\sigma^2}}$. Here, $W = \frac{X}{\sigma} \sim N(0, 1)$.

Concept: 3. Location-Scale Family

The family is a **location-scale family** if the data can be standardized by both a location parameter θ_1 and a scale parameter θ_2 :

$$W_i = \frac{X_i - \theta_1}{\theta_2}$$

where $W_i \stackrel{iid}{\sim} F$, free of θ_1, θ_2 .

Example: $X \sim N(\mu, \sigma^2)$. The standard pivot is $W = \frac{X - \mu}{\sigma} \sim N(0, 1)$.

101 Advanced Pivot Examples

Example: Example 3: Exponential Distribution (Scale Family)

Let $X_1, \dots, X_n \stackrel{iid}{\sim} \text{Exp}(\lambda)$. As seen in the scale family, $W_i = \lambda X_i \sim \text{Exp}(1)$. We choose an appropriate function of W_i : their sum.

$$T_n(\lambda) = \sum_{i=1}^n (\lambda X_i) = \lambda \sum_{i=1}^n X_i \sim \text{Gamma}(n, 1)$$

Step 2: Find (a, b) from the $\text{Gamma}(n, 1)$ distribution.

$$a = \Gamma_{1-\alpha/2; n, 1}, \quad b = \Gamma_{\alpha/2; n, 1}$$

Step 3: Solve for λ .

$$\begin{aligned} a = \lambda \sum X_i &\implies \lambda = \frac{a}{\sum X_i} = \frac{a}{n\bar{X}_n} = S_a \\ b = \lambda \sum X_i &\implies \lambda = \frac{b}{\sum X_i} = \frac{b}{n\bar{X}_n} = S_b \end{aligned}$$

Step 4: Monotonicity. $T_n(\lambda) = \lambda \sum X_i$ is a **strictly increasing** function of λ . Therefore, the bounds do not reverse.

$$P_\lambda \left(\frac{a}{\sum X_i} \leq \lambda \leq \frac{b}{\sum X_i} \right) = 1 - \alpha$$

Example: Homework: $U(\theta, \theta + 1)$ Location Family Pivot

Let $X_1, \dots, X_n \stackrel{iid}{\sim} U(\theta, \theta + 1)$. This is a location family. $W_i = X_i - \theta \sim U(0, 1)$. How do we combine W_i into a good scalar pivot? A brilliant trick from probability theory links Uniform and Exponential distributions: If $W_i \sim U(0, 1)$, then $-\log(W_i) \sim \text{Exp}(1)$. Therefore, the sum of n standard exponentials is a Gamma distribution:

$$T_n(\theta) = \sum_{i=1}^n (-\log W_i) = \sum_{i=1}^n -\log(X_i - \theta) \sim \text{Gamma}(n, 1)$$

This forms a perfect, parameter-free pivot from which an interval can be derived!

Example: Example 4: Normal Mean with Unknown Variance (The T-Pivot)

Let $X_1, \dots, X_n \stackrel{iid}{\sim} N(\mu, \sigma^2)$, where σ^2 is **unknown**.

We have two independent known distributions:

1. $Z = \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma} \sim N(0, 1)$
2. $V = \frac{\sum_{i=1}^n (X_i - \bar{X}_n)^2}{\sigma^2} \sim \chi_{(n-1)}^2$

By definition, the ratio $\frac{Z}{\sqrt{V/(n-1)}}$ follows a Student's t-distribution with $n - 1$ degrees of freedom. Notice how the unknown σ perfectly cancels out:

$$\begin{aligned} T_n(\mu) &= \frac{\frac{\sqrt{n}(\bar{X}_n - \mu)}{\sigma}}{\sqrt{\frac{\sum (X_i - \bar{X}_n)^2}{\sigma^2(n-1)}}} \\ &= \frac{\sqrt{n}(\bar{X}_n - \mu)}{\sqrt{\frac{1}{n-1} \sum (X_i - \bar{X}_n)^2}} = \frac{\sqrt{n}(\bar{X}_n - \mu)}{S_n^*} \sim t_{(n-1)} \end{aligned}$$

Where S_n^* is the unbiased sample standard deviation.

Inversion: Since $t_{(n-1)}$ is symmetric around 0, we choose $a = -t_{\alpha/2; n-1}$ and $b = t_{\alpha/2; n-1}$.

$$-t_{\alpha/2; n-1} \leq \frac{\sqrt{n}(\bar{X}_n - \mu)}{S_n^*} \leq t_{\alpha/2; n-1}$$

Solving for μ (and noting that $-\mu$ causes the bounds to reverse, returning to the symmetric form):

$$P_\mu \left(\bar{X}_n - t_{\alpha/2; n-1} \frac{S_n^*}{\sqrt{n}} \leq \mu \leq \bar{X}_n + t_{\alpha/2; n-1} \frac{S_n^*}{\sqrt{n}} \right) = 1 - \alpha$$